AI Creative Center ## Complete User Manual **Local, offline AI studio for images · video · music · voices · 3D · film** *Master edition — documents every module and function across all editions* Manual revision: 1.0 · Applies to the current desktop release

About this manual

AI Creative Center is a fully local, offline creative suite. Everything — image and video generation, music, voices, 3D models, complete films — runs on your own GPU; nothing is uploaded to the cloud. The only optional network use is a one-time download of model weights (and, in the *Studio Pro* edition, importing models from Civitai).

This is the **master manual**: it documents every feature across every edition. Each function is tagged with the **minimum edition** required to use it, so you always know what your edition unlocks.

How to read a function entry

Every function in this manual follows the same layout:

> #### Function name
>
> **Edition:** *badge* · **Capability:** `cap.id`
>
> **What it does** — a one-line summary. **Where to find it** — the exact tab / sub-tab / menu path. **Step by step** — a numbered, do-this-then-that procedure. **Parameters** — a table listing every control, its default, its range, and when to change it. **Specializations** — sub-modes, variants, and Classic vs. Pro interface differences. **Tips for best results** — practical advice to get professional output.

Edition badges at a glance

Badge	Edition	Highlights
Demo	Free	Image generation (capped), invisible provenance signature. Outputs carry a visible brand watermark.
Supporter	$3	Adds image upscaling; higher image caps. Brand watermark still applied.
Full	€15	Unlocks video, music, audio/voice, 3D, Film Studio, quality tools, LoRA auto-training. No brand watermark.
Unlocked 18+	€25	Everything in *Full* plus the Unlocked (18+) mode (uncensored models/encoders).
Studio Pro	€45	Everything, plus model import (own files + Civitai), per-node overrides, heavy FLUX LoRA training, and the local REST API.

A trailing **+** (e.g. *Full*+) means "this edition and all higher ones".

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Getting started

Your on-ramp to AI Creative Center: what it is, what your machine needs, how to install and launch it, how editions unlock features, and how to find your way around the interface.

Welcome to **AI Creative Center** — a complete, self-contained creative studio that generates images, video, music, voices, 3D models, and entire films on your own computer. This chapter gets you from a fresh install to confidently navigating the app. When you're ready to create, jump to the module chapters: Images, Video, Audio & voice, Music, 3D, and Cinema — AI Film Studio.

What AI Creative Center is

AI Creative Center is a **100% local, offline** desktop application. Every generation — every image, frame of video, note of music, and 3D mesh — is computed on your own GPU. Nothing you create is uploaded to a cloud service, and no account or internet connection is needed to *use* the app.

There are only two optional moments where the app touches the network, and both are under your control:

1. **One-time model-weight downloads** — the first time you use a module, its AI model weights may need to be fetched. Once downloaded, they live on your disk and are reused offline forever after. The setup wizard handles this; see Wizards.
2. **Civitai import (Studio Pro only)** — the *Studio Pro* edition can search and download community models from Civitai. This is an opt-in feature of the Import models tab, not something the app does on its own.

Everything else — including the language you read this in, the models you run, and the files you produce — stays on your machine.

System requirements & GPU / VRAM notes

AI Creative Center runs on Windows with an **NVIDIA GPU**. The single most important factor for what you can do — and how fast — is your GPU's **VRAM**.

- **Image generation** is the lightest workload and runs on modest cards.
- **Video, 3D, and film** are the heaviest and reward more VRAM (24 GB is comfortable for the largest models).
- The app measures your GPU's memory without ever creating a CUDA context in the interface process, so the main window stays stable even when heavy sub-processes are running.

Running on limited VRAM (quantization)

You do **not** need a top-tier card to run large models. AI Creative Center supports several quantized weight formats that trade a small amount of quality for dramatically lower memory use, and it can **auto-select the right format for your VRAM**:

Format	Typical VRAM	Quality vs. full precision	Good for
NVFP4 (mixed)	~14 GB	Almost identical	32 GB cards (fastest)
FP8 scaled (mixed)	~16–18 GB	Almost identical	Universal compromise
GGUF Q8_0	14–16 GB	98–99%	16 GB cards
GGUF Q6_K	11–13 GB	~95%	12 GB cards
GGUF Q5_K_M	9–11 GB	~90%	8 GB cards (sweet spot)
GGUF Q4_K_M	7–9 GB	75–85%	8 / 6 GB cards

VRAM-saving techniques (text-encoder offload, VAE offload, block-swap, model unloading between jobs) stack on top of quantization. For a full walkthrough of running the largest image model on constrained hardware, see `docs/FLUX2_LOW_VRAM.md`. You can also cap the memory the app is allowed to use and let it place models automatically — see **VRAM calibration** in Advanced settings.

Installation & first launch

1. Run the installer for your platform and follow the prompts. The installer is **selective**: it can link the models you already have on disk and download only the weights your edition actually needs, keeping the footprint small.
2. Launch AI Creative Center. On first run, the **setup wizard** opens automatically to connect your local models and fetch any missing weights. Follow it end to end — it is the fastest path to a working install. Full details are in Wizards.
3. When the wizard finishes, the main window opens and you are ready to create.

You can re-open the wizard at any time from the **Tools ▸ Wizard** menu.

Editions & upsell

AI Creative Center ships as a single application whose features unlock according to your **edition**. Higher editions include everything below them.

Edition	Price	What it unlocks
Demo (`free`)	€0	Image generation only (capped at 768 px and 20 generations), invisible provenance signature. Outputs carry a visible **"AI Tools Forge"** brand watermark.
Supporter	$3	Adds image upscaling; image caps raised to 1024 px / 200 generations. Brand watermark still applied.
Full (`plus`)	€15	Unlocks video, audio/voice, music, 3D, Film Studio, quality tools, LoRA auto-training, and the Advanced tab. Brand watermark removed.
Unlocked 18+ (`premium`)	€25	Everything in *Full* plus the **Unlocked (18+) mode** (`cap.adult_mode`): uncensored models and text encoders.
Studio Pro (`studio`)	€45	Everything, plus model import (`cap.import_models`), Civitai import (`cap.civitai_import`), per-node model overrides (`cap.node_overrides`), the heavy FLUX LoRA training tab, and the local REST API.

Throughout this manual, a trailing **+** on an edition badge (e.g. *Full*+) means "this edition and every higher one."

How locked features appear

Locked features are never hidden away — they are shown in the interface with an **upsell** prompt so you always know what the next edition unlocks:

- A tab or panel you haven't unlocked shows a teaser: "*This feature is ready: unlock it and use it right away, in the same window.*"
- A locked control shows a strip reading 🔒 **{feature} — Available in the {tier} plan** with an **Unlock** → button.
- Choosing to unlock opens a dialog explaining that the feature is **already installed and ready** — switching to the indicated plan enables it immediately, with an 🔓 **Unlock on Patreon** button.

Because features are already installed, upgrading is instant: no re-download, no reinstall.

Checking your edition

Your active edition drives which tabs and functions are visible. Feature gating is applied at startup, so if a module is missing it is because your edition doesn't include it — the upsell strip will tell you the minimum tier. (In developer builds, everything is visible.)

The interface

When the main window opens you'll see three areas: the **tab navigation** across the top, the **work area** (with a live canvas/preview on the left), and the **generation queue** dock on the right.

The tab bar

Navigation is a set of large, wrapping buttons (each with an emoji) that switch between modules. In order, the modules are:

Emoji	Module	What it does
🖼️	**Images**	Generate, inpaint/outpaint, recolor, upscale, segmentation, face adapter, compose, face-swap → Images
🎙️	**Audio / Voice**	TTS, voice cloning, the shared voice library → Audio & voice
🎬	**Video**	Text/image/audio-to-video, presets, LoRA, sampler/seed, storyboard → Video
🎛️	**ControlNet**	Guided video generation (canny/depth/pose) → Video
⏩	**Extend video**	Extend an existing clip → Video
✨	**AI Video Edit**	In-place video editing, tracking, inpaint → Video
💎	**Enhance quality**	Photo restore/upscale/colorize and video super-resolution → Enhance quality
🧊	**3D Model**	Image → textured 3D model (TRELLIS.2) → 3D
🦴	**3D Rig**	Add a skeleton to a mesh → 3D
🔄	**Converter**	Convert 3D formats → 3D
🕺	**Animation**	Animate a rig from your webcam (mocap) → 3D
🎓	**LoRA Train Img.**	Train an image LoRA → Training
🎞️	**LoRA Wizard Vid.**	Auto-dataset video LoRA → Training
🖌️	**LoRA Wizard Img.**	Auto-dataset photo LoRA → Training
🎶	**Music**	Songs, instrumentals, and video sonification → Music
🌍	**3D World**	Explorable 3D scene from a photo → 3D
📦	**Model weights**	Download and manage model weights → Wizards
⚙️	**Advanced**	Sampler/scheduler settings, long-video engine, tools, and (Studio Pro) Import Models → Advanced

The **Enhance quality** tab further splits into 🖼️ **Photo** and 🎬 **Video** sub-tabs, and the **Advanced** tab splits into 🛠️ **Tools** and 📦 **Import Models**.

The tabs you see depend on your edition — the Demo edition, for example, shows only the Images module.

The live canvas / preview

The Images module's left-hand **canvas** doubles as your live preview and your editing surface: it shows the current image, and it's where you paint masks for inpainting, isolate subjects, and feed results into other modules (for example "Use generated image" in the 3D tab).

The generation queue dock

Generations run through a **serial queue** — one job at a time, so jobs never fight over your GPU. The **Generation queue** dock on the right lists every job with a status icon:

Icon	Status
⏳	queued
▶	running
✅	done
❌	error
⛔	cancelled

You can **Cancel selected** to stop a queued or running job and **Clear finished** to tidy the list. When the queue is empty it reads "*No generations queued.*" Toggle the dock's visibility with the 📋 button in the navigation bar. Job labels are descriptive (for example *Image*, *Video*, *Super-resolution (SeedVR2)*, *3D model (TRELLIS)*, *Lip-sync*).

The Tools menu

The **Tools** menu (labeled **Strumenti**) collects the app-wide utilities:

- **Wizard** — re-run the setup wizard (Wizards).
- **Settings** — the model/engine settings dialog (below).
- 🖼 **Navigator Gallery...** (Ctrl+G) — browse past generations with metadata and parameter reuse.
- 🔍 **Analyze settings...** — check your current parameters for conflicts.
- 🎬 **Workflow presets...** — 35 ready-made workflows.
- 🎬 **AI Film Studio** — open the film studio (Cinema).
- 🎨 **Interface** — switch the interface variants (below).

Classic vs. Pro interface variants

From the **Tools** ▸ 🎨 **Interface** menu you can switch two modules between a **Classic** and a **Pro** ✨ variant, independently:

- **Models Tab** — Classic / Pro ✨
- **AI Film Studio** — Classic / Pro ✨

The **Pro** variants add refinements on top of the Classic ones: the Pro Import Models tab introduces a segmented layout, lock badges, and lazy-loading thumbnails (Import models), while the Pro Film Studio adds workflow improvements (Cinema). Both variants **respect the feature matrix** identically — switching interface style never changes what your edition unlocks. The switch happens live, without restarting the app.

Languages

AI Creative Center's interface is available in **15 languages**:

Arabic (العربية) · German (Deutsch) · Greek (Ελληνικά) · English · Spanish (Español) · French (Français) · Hebrew (עברית) · Hindi (हिन्दी) · Hungarian (Magyar) · Italian (Italiano) · Korean (한국어) · Portuguese (Português) · Russian (Русский) · Ukrainian (Українська) · Simplified Chinese (中文).

To switch language, click the flag button in the status bar (bottom of the window) and pick your language from the popup. A confirmation appears — "*The language will be applied the next time the application starts.*" — so restart the app to see the new language everywhere. Your choice is remembered between sessions.

Automatic state persistence

AI Creative Center remembers your work between sessions so you never start from a blank slate:

- **Tab values** — the settings of every control (dropdowns, text fields, numbers, checkboxes, sliders) across all tabs are saved and restored automatically.
- **Window geometry and scroll position** are restored.
- **Favorites and custom presets** — starred LoRAs and models, your saved per-LoRA strengths, and your custom video presets come back exactly as you left them.

This is fully automatic; there is no "save layout" button to remember. (For reproducible *generations*, use **Fixed seed** in Advanced settings.)

The Settings dialog

Open it from **Tools** ▸ **Settings**. It has three sections:

Generation engine (native)

Choose how models are loaded. **ComfyUI-native** uses dynamic VRAM management that avoids out-of-memory errors; the alternatives use native pipelines.

Control	Type	Default	Options	What it does
Image engine:	Dropdown	ComfyUI-native	ComfyUI-native (dynamic VRAM, no-OOM) · Diffusers (native pipeline)	How image models are loaded.
Video engine:	Dropdown	ComfyUI-native	ComfyUI-native (dynamic VRAM, no-OOM) · LTX-2 official (ltx2_infer)	How video models are loaded.

Provenance

Control	Type	Default	What it does
Invisible provenance signature (recommended)	Checkbox	On	Embeds an invisible, imperceptible signature into every generated photo and video, verifiable with the ATF Detector, without visibly altering the output. See Provenance signature.
Signature text (max 5)	Text (≤5 chars)	empty	An optional custom code/handle embedded in the signature. Empty = a valid signature without text. The 5-character limit keeps the mark robust to screenshots and compression.

Model overrides (ComfyUI template) — Studio Pro

> **Capability:** `cap.node_overrides`

For each loader node in the current template you can pick a specific model file from the files available on your system. This is a *Studio Pro* power-user feature; if the template has no loader nodes you'll see "*No loader node in the current template.*"

Where your settings live: the app stores your preferences in `data/user_prefs.json` (favorites, per-LoRA strengths, custom presets,

engine and signature choices), tab/window state in `data/gui_state.json`, and your language in `data/gui_lang.json`.

Next: dive into Images to make your first generation, or read Advanced settings to understand the sampler, VRAM, and long-video controls. The Appendix has the full capability→edition table, file locations, and troubleshooting.

Images

Generate images from text, edit them with inpaint/outpaint, recolor and upscale, cut objects out with AI selection, and impose real faces — all around a single shared canvas.

The **Images** tab (`tab.images`) is the heart of still-image creation in AI Creative Center. It is organised as four sub-tabs down the left — **Generate image**, **Inpaint / Outpaint**, **Color**, and **Definition** (upscale) — plus an always-visible **mask canvas** on the right that every operation draws from. Below the canvas sits the **AI selection** panel (SAM segmentation), and the Generate sub-tab adds the **Face adapter (PuLID)** and **Compose multiple photos** panels.

How image generation works

An image model turns your text **prompt** into pixels through a number of **diffusion steps**. You choose an **architecture** (FLUX, Z-Image or Ernie), a specific **model** file, and optionally a **text encoder** and **VAE** — or leave those on **auto** and let the app pick the matching pair. Two extra dials shape the result: **CFG** (how strictly the image obeys the prompt) and the **seed** (the random starting point; fix it to reproduce a result exactly).

The editing sub-tabs work differently from generation:

- **Inpaint / Outpaint** use a **mask** (drawn on the canvas) to mark the area to regenerate or the borders to extend, using a Fill-type model.
- **Integrate** and **Compose** are *edit-by-reference* operations (Kontext / FLUX.2): the extra images are handed to the model as *references*, not pasted, and the model harmonizes the scene. **No mask is needed** for these — only for Inpaint/Outpaint.
- **Recolor** changes the colour of a described subject from the prompt alone.

All model weights are downloaded once from Hugging Face; nothing runs in the cloud. See Import models for managing them.

Generate an image

> **Edition: Demo+** · **Capability:** `cap.gen_images` *(Demo is capped at 768 px and 20 generations; Supporter at 1024 px and 200; Full and above are unlimited. Demo/Supporter outputs carry the "AI Tools Forge" watermark; removed from Full up.)*

What it does — Creates a brand-new image from a text prompt using the model and architecture you select. **Where to find it** — **Images** → **Generate image** (`f1.subtab.gen`). **Step by step** 1. Set **Content** to *Standard* or *18+* — this filters the model and LoRA lists. (The adult option requires **Unlocked (18+) mode**; see edition note below.) 2. Choose an **Architecture** (`f1.gen.arch_lbl`): FLUX, Z-Image or Ernie. 3. Pick an **Engine** (`f1.gen.engine_lbl`): *Comfy (fast, dynamic VRAM)* (recommended) or *Diffusers*. 4. Choose a **Model** (`f1.model_lbl`). Use the precision filter beside it to narrow by fp16/fp8/fp4/gguf; colours flag VRAM feasibility. Leave *auto* to let the app choose. 5. Leave **Text encoder**, **CLIP type** and **VAE** on *auto* unless you have a reason to override. 6. Type your **Prompt img** (`f1.gen.prompt_lbl`) and optionally a **Negative** prompt. 7. Set **Steps**, **CFG** and the **Seed** (untick 🎲 **random** to fix it). 8. Click 🖼️ **Generate image** (`f1.gen.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Content (`f1.gen.content_lbl`)	Dropdown	Standard	Standard / 18+	Filters models & LoRAs. The 18+ option requires the Unlocked (18+) mode.
Architecture (`f1.gen.arch_lbl`)	Dropdown	FLUX	FLUX / Z-Image / Ernie	Determines compatible models, encoders and LoRAs.
Engine (`f1.gen.engine_lbl`)	Dropdown	Comfy	Comfy / Diffusers	Comfy scales VRAM dynamically (recommended); Diffusers is the classic pipeline.
Model (`f1.model_lbl`)	Dropdown	First available (or ⚡ Auto)	Installed models	The generation checkpoint. Colours = VRAM feasibility.
precision filter	Dropdown	All	All / FP16 / FP8 / FP4 / GGUF	Narrows the model list by weight precision.
Text encoder (`f1.gen.encoder_lbl`)	Dropdown	auto	auto + compatible encoders	Interprets the prompt; auto matches the model.
CLIP type (`f1.gen.clip_lbl`)	Dropdown	auto	auto + types	CLIP variant; auto is fine.
VAE (`f1.gen.vae_lbl`)	Dropdown	auto	auto + VAEs	Decodes the latent; auto uses the model's own.
Sampler (`f1.sampler_lbl`)	Dropdown	auto	auto / euler / euler_ancestral_cfg_pp / dpmpp_2m / res_2s	Sampling algorithm; euler/dpmpp are reliable.
Scheduler (`f1.scheduler_lbl`)	Dropdown	auto	auto / karras / bong_tangent / beta / simple	Strongly affects image look; bong_tangent/kl_optimal often best on FLUX.
🪄 **Enhance prompt (LLM Qwen3)** (`f1.gen.enhance`)	Checkbox	Off	On / Off	A local LLM expands short prompts into detailed descriptions before generating.
Steps (`f1.steps_lbl`)	Spinbox	20	1–80	More steps = more detail, slower.
CFG (`f1.cfg_lbl`)	Spinbox	3.5	0.0–20.0, step 0.1	Prompt adherence; too high causes artifacts.
Seed (`f1.seed_lbl`)	Spinbox + checkbox	0, random on	0–2,147,483,647	Fix the seed for reproducible results.
Prompt img (`f1.gen.prompt_lbl`)	Text	(empty)	free text	Image-specific prompt; empty falls back to the global prompt.
Negative (`f1.neg_lbl`)	Text	*pc game, cartoon, ugly, deformed, low quality, blurry, watermark, text…*	free text	Elements to avoid.

Specializations — **Classic vs. Pro:** the Pro interface (`ImportModelsTabPro`, and the Pro image layout) reorganizes these controls for denser workflows but exposes the same parameters; any UI change is mirrored across both variants. A 🧱 **Prompt Builder** button next to the prompt opens a structured field-to-prompt helper. **Tips for best results**

- Leave encoder/VAE on **auto** — the app pairs them to the architecture automatically.
- Keep **CFG** near the model's recommended value (FLUX ≈ 3.5); raise steps before raising CFG for more detail.
- Turn on **Enhance prompt** for short prompts; it fills in visual detail (first use downloads Qwen3-4B, ~3 GB).

Face adapter (PuLID)

Edition: Demo+ (with image generation) · **Capability:** `cap.gen_images`

What it does — Injects the identity of a face from a photo into the generated image, so the result looks like that person (FLUX architecture only). **Where to find it** — **Images → Generate image**, the **Face adapter (PuLID)** group (`f1.pulid.group`). **Step by step** 1. Tick **Enable PuLID (face from photo)** (`f1.pulid.enable`). 2. In **Face photo** (`f1.pulid.face_lbl`), click **Browse…** and choose a clear face image (you can pass several paths separated by `|`). 3. Set **Strength** (0.6–0.9 recommended), and optionally **Start step** and **True CFG**. 4. Generate as usual.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Enable PuLID (`f1.pulid.enable`)	Checkbox	Off	On / Off	Turns on face-identity injection (FLUX only).
Face photo (`f1.pulid.face_lbl`)	File picker	(empty)	image file(s)	The identity source.
Strength (`f1.pulid.strength_lbl`)	Spinbox	0.8	0.0–3.0, step 0.05	How strongly the face identity is applied (0.6–0.9 recommended).
Start step (`f1.pulid.start_lbl`)	Spinbox	0	0–10	When to start injecting identity. 1–4 lets pose/composition form first → more faithful pose.
True CFG (`f1.pulid.truecfg_lbl`)	Spinbox	1.0	1.0–10.0, step 0.5	1.0 = off/fast; >1 improves prompt adherence and reduces identity dominance (slower).

Specializations — Only available with the **FLUX** architecture; the panel disables for Z-Image/Ernie (`f1.pulid.only_flux`). REST: `POST /image/generate` with `pulid_face` and `pulid_strength` (default 0.8). See REST API. **Tips for best results**

- Use a sharp, front-facing, well-lit face photo; several photos of the same person improve robustness.
- If the pose looks locked, raise **Start step** to 2–4 so the prompt sets the composition first.

Inpaint & Outpaint

> **Edition: Full+** · **Capability:** `cap.inpaint`

What it does — Regenerates a masked area of an image (**inpaint**) or extends the image beyond its borders (**outpaint**) using a Fill-type model, keeping the rest intact. **Where to find it** — **Images → Inpaint / Outpaint** (`f1.subtab.inpaint`). Draw the mask on the canvas first. **Step by step** 1. Load your image onto the canvas and use the **Brush** tool to paint the area to change (for outpaint, set a larger canvas aspect so there is empty border). 2. Open the **Inpaint / Outpaint** sub-tab. 3. Choose an **Inpaint model** (`f1.inp.model_lbl`) — e.g. *FLUX_DEV_Fill*. 4. Type the **Inpaint prompt** (what to draw in the mask) or the **Outpaint prompt** (what to generate at the edges). 5. Set **Strength (retention)**, **Guidance (prompt adherence)** and **Edge feather** as needed. 6. Click 🎨 **Generate inpaint** (`f1.inpaint.btn`) or ⬜ **Generate outpaint** (`f1.outpaint.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Inpaint prompt (`f1.inpaint.prompt_lbl`)	Text	(empty)	free text	What to draw in the masked area.
Outpaint prompt (`f1.outpaint.prompt_lbl`)	Text	(empty)	free text	What to generate beyond the edges.
Negative (`f1.neg_lbl`)	Text	default negative	free text	What to avoid in the regenerated area.
Strength (retention) (`f1.inpaint.strength_lbl`)	Slider	0.75	0.00–1.00	How much of the ORIGINAL is preserved. Internally `denoise = 1 − 0.4×strength` (min 0.5).
Guidance (prompt adherence) (`f1.inpaint.guidance_lbl`)	Spinbox	30.0	1.0–30.0, step 0.5	How strongly the edit follows the prompt. Auto-adjusts per model (Fill dev ≈30, distilled ≈2.5, FLUX.2 Klein ≈4).
Edge feather (px) (`f1.inpaint.feather_lbl`)	Spinbox	0 = Auto	0–16	Blend between edit and original. Auto = proportional to the photo.
Inpaint model (`f1.inp.model_lbl`)	Dropdown	FLUX_DEV_Fill	Installed Fill/compatible models	The editing checkpoint.
LoRA (`f1.inpaint.lora_lbl`)	Dropdown	(none)	Installed LoRAs	Optional style/subject LoRA during the edit.
Steps (`f1.steps_lbl`)	Spinbox	20	1–80	Diffusion steps (auto-set from model).
Seed (`f1.seed_lbl`)	Spinbox + checkbox	0, random on	0–2,147,483,647	Fix for reproducibility.
✨ **Enhance prompt (AI)** (`f1.edit.enhance`)	Checkbox	Off	On / Off	A local LLM enriches the edit prompt. Invisible, optional.

Specializations — REST: `POST /image/inpaint` / `POST /image/outpaint` with `arch` (`inpaint` | `flux2_inpaint`), `steps` (20), `guidance` (30.0). **Retention** vs **Guidance** are distinct: retention controls how much original survives; guidance controls prompt obedience. **Tips for best results**

> - Paint a mask slightly larger than the object and use a small **Edge feather** for seamless blends.
> - For subtle edits, raise **Strength (retention)** toward 1.0; to fully replace content, lower it.

Integrate images (edit-by-reference)

> **Edition: Full+** · **Capability:** `cap.inpaint`

What it does — Blends extra images pasted on the canvas into the scene coherently (light, shadow, perspective) using Kontext (FLUX.1) or FLUX.2 — no mask required. **Where to find it** — **Images → Inpaint / Outpaint**, the 🧩 **Integrate images** group (`f1.integrate.group`). **Step by step** 1. Paste the extra image(s) onto the canvas as layers. 2. In **Integrate engine**, choose *Kontext (FLUX.1)* (recommended) or *FLUX.2 (experimental)*. 3. Pick the matching **model**. 4. Optionally describe how to blend them in **Integrate prompt**. 5. Set **Guidance** and click 🧩 **Integrate images** (`f1.integrate.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Integrate engine (`f1.integrate.engine_lbl`)	Dropdown	Kontext (FLUX.1)	Kontext / FLUX.2	Editing engine. Kontext is recommended.
Integrate model	Dropdown	First available	Kontext or FLUX.2 models	The editing checkpoint for the chosen engine.
Integrate prompt (`f1.integrate.prompt_lbl`)	Text	(empty)	free text	How to blend the images; empty = automatic.
Guidance (`f1.integrate.guidance_lbl`)	Slider	2.5	1.0–10.0	Edit strength (FluxGuidance). Kontext ≈2.5, FLUX.2 ≈4.0.
use puzzle_lora (`f1.integrate.puzzle`)	Checkbox	Off	On / Off	Experimental FLUX.1-only LoRA; leave off (tends to add a frame).
Steps (`f1.steps_lbl`)	Spinbox	20	1–80	Diffusion steps.
Seed	Spinbox + checkbox	0, random on	0–2,147,483,647	Fix for reproducibility.

Tips for best results

> - Kontext harmonizes without any mask — describe the target scene, not the cut-out.
> - Keep **puzzle_lora** off; it is experimental and usually degrades the blend.

Compose multiple photos

> **Edition: Full+** · **Capability:** `cap.inpaint`

What it does — Combines 2+ photos into one image *without* pasting them, so hidden parts aren't lost — using multiple references or a side-by-side stitch. **Where to find it** — **Images → Generate image** (and Inpaint tab), the 🧩 **Compose multiple photos** group (`f1.compose.group`). **Step by step** 1. Click ➕ **Add photo** (`f1.compose.add`) to add each source (order matters — the 1st is the base). 2. Optionally select a photo and click 🎭 **Mask** to load a PNG mask marking which part to use. 3. Choose a **Method**: *Multiple references (recommended)* or *Side-by-side (stitch)*. 4. Describe the final scene in the prompt. 5. Click 🧩 **Compose** (`f1.compose.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Photos list	List	(empty)	image files	Sources to combine; the 1st is the base.
➕ Add photo / ➖ Remove / 🎭 Mask	Buttons	—	—	Manage the source list and optional per-photo masks.
Method (`f1.compose.method_lbl`)	Dropdown	Multiple references	Multiple references / Side-by-side (stitch)	References = the model blends each photo naturally; stitch = photos placed next to each other as one reference.
Compose prompt	Text	(empty)	free text	Describe the final composition.

Specializations — Works with FLUX.1 and FLUX.2. Masks are optional and only indicate *which part* of a photo to use. **Tips for best results**

> - Use **Multiple references** for natural blends (e.g. "*the person from photo 1 in the landscape of photo 2, same lighting*").
> - Use **Side-by-side** only when you want the photos literally adjacent.

Recolor / restyle

> **Edition: Full+** · **Capability:** `cap.inpaint`

What it does — Changes the colour or style of a described subject from the prompt alone, leaving the rest of the image unchanged. **Where to find it** — **Images → Color** (`f1.subtab.color`). **Step by step** 1. Load the image on the canvas. 2. Enter the target **Color** (name or hex, or use 🎨 **choose**). 3. Name the **Subject** to recolour (e.g. "*the shirt*"), or use AI selection / a mask. 4. Choose an **Engine** (Kontext or FLUX.2) and set **Steps**. 5. Click 🎨 **Recolor / Restyle** (`f1.color.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Color (`f1.color.color_lbl`)	Text + picker	(empty)	name / hex	Desired colour, e.g. *navy blue*, `#cc0000`.
Subject (`f1.color.subject_lbl`)	Text	(empty)	free text	What to recolour.
Engine (`f1.engine_lbl`)	Dropdown	Kontext	Kontext / FLUX.2	Recolor engine.
Steps (`f1.steps_lbl`)	Spinbox	20	1–80	Diffusion steps.

Specializations — REST: `POST /image/recolor` with `target`, `subject`, `arch` (default `kontext`), `guidance` (2.5). See REST API.

AI upscale (increase definition)

> **Edition: Supporter+** · **Capability:** `cap.upscale_image`

What it does — Raises an image's resolution with an ESRGAN model, optionally adding a FLUX.2 refinement pass for extra detail. **Where to find it** — **Images** → **Definition** (`f1.subtab.def`), the 🔍 **Increase definition** group (`f1.upscale.group`). **Step by step** 1. Load the image on the canvas. 2. Choose an **Upscale model** and a **Factor** (2×/3×/4×). 3. Pick a **Mode**: *Model only (fast)* or *Model + AI refinement (more detail)*. 4. For refinement, choose a **Refinement model** and set **Refinement strength** and refine **Steps**. 5. Click 🔍 **Increase definition** (`f1.upscale.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Upscale model (`f1.upscale.model_lbl`)	Dropdown	First preferred (Nomos/UltraSharp…)	Installed ESRGAN models	The super-resolution network.
Factor (`f1.upscale.factor_lbl`)	Dropdown	4×	2× / 3× / 4×	Magnification.
Mode (`f1.upscale.mode_lbl`)	Dropdown	Model only (fast)	fast / refine	Refine adds a FLUX.2 pass for more detail.
Refinement model (`f1.upscale.refine_model_lbl`)	Dropdown	First FLUX.2	FLUX.2 models	Model for the AI refinement pass (refine mode only).
Refinement strength (`f1.upscale.refine_strength_lbl`)	Slider	0.30	0.05–0.70	High redraws more, low preserves the original.
Steps (`f1.steps_lbl`)	Spinbox	12	4–40	Refinement steps (refine mode only).

Specializations — REST: `POST /image/upscale` with `model` and `target_scale` (default 4.0). A 🔧 **Restore / Colorize photo…** button links to the Enhance quality Photo tab with the current image pre-loaded. **Tips for best results**

> - Use *fast* mode for a clean resolution bump; switch to *refine* only when you want extra invented detail.
> - Keep refinement strength ≤ 0.30 to avoid altering faces and text.

Object segmentation (AI selection with SAM)

> **Edition: Full+** · **Capability:** `cap.sam`

What it does — Selects an object as an editable mask, either from a text description (GroundingDINO + SAM) or by clicking/boxing on the canvas (SAM). **Where to find it** — **Images**, the **AI selection** panel (`f1.seg.group`) below the canvas. **Step by step** 1. To select by text: type the object in **Object to select** (e.g. *person, car, sky*) and click 🎯 **Select from text** (`f1.seg.btn`). 2. To select by pointer: choose the **SAM model (pointer)** and use the **SAM Point** canvas tool to click on the object. 3. Adjust **Threshold** if the selection is too wide or too narrow. 4. The resulting mask is editable by hand with the brush/eraser, then usable by inpaint, recolor, etc.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Object to select (`f1.seg.text_lbl`)	Text	(empty)	free text	Object description for text selection.
Detector (`f1.seg.detector_lbl`)	Dropdown	GroundingDINO + SAM-HQ	+SAM-HQ (refined edge) / DINO only (faster)	How the text selection is found and refined.
Threshold (`f1.seg.threshold_lbl`)	Spinbox	0.50	0.10–0.95, step 0.05	High = tighter/precise; low (0.20–0.35) = wider. Lower it if a click grabs only a patch.
SAM model (pointer) (`f1.seg.sam_lbl`)	Dropdown	First vanilla SAM	Installed SAM models	Used only by pointer/box selection on the canvas.

Specializations — REST: `POST /image/segment` with `text`, `threshold` (0.3), `use_sam` (true). Pointer selection needs a *vanilla* SAM model, not SAM-HQ. **Tips for best results**

> - For text selection, use a single concrete noun ("*the dog*"); combine with +SAM-HQ for a clean edge.
> - If the pointer grabs too little, lower the **Threshold**.

Face-swap (InsightFace)

> **Edition: Full+** · **Capability:** `cap.restore`

What it does — Replaces the face in an image with a face from another photo using InsightFace. **Where to find it** — Available from the Enhance quality Photo sub-tab (**Face-swap** operation), reachable via the 🔧 **Restore / Colorize photo…** button on the Definition sub-tab. **Step by step** 1. Click 🔧 **Restore / Colorize photo…** on the Definition sub-tab (or open **Enhance quality** → **Photo**). 2. Choose **Face-swap** and pick the **Source face** photo. 3. Process the image.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Source face	File picker	(empty)	image file	The face to impose onto the target.

Specializations — REST: `POST /image/faceswap` with `target_path` and `source_face_path`.

The mask canvas

> **Edition: Full+** (masks feed inpaint/segmentation) · **Capability:** `cap.inpaint` / `cap.sam`

What it does — The shared drawing surface where you load images, paint and erase masks, arrange layers, and set the working aspect ratio. **Where to find it** — Right side of the **Images** tab, always visible. **Step by step** 1. Load an image with 📁, or paste extra layers via the layers panel. 2. Pick a tool from the vertical toolbar and (for the brush) set the **Brush** size. 3. Paint the mask (red) with **Brush**; remove with **Erase**; use **Fill**, **Lasso** or **SAM Point** for faster selections. 4. **Undo/Redo** with ↶/↷; **Clear Mask** to start over; **Save Mask** to persist it. 5. Use the aspect-ratio menu to set the output size (or a custom W×H).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Brush	Slider	40 px	3–200 px	Brush/eraser diameter.
Tools	Buttons	Move	Move / Brush / Erase / Fill / Lasso / SAM Point	Move layers, paint/erase the mask, flood-fill, freehand lasso, or click-to-select with SAM.
Aspect ratio	Menu	—	1:1, 16:9, 9:16, 4:3, 3:4, 3:2, 2:3, 5:4, 4:5, 2:1, 21:9, 2.39:1, custom	Sets the working canvas dimensions (multiples of 32).
Layers panel	List + buttons	Base photo	Add / Up / Down / Delete	Manage stacked images used as references.

Tips for best results

> - The mask you draw here is what Inpaint/Outpaint and Recolor operate on — a clean, slightly-oversized mask gives the best edits.
> - Set the aspect ratio *before* generating so the output matches your target format.

Troubleshooting

- **"Template missing for ."** The selected architecture has no image template; switch to FLUX (always available) or install the required workflow. See Import models.
- **PuLID panel is greyed out.** PuLID works only with the FLUX architecture — switch **Architecture** to FLUX.
- **Pointer selection does nothing / warns "need a vanilla SAM."** Load a vanilla SAM model in the **SAM model (pointer)** dropdown; SAM-HQ is only used for text-based edge refinement.
- **Inpaint result looks noisy or over-cooked.** Lower **Guidance** toward the model's recommendation (distilled ≈2.5, FLUX.2 Klein ≈4, FLUX Fill dev ≈30) and raise **Strength (retention)** to keep more of the original.
- **Integrate adds an unwanted frame around the subject.** Turn **use puzzle_lora** off — it is experimental and FLUX.1-only.
- **Colours bleed during Recolor.** Name the subject precisely, or draw/segment a mask so only that region changes.
- **Outputs carry an "AI Tools Forge" watermark.** That is the Demo/Supporter brand watermark; it is removed from the **Full** edition upward (`cap.watermark`).

Video

Turn a prompt, a photo, or an audio track into moving pictures with LTX-2 — then guide motion with ControlNet, extend clips to any length, edit them with AI, re-shoot sections, and lip-sync speech.

The **Video** tab (`tab.video`) generates video from text (t2v), from a starting image (i2v), or from an audio track (a2v). It shares its prompt, model selector, LoRA panel and sampler/seed group with several companion tabs that specialize in editing and post-production: **ControlNet**, **Extend video**, **AI Video Edit**, plus the in-tab **V2V** panel (Normal / Talk / Sing / ReTake) and **Storyboard** for multi-scene shots.

How video generation works

You write a **prompt** (and optionally a **negative prompt**), pick a **preset** that sets resolution, FPS, duration and step count, choose a **model**, and press generate. LTX-2 samples the video over a number of **steps** using the **LTXVScheduler** with **CFG 1.0** — the guidance is provided by NAG, so you don't raise CFG the way you would for images.

- **Text-to-video (t2v):** prompt only.
- **Image-to-video (i2v):** add a guide image; the **i2v method** and **image strength** control how the video starts from it.
- **Audio-to-video (a2v):** attach an audio track from the Audio & voice tab; the model animates in time with it.

Type `\` in any prompt field to open the tag palette (pauses, emotions, speed). Advanced sampler, scheduler and SHIFT controls live in the embedded Sampler/Seed group and the Advanced settings tab.

Generate a video (text / image / audio)

> **Edition: Full+** · **Capability:** `cap.video`

What it does — Produces a video clip from your prompt, optionally conditioned on a starting image and/or an audio track. **Where to find it** — **Video** tab (`tab.video`). **Step by step** 1. Type your **Prompt** (`vid.prompt.lbl`) and, if needed, a **Negative prompt** (`vid.prompt.neg_lbl`). 2. Pick a **Preset** (`vid.preset.lbl`) — it fills resolution, FPS, duration and steps for you. 3. Set **Content** (Standard / 18+) and choose a **Model** (`vid.model_lbl`); use the precision filter and VRAM colours to fit your GPU. 4. Adjust **Resolution**, **W/H**, **FPS** and **Duration** if the preset isn't exactly what you want. 5. For image-to-video, load a guide image, choose an **i2v method** and set **Image strength**. 6. For audio-to-video, set the audio source in the Audio & voice tab. 7. Optionally add **LoRA (LTX-2)**, a **Storyboard**, or automatic face restoration. 8. Press 🎬 **Generate**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Prompt (`vid.prompt.lbl`)	Text	(empty)	free text	Describes the video. Type \ for the tag palette.
Negative prompt (`vid.prompt.neg_lbl`)	Text	*pc game, cartoon, ugly, deformed, low quality, blurry, watermark, text*	free text	What to avoid.
Preset (`vid.preset.lbl`)	Dropdown	—	See preset table below	Applies resolution/FPS/duration/ste
Content (`vid.content_lbl`)	Dropdown	Standard	Standard / 18+	Filters models. The 18+ opti requires the Unlocked (18+) mode.
Model (`vid.model_lbl`)	Dropdown	Auto	Installed LTX models	Generation model; colours flag VRAM feasibility.
Resolution (`vid.res_lbl`)	Dropdown	Vertical 9:16 (704×1280)	9:16, 16:9, 1:1, 4:3, 3:4, 21:9, 2.39:1, 2.35:1, 1.85:1, small variants, Custom	Aspect/resolution; Custom enables manual W/H.
W (`vid.w_lbl`)	Spinbox	704	256–2048, step 32	Width (multiple of 32).
H (`vid.h_lbl`)	Spinbox	1280	256–2048, step 32	Height (multiple of 32).
FPS (`vid.fps_lbl`)	Spinbox	24	1–60	24 = cinematic, 30 = TV, 60 fluid.
Duration (`vid.dur_lbl`)	Spinbox	8.0 s	1.0–60.0 s	Clip length.
Image strength (soft reference) (`vid.img_strength_lbl`)	Slider	0.50	0.00–1.00	How faithful the video stays the guide frame. 0.45–0.60 = natural motion; 0.80+ = tigh but rigid.
Image→video method (`vid.i2v_method_lbl`)	Dropdown	AddGuide multi-keyframe (default)	AddGuide multi-keyframe / Inplace soft	AddGuide = attention guide supports multiple keyframes Inplace = writes the image into frame 1 with a noise mask (physical start, natural motion).
Video engine (`vid.engine_lbl`)	Dropdown	LTX-2 native (all features)	LTX-2 native / Comfy (dynamic VRAM, t2v/i2v)	Native = every feature (audio/keyframe/controlnet) Comfy = dynamic VRAM bu t2v/i2v only.
Fast cache (TeaCache) (`vid.teacache_lbl`)	Spinbox	0.0 (off)	0.0–0.15, step 0.01	Experimental; keep 0 — any value >0 degrades LTX 2.3 video.

Specializations — REST equivalents are `POST /video/generate` (t2v/i2v/a2v) with `i2v_method` (`addguide`|`inplace`), `ref_strength`, `audio_path`, `audio_scale`, etc. See REST API. **Downscale/Upscale/Refine HR** checkboxes are currently inert (generation is native full-resolution); use the Enhance quality Video tab for super-resolution.

Available presets (from the built-in preset list):

Preset	Steps	Resolution	FPS	Duration	Notes
⚡ **FAST (draft ~1 min)**	8	704×384	16	4.0 s	Low-res draft to validate the idea.
🎯 **PRO (balanced)**	24	1280×704	24	6.0 s	Best speed/quality compromise.
💎 **HQ (max quality)**	44	1280×704	24	8.0 s	No cache, highest quality, slowest.
🎯 **Best quality**	40	1280×704	24	8.0 s	Max quality, 40 steps.
⚡ **Quick preview**	8	480×832	16	4.0 s	Fast draft (~1 min).
📺 **Broadcast 16:9**	30	1280×704	25	6.0 s	TV/YouTube, 25 fps PAL.
📱 **Social 9:16**	30	704×1280	30	6.0 s	Reels/TikTok/Shorts vertical.
🎬 **Cinemascope**	35	1280×544	24	10.0 s	2.35:1 cinematic.
📽️ **Anamorphic 2.39:1**	35	1280×544	24	8.0 s	Anamorphic cinema format.
🚀 **GGUF fast (low RAM)**	8	704×1280	24	5.0 s	Optimized for quantized GGUF.
🌟 **Balanced**	24	704×1280	24	8.0 s	Everyday quality/speed balance.

Tips for best results

- Start from 🎯 **PRO** for a balance, ⚡ **FAST** to test an idea, 💎 **HQ** for the final render.
- For i2v, keep **Image strength** around 0.45–0.60 for natural motion; raise it only if the subject drifts.
- Leave **TeaCache** at 0 — on LTX 2.3 any value visibly degrades the video.

First frame from a face (PuLID → i2v)

Edition: Full+ · **Capability:** `cap.video`

What it does — Generates the first frame carrying a face identity (PuLID-FLUX), then uses it as the i2v starting frame so the person's face is preserved through the video. **Where to find it** — **Video** tab, the **Face → first frame (PuLID)** control (`vid.pulid_kf_lbl`). **Step by step** 1. In **Face → first frame (PuLID)**, browse to one or more face photos (`vid.pulid_kf.browse`). 2. Click 👱 **First frame from face** (`vid.pulid_kf.gen`) — the identity-carrying frame is generated and loaded. 3. Press **Generate video**: i2v animates the frame while keeping the face.

Tips for best results

- Provide several clean photos of the same person for a more robust identity.

Automatic face restoration, face-swap and smoothing

Edition: Full+ · **Capability:** `cap.restore`

What it does — Post-processes the generated video automatically: restores faces (CodeFormer/GFPGAN), swaps in a face from a photo (InsightFace), and/or smooths motion with RIFE. **Where to find it** — **Video** tab: the 😀 **Automatic face restoration** group (`vid.fr_auto.group`), the **Face-swap** option (`vid.faceswap.auto`), and the **Smooth (RIFE)** option (`vid.rife.auto`). **Step by step** 1. To fix "plastic" faces, tick **Restore faces at the end of generation (CodeFormer / GFPGAN)** (`vid.fr_auto`), choose a **Model**, and set **Fidelity** (~0.70). 2. To impose a specific face, tick **Face-swap: impose a face from a photo** (`vid.faceswap.auto`) and **Choose face...** (`vid.faceswap.face.btn`). 3. To smooth motion, tick **Smooth (RIFE): less judder, more fps** (`vid.rife.auto`) and pick an **Interpolation** factor.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Restore faces at the end (`vid.fr_auto`)	Checkbox	Off	On / Off	Per-frame face restoration preserving identity (no re-diffusion).
Model (`vid.fr_auto_model_lbl`)	Dropdown	First available	CodeFormer / GFPGAN	Face-restoration network.
Fidelity (`vid.fr_auto_fidelity_lbl`)	Slider	0.70	0.00–1.00	1.0 preserves the original more; 0.0 reconstructs more.
Face-swap: impose a face (`vid.faceswap.auto`)	Checkbox	Off	On / Off	Replaces every frame's face with the chosen photo (InsightFace).
Smooth (RIFE) (`vid.rife.auto`)	Checkbox	Off	On / Off	Interpolates in-between frames (anti-judder).
Interpolation (`vid.rife.factor_lbl`)	Dropdown	x2	x2 / x4	x2 doubles the fps, x4 quadruples them.

Tips for best results

- Face restoration adds a second pass — the final video is the restored one. See also Enhance quality.

The Sampler / Seed group

Edition: Full+ · **Capability:** `cap.video`

What it does — Exposes the sampling engine (quality preset, steps, CFG, sampler, scheduler, SHIFT), the text encoder, prompt enhancement, and a fixed-seed option for reproducibility. **Where**

to find it — **Video** tab, the embedded Sampler/Seed group (see also Advanced settings).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Quality	Dropdown	Custom	Custom / Fast (8, CFG 1.0) / Pro (30, CFG 3.0) / HQ (50, CFG 4.0)	One-click quality tier that writes Steps/CFG.
Steps	Spinbox	32	1–60	Sampling steps; 32 is the reference quality value.
CFG	Spinbox	1.0	0.0–15.0, step 0.1	Keep 1.0 — NAG provides the guidance on LTX.
Sampler	Dropdown	euler_ancestral_cfg_pp	euler_ancestral_cfg_pp / euler_cfg_pp / euler / dpmpp_2m	Sampling algorithm verified for LTX-2.3.
Scheduler	Dropdown	LTXVScheduler (shift)	LTXVScheduler / linear_quadratic / bong_tangent / karras / beta	The video scheduler; LTXVScheduler is the correct default.
Base Shift	Spinbox	0.95	0.10–6.00, step 0.05	LTXVScheduler shift lever — a real, zero-cost quality dial.
Max Shift	Spinbox	2.05	0.50–12.00, step 0.05	Upper shift bound.
Text encoder	Dropdown	auto	auto / Gemma full / Gemma abliterated	auto picks the full encoder for standard content and the abliterated encoder in Unlocked (18+) mode.
Enrich prompt (local AI)	Checkbox	Off	On / Off	LLM enriches the prompt before generation.
Fixed seed + **Seed**	Checkbox + spinbox	Off, 777	0–2,147,483,647	Fix the seed for reproducible runs.

Tips for best results

- The single biggest quality lever at zero cost is **Base Shift** — nudge it if detail degenerates on native resolution.

Storyboard / keyframes (multi-scene)

Edition: Full+ · **Capability:** `cap.video`

What it does — Builds one continuous video from several keyframe images anchored at different times; the model interpolates between scenes, with no editing. **Where to find it** — **Video** tab, the 🎬 **Multi-scene storyboard** group (`vid.sb.group`). **Step by step** 1. Tick **Enable storyboard (multiple scenes)** (`vid.sb.enable`). 2. For each scene, click 📷 **Image** to pick a keyframe, set the time (**at … s**) and the **strength**. 3. Click **+ Add scene** (`vid.sb.add`) for more scenes (at least 2 with images are required). 4. Generate — you get a single continuous clip.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Enable storyboard (`vid.sb.enable`)	Checkbox	Off	On / Off	Turns on multi-scene interpolation.
at (time) (`vid.sb.at`)	Spinbox	—	0.0–3600.0 s, step 0.5	When to anchor this keyframe.
strength (`vid.sb.force`)	Spinbox	0.8	0.0–1.0, step 0.05	1 = rigid (little motion), 0 = free.

Specializations — REST: `POST /video/keyframes` with a `keyframes` list of `{path, frame, strength}`. See REST API. **Tips for best results**

- Space keyframes evenly and keep strengths around 0.7–0.8 so motion flows between scenes.

ControlNet tab — copy motion from a reference video

Edition: Full+ · **Capability:** `cap.controlnet`

What it does — Copies the *motion/shape* of a reference video into your generated video; the content comes from the Video-tab prompt, the motion from the reference. **Where to find it** — the **ControlNet** tab (`tab.controlnet`). **Step by step** 1. Tick ✅ **Enable ControlNet (copy the motion from the video below)** (`cn.enable`). 2. Choose a **Type** and load the **reference video** (`cn.video.pick.btn`). 3. In the **Video** tab, write the prompt describing the *content* you want. 4. Set **Control strength** and **Guide attention**, then press 🎬 **Generate**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Enable ControlNet (`cn.enable`)	Checkbox	Off	On / Off	The generated video follows the reference motion.
Type (`cn.type_lbl`)	Dropdown	Canny	Canny (edges) / Depth (3D geometry) / Pose (human skeleton, DWPose)	Canny for rigid objects, Depth for full scenes, Pose for body motion.
Control strength (`cn.strength_lbl`)	Spinbox	0.8	0.0–1.0, step 0.05	1.0 strictly follows the reference; lower = more creative freedom.
Guide attention (`cn.attn_lbl`)	Spinbox	1.0	0.1–1.0, step 0.05	`conditioning_attention_strength`: lower (0.5–0.8) allows freedom while preserving motion.

Specializations — REST: `POST /video/controlnet` with `control_type` (`canny` | `depth` | `pose`) and `control_strength` (0.8). See REST API. **Tips for best results**

> - Use **Pose** for dance/sport, **Depth** for environments, **Canny** for products and rigid shapes.
> - The reference resolution/duration don't matter — the workflow resamples to your Video-tab settings.

Extend video tab — unlimited-length continuation

> **Edition: Full+** · **Capability:** `cap.extend_retake`

What it does — Adds time to an existing video with an autoregressive, memory-bounded chain — no length limit; up to 20 s is one pass, beyond that it stitches overlapping segments. **Where to find it** — the **Extend video** tab (`tab.extend_video`). **Step by step** 1. Load a video (`ext.pick.btn`) or use the last generation. 2. Write the **Continuation prompt** (what happens next). 3. Set **Total duration** to add, and — only if over 20 s — the **Segment** length, **Overlap** and **Context**. 4. Set **Steps** and click ➕ **Extend video** (`ext.run.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Total duration (`ext.duration_lbl`)	Spinbox	20.0 s	1.0–3600.0 s	Time to add. No limit — generated in chained chunks.
Segment (over 20 s) (`ext.seg_lbl`)	Spinbox	18.0 s	2.0–20.0 s	Chunk length when exceeding 20 s. ≤20 s → single pass (no seam).
Overlap (frames) (`ext.overlap_lbl`)	Spinbox	8	0–48	Overlap frames between chunks for motion continuity.
Context (frames) (`ext.context_lbl`)	Spinbox	16	8–64	Tail frames used as context for the next chunk. Higher = more anchored to the past.
Steps (`ext.steps_lbl`)	Spinbox	8	1–60	Sampling steps per chunk.

Specializations — REST: `POST /video/extend` with `add_seconds` (5.0), `seg_seconds` (0 = default), `steps` (8). SSML tags are supported in the prompt. **Tips for best results**

> - Keep additions ≤20 s for a single seamless pass; longer requests are stitched automatically.

AI Video Edit tab — instruction, inpaint, style, replace character

> **Edition: Full+** · **Capability:** `cap.controlnet` / `cap.extend_retake`

What it does — Edits an existing video in five modes: global edit-by-instruction, object inpaint (SAM2 tracking), anime ↔ real style transfer, and character replacement. **Where to find it** — the ✨ **AI Video Edit** tab (`tab.video_edit_ai`). **Step by step** 1. Choose a video (`vedit.pick`). 2. Pick a **Mode** (`vedit.mode_lbl`). 3. Fill the mode-specific fields (object, reference image, etc.). 4. Write the **Prompt** and set the edit parameters. 5. Click ✨ **Generate edit** (`vedit.go`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Mode (`vedit.mode_lbl`)	Dropdown	Edit by instruction	Edit by instruction (edit-anything) / Replace object (inpaint, experimental) / Style: anime→real / Style: real→anime / Replace character	The kind of edit to apply.
Object (by words) (`vedit.obj_lbl`)	Text	(empty)	free text	Object to select (Inpaint) or character to limit to (Replace character).
Prompt (`vedit.prompt_lbl`)	Text	(empty)	free text	What to generate/change.
Edit strength (`vedit.strength_lbl`)	Dropdown	1.0	0.6 / 0.8 / 1.0	How strongly to apply the edit.
Replacement image (character) (`vedit.ref_lbl`)	File picker	(empty)	image	First video frame with the other character (Replace character).
Identity strength (reference) (`vedit.ref_strength_lbl`)	Spinbox	0.8	0.1–1.0, step 0.05	How much the reference identity counts vs. the video's motion.
CFG (prompt adherence) — inpaint (`vedit.cfg_lbl`)	Spinbox	6.0	1.0–10.0, step 0.5	Prompt adherence for inpaint.
Guide strength (anchor to original) (`vedit.guide_lbl`)	Spinbox	0.4	0.1–1.0, step 0.1	How tightly the edit anchors to the original.
Mask dilation (px) (`vedit.dilate_lbl`)	Spinbox	24	0–64	Grows the tracked mask; more room for the new object.
Mask feather (px) (`vedit.blur_lbl`)	Spinbox	6.0	0.0–30.0, step 1.0	0 = hard edge; 6–12 = smooth transition.
✨ **Enhance prompt (AI)** (`vedit.enhance`)	Checkbox	Off	On / Off	LLM enriches the edit prompt. Invisible, optional.

Specializations — REST: `POST /video/v2v_edit` with `edit_lora`, `control_strength`. **Replace character** keeps the original audio and preserves scene/motion via canny; long videos are processed in segments. **Tips for best results**

> - **Replace object (inpaint)** is experimental — SAM2 tracks the object across the clip and regenerates its area.
> - For **Replace character**, load an image that is the first video frame with only the character changed.

V2V panel — Normal, Talk (lip-sync), Sing, ReTake

> **Edition: Full+** · **Capability:** `cap.extend_retake`

What it does — In-tab video-to-video operations: re-shoot a section (**ReTake**), lip-sync speech (**Talk**), make a face sing a song (**Sing**), or pass through (**Normal**). **Where to find it** — **Video** tab, the 🎬 **Video editing (V2V)** panel (`vid.v2v.group`), **Mode** selector (`v2v.mode_label`).

Specializations

- **ReTake** (`v2v.mode_retake`) — regenerate a section. Set the **Section to regenerate** with **Start/End** sliders (constraint: at least 3 s of video before and after). Choose section **Audio**: **Keep original** (`v2v.audio_keep`) or **Regenerate** (`v2v.audio_regen`). REST: `POST /video/retake` with `start_time` / `end_time`.
- **Talk** (`v2v.mode_talk`) — lip-sync. With a video chosen, it lip-syncs an existing clip; with no video, it generates speech from the Frame-1 image (i2v). Pick the **Face mask**: **Mouth/lips (faithful lip-sync)** (`v2v.region_mouth`) or **Whole face (more natural)** (`v2v.region_face`), with a **Mask offset (px)**. The voice follows the Audio-tab decision tree (uploaded file → selected TTS voice + text → dialogue in the main prompt). REST: `POST /video/lipsync` with `audio_scale` (3.0), `separate_vocals`, `talk_region`, `talk_mask_offset`.
- **Sing** (`v2v.mode_sing`) — from the Frame-1 photo, generate a video of the face singing a real song with lip-sync and beat-driven motion. **Choose song...** (`v2v.sing_pick_audio`), tick **Isolate vocals (recommended)** (`v2v.sing_isolate_vocals`), and set **Sync strength** (1.5–3.0 recommended; default 2.0).
- **Normal** (`v2v.mode_normal`) — standard pass-through mode.

Parameters (ReTake / Talk / Sing highlights)

Control	Type	Default	Range / Options	What it does & when to change
Mode (`v2v.mode_label`)	Radio	Normal	Normal / Talk / Sing / ReTake	Selects the V2V operation.
Start / End (`v2v.start_label` / `v2v.end_label`)	Sliders	—	within the clip (3 s guard each side)	The section to re-shoot (ReTake).
Section audio	Radio	Keep original	Keep original / Regenerate	Whether to keep or regenerate the section's audio.
Face mask (`v2v.face_mask_label`)	Dropdown	Mouth/lips	Mouth/lips / Whole face	Region for lip-sync (Talk).
Mask offset (px) (`v2v.mask_offset_label`)	Spinbox	0	0–128	Shifts the face mask (Talk).
Isolate vocals (`v2v.sing_isolate_vocals`)	Checkbox	On	On / Off	Separates voice from music (demucs) for accurate lip-sync (Sing).
Sync strength (`v2v.sing_audio_scale`)	Slider	2.0	0.5–3.0	Lip-sync strength on the audio (Sing).

Tips for best results

> - For lip-sync, choose **Mouth/lips** for accuracy or **Whole face** for a more natural look.
> - In **Sing**, keep **Isolate vocals** on and **Sync strength** at 1.5–3.0.

LoRA (LTX-2) panel

> **Edition: Full+** · **Capability:** `cap.video`

What it does — Applies one or more LTX-2 video LoRAs (style, subject, effects) filtered to the current video model's architecture.

Where to find it — **Video** tab, the **LoRA (LTX-2)** group (`vid.lora.group`). **Step by step** 1. Search/filter the LoRA list (adult-LoRA visibility follows the video Content selector). 2. Tick the LoRAs you want and set each one's strength (default 0.6, range 0.0–2.0). 3. Generate — triggers are added automatically if the auto-trigger option is on.

Tips for best results

> - Keep total LoRA strength moderate; stacking several strong LoRAs can wash out the model.

Troubleshooting

- **The video degenerates into "dots" or dematerializes on native resolution.** This is a known LTXVScheduler shift issue on native token counts — adjust **Base Shift** in the Sampler/Seed group, and don't force high CFG (keep 1.0).
- **CFG has no effect / high CFG looks over-cooked.** On LTX, guidance comes from NAG — leave **CFG at 1.0** and change the preset/steps instead.
- **Downscale/Upscale/Refine HR checkboxes do nothing.** They are temporarily inert; generation is native full-resolution. Use the Enhance quality Video tab for super-resolution.
- **ControlNet output ignores the motion.** Raise **Control strength** toward 1.0 and make sure the reference has clearly readable motion for the chosen type.
- **Extend produces a black seam.** Keep the addition ≤20 s (single pass); for longer clips the overlap/context settings smooth the joins.
- **Faces look "plastic."** Enable automatic face restoration (Fidelity ~0.70) or run the Enhance quality face restorer as a final pass.
- **A2V/Talk has no dialogue.** Set the audio source correctly in the Audio & voice tab, or write the spoken line into the main prompt for native generation.

Enhance quality

Rescue and upgrade your media after it's made: restore, upscale, colorize and face-swap photos, and apply AI super-resolution, face restoration and smooth frame interpolation to video.

The **Enhance quality** tab (`tab.quality`, shown as "✨ *Migliora qualità*") is a post-processing workbench. It never regenerates your content from scratch — it takes an existing image or video and improves its resolution, colours, faces or fluidity while keeping the original intact. The tab is split into two nested sub-tabs:

- 🖼 **Photo** (`tab.quality.photo`) — restore, upscale, colorize black-and-white, or face-swap a single image.
- 🎬 **Video** (`tab.quality.video`) — AI super-resolution (SeedVR2 / FlashVSR / ESRGAN), per-frame face restoration, RIFE frame interpolation, and an EXR/HDR export path for VFX.

Every operation here runs locally and preserves your source file; results are written as new files.

Photo sub-tab

Load an image, choose one operation, tune its settings, and press 🔧 **Process image**. The four operations are mutually exclusive — the settings groups for the others grey out automatically.

Restore a photo (ESRGAN super-resolution)

> **Edition: Supporter+** · **Capability:** `cap.upscale_image` / `cap.restore`

What it does — Increases a photo's resolution and sharpness with an ESRGAN super-resolution model, optionally applying a light denoise to clean up compression artifacts. **Where to find it** — **Enhance quality** → 🖼 **Photo**, the **Operation** group with 🔧 **Quality restore (ESRGAN super-resolution)** selected. **Step by step** 1. Load a source image: drag it onto the field, click 📁 **Browse...** (`res.input.browse.btn`), or click 🖼 **Use canvas** (`res.input.canvas.btn`) to pull the current Images canvas image. 2. In **Operation**, choose 🔧 **Quality restore (ESRGAN super-resolution)** (`res.op.restore`, the default). 3. Under **Restore settings (ESRGAN)**, pick an **Upscale model** and a **Target scale**. 4. Optionally tick **Denoising** and set its strength. 5. Click 🔧 **Process image** (`res.run.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Upscale model (`res.restore.model_lbl`)	Dropdown	First "ultra/real/nomos" model	Installed ESRGAN models	The super-resolution network. Colours indicate VRAM feasibility.
Target scale (`res.restore.scale_lbl`)	Spinbox	2.0×	1.0×–4.0×, step 0.5	How much to enlarge. 4× for very small sources; 2× for a gentle boost.
Denoising (`res.restore.denoise`)	Checkbox	Off	On / Off	Removes compression artifacts; may soften detail slightly.
Denoise strength (`res.restore.denoise_str_lbl`)	Spinbox	0.20	0.05–0.50, step 0.05	Strength of the denoise pass; active only when the checkbox is on.

Tips for best results

> - ESRGAN needs no extra download beyond the upscale models you already have.
> - Start at 2× with denoise off; enable denoise only if you see JPEG blocking.

Colorize a black-and-white photo (FLUX.2)

> **Edition: Full+** · **Capability:** `cap.restore`

What it does — Adds realistic, period-accurate colour to a black-and-white photo using a FLUX.2 diffusion model guided by your description. **Where to find it** — **Enhance quality** → 🖼 **Photo**, **Operation** → 🎨 **B&W colorization (FLUX.2 — requires a FLUX.2 model)** (`res.op.colorize`). **Step by step** 1. Load the black-and-white image. 2. Select 🎨 **B&W colorization** in **Operation**. 3. Under **Colorization settings (FLUX.2)**, choose a **FLUX.2 model**. 4. Describe the **Subject** and adjust **Color style**, **Steps**, **Strength (denoise)** and **Adherence (guidance)** as needed. 5. Click 🔧 **Process image**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
FLUX.2 model (`res.color.model_lbl`)	Dropdown	First FLUX.2 model	Installed FLUX.2 models	The colorization model. Requires a FLUX.2 checkpoint.
Subject (`res.color.subject_lbl`)	Text	(empty)	free text	Main subject, e.g. *"man in uniform"* — helps the model choose plausible colours.
Color style (`res.color.style_lbl`)	Text	*natural realistic colors, period-accurate*	free text	Desired palette and look.
Steps (FLUX.2) (`res.color.steps_lbl`)	Spinbox	20	5–50	Diffusion steps: more = more quality, slower.
Strength (denoise) (`res.color.denoise_lbl`)	Spinbox	0.85	0.30–1.00, step 0.05	High = bolder colours but less faithful to structure; low = structure better preserved.
Adherence (guidance) (`res.color.guidance_lbl`)	Spinbox	2.5	1.0–6.0, step 0.5	How strictly the result follows your subject/style description.

Specializations — Choose ✨ **Restore + Colorization (sequential)** (`res.op.both`) to run ESRGAN restore and FLUX.2 colorization back-to-back in one click. **Tips for best results**

> - Name the era and subject in **Subject** (e.g. "*1940s family portrait*") for believable, period-accurate tones.
> - If colours bleed or look invented, lower **Strength** toward 0.5 to keep the original structure.

Face-swap a photo (InsightFace)

> **Edition: Full+** · **Capability:** `cap.restore`

What it does — Replaces the face in the target photo with a face from another photo using InsightFace. **Where to find it** — **Enhance quality** → 🖼 **Photo**, **Operation** → **Face-swap (impose a face)** (`res.op.faceswap`). **Step by step** 1. Load the target image (the one whose face you want to replace). 2. Select **Face-swap** in **Operation**. 3. Under **Face-swap**, click **Choose face...** (`res.faceswap.source.btn`) and pick the source face photo. 4. Click 🔧 **Process image**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Source face (`res.faceswap.source_lbl`)	File picker	(empty)	image file	The face to impose onto the target.

Specializations — The REST equivalent is `POST /video/faceswap` with `video_path` (image or video) and `source_face_path`. See REST API. **Tips for best results**

> - Use a clear, front-facing source face with even lighting for the cleanest swap.

Video sub-tab

Load a video, pick a super-resolution **Engine**, tune it, and press ✨ **Enhance video**. The Video sub-tab also hosts face restoration, RIFE interpolation and EXR export. The original audio is preserved throughout.

AI video super-resolution (SeedVR2 / FlashVSR / ESRGAN)

> **Edition: Full+** · **Capability:** `cap.restore`

What it does — Upscales a video to a higher resolution with your choice of three engines: SeedVR2 (AI video model, best quality), FlashVSR (node-based AI), or ESRGAN (fast per-frame). **Where to find it** — **Enhance quality** → 🎬 **Video**, the **Restore / super-resolution** group (`def.restore.group`). **Step by step** 1. Click 📁 **Load video...** (`def.pick.btn`) and choose your clip. 2. Pick an **Engine** (`def.engine_lbl`). The relevant controls enable for that engine. 3. For **SeedVR2**: choose the model, **Target resolution**, and tune **BlockSwap**, **Batch** and **Temporal overlap** for your VRAM. 4. For **ESRGAN / FlashVSR**: choose the ESRGAN model and **Scale**. 5. Optionally tick 🔗 **Then restore faces (in sequence)** to chain face restoration (and RIFE) automatically. 6. Click ✨ **Enhance video** (`def.run.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Engine (`def.engine_lbl`)	Dropdown	SeedVR2	**SeedVR2 (AI, recommended)** · **ESRGAN per-frame (fast)** · **FlashVSR (requires node)**	The super-resolution method. SeedVR2 = best; ESRGAN = fastest.
SeedVR2 model (`def.seedvr2_model_lbl`)	Dropdown	First available	SeedVR2 variants	Model variant (SeedVR2 only). Downloads on first use.
Target resolution (`def.resolution_lbl`)	Spinbox	1080 px	256–8192, step 64	Target vertical resolution (SeedVR2). 1080 = FullHD.
BlockSwap (VRAM) (`def.blocks_lbl`)	Spinbox	0	0–36	>0 lowers VRAM use (slower); 0 = everything on GPU (fine on 32 GB).
Batch (4n+1) (`def.batch_lbl`)	Spinbox	5	1–121, step 4	Frames per batch (1, 5, 9…). Higher = more temporal coherence and VRAM.
Temporal overlap (`def.overlap_lbl`)	Spinbox	3	0–16	Overlap frames between SeedVR2 chunks; reduces judder at cuts on long clips.
ESRGAN model (`def.esrgan_model_lbl`)	Dropdown	First available	Installed ESRGAN models	Per-frame upscaler (ESRGAN engine).
Scale (ESRGAN) (`def.scale_lbl`)	Spinbox	2.0×	1.0×–8.0×, step 0.5	Upscale factor for ESRGAN / FlashVSR.
🔗 **Then restore faces (in sequence)** (`def.chain_face`)	Checkbox	Off	On / Off	After super-resolution, auto-runs face restoration then RIFE — no manual re-run.

Specializations — REST: `POST /video/enhance` with `engine` (`seedvr2` | `flashvsr`), `seedvr2_model`, `resolution` (default 1080), `batch_size` (default 5), `block_swap` (default 0), `scale`. See REST API. **Tips for best results**

- **SeedVR2** gives the most natural motion-aware upscale; keep **Batch** at 5 unless you have spare VRAM.
- On long videos with visible seams at cuts, raise **Temporal overlap** to 4–6.
- SeedVR2 downloads its model on first use and always keeps the original audio.

Restore faces in a video (CodeFormer / GFPGAN)

Edition: Full+ · **Capability:** `cap.restore`

What it does — Reconstructs faces frame-by-frame to fix the "plastic" look and identity drift of generated video, while preserving identity (it does not re-run diffusion). **Where to find it — Enhance quality** → 🎬 **Video**, the 😊 **Face restoration (CodeFormer / GFPGAN)** group (`def.face.group`). **Step by step** 1. Load the video. 2. If the status shows the node isn't installed, click 📥 **Install face-restore** (`def.face.install.btn`) and restart when prompted. 3. Choose a **Model** and **Face detector**. 4. Set **Fidelity** (≈0.70 recommended). 5. Click 😊 **Restore video faces** (`def.face.run.btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Model (`def.face.model_lbl`)	Dropdown	First available	CodeFormer (quality) / GFPGAN (speed)	The face-restoration network.
Face detector (`def.face.detection_lbl`)	Dropdown	First available	Available detectors	Locates faces before restoring them.
Fidelity (`def.face.fidelity_lbl`)	Slider	0.70	0.00–1.00	1.0 preserves the original more; 0.0 reconstructs more (sharper, may drift). ~0.7 is a good balance.

Tips for best results

- Apply face restoration as the *last* step, after super-resolution.
- This can also be triggered automatically from the Video tab's automatic face-restore option.

Increase FPS with frame interpolation (RIFE)

Edition: Full+ · **Capability:** `cap.restore`

What it does — Generates in-between frames with RIFE to raise a video's frame rate, reducing judder and adding fluidity. **Where to find it — Enhance quality** → 🎬 **Video**, the **FPS increase (RIFE, optional)** group (`def.rife.group`). **Step by step** 1. Load the video. 2. Tick **Interpolate frames (more fluidity)** (`def.rife.check`). 3. Choose an **FPS multiplier**. 4. Run it via ✨ **Enhance video** (or automatically as part of the chain).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Interpolate frames (`def.rife.check`)	Checkbox	Off	On / Off	Enables RIFE interpolation.
FPS multiplier (`def.rife_mult_lbl`)	Dropdown	2×	2× / 3× / 4×	2× doubles the frame rate, 4× quadruples it.
RIFE model (`def.rife_ckpt_lbl`)	Dropdown	(fixed)	—	Non-selectable: the native RIFE uses a single Practical-RIFE model.

Specializations — REST: `POST /video/rife` with `factor` (2, 3 or 4). See REST API. **Tips for best results**

> - Use 2× to turn a 24 fps clip into a smooth 48 fps; reserve 4× for slow-motion looks.

Export HDR / EXR frames (VFX)

> **Edition: Full+** · **Capability:** `cap.restore`

What it does — Exports every frame of a video as OpenEXR float32 for a professional VFX pipeline (Nuke, Resolve, Blender). **Where to find it** — **Enhance quality** → 🎬 **Video**, the **HDR / EXR export (VFX)** group (`def.exr.group`). **Step by step** 1. Load the video. 2. Choose whether to **Linearize** (de-gamma to linear light). 3. Click **Export EXR frames** (`def.exr.run`) — frames are written to a `<video>_exr` folder.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Linearize (de-gamma BT.709 → linear light) (`def.exr.linear`)	Checkbox	On	On / Off	On = standard compositing (linear light); Off = display-referred (gamma) values.

Tips for best results

> - For a *true* HDR result, generate the video with the HDR IC-LoRA in the Video tab first, then export here.

Troubleshooting

- **Colorization says a FLUX.2 model is required.** Install a FLUX.2 checkpoint via Import models; ESRGAN restore does not need one, but colorization does.
- **Face restoration is greyed out or errors.** The `facerestore_cf` node isn't installed — click 📥 **Install face-restore** and restart when prompted. Its models (~350 MB) download on first use.
- **SeedVR2 runs out of memory.** Raise **BlockSwap** above 0 and/or lower **Batch** toward 1; both reduce VRAM at the cost of speed.
- **The RIFE model dropdown won't let me choose.** That's expected — the native RIFE uses one fixed Practical-RIFE model.
- **Upscaled video looks over-sharp.** Prefer SeedVR2 over ESRGAN for moving footage; ESRGAN is per-frame and can accentuate flicker.

Audio & Voice

Turn text into natural speech, clone a voice from a short sample, and attach a spoken track to any video — all locally, with your choice of seven speech engines.

The **Audio / Voice** tab (`tab.audio_voice`) is where you decide what your videos *say* and *how* they say it. It combines three things in one place: a **source selector** that chooses where a video's audio comes from, a full **multi-engine text-to-speech (TTS) studio**, and a shared **voice library** that stores custom and cloned voices for reuse across the whole application (Video, Music Video, and the AI Film Studio all draw voices from this same library).

How audio works in AI Creative Center

When you generate a video, the model can produce sound in several ways. The **Audio source** radio group at the top of the tab picks exactly one of them:

- **Native LTX audio** — the video model (LTX-2) invents the audio itself, driven by your video prompt. Good for ambient sound and simple spoken lines described in the prompt.
- **Audio I provide** — you supply a ready-made track: a file from disk, a live microphone recording, or a clip you synthesized with the built-in TTS and sent over with **Use as video audio**.
- **Clone voice** — you pick a saved voice (with its reference sample) from the library. With text in the TTS box, that text is synthesized in the chosen voice; with the text left empty, only the voice *timbre* is handed to LTX, which then generates the spoken lines from your video prompt.
- **Mute** — no audio at all.

The TTS studio below the source selector works independently: you can synthesize, preview, and reuse speech at any time, whether or not you are making a video.

Choose the audio source for a video

> **Edition: Full+** · **Capability:** `cap.audio_tts_clone`

What it does — Selects where the audio of your generated video comes from: the model itself, a file you provide, a cloned voice, or silence. **Where to find it** — **Audio / Voice** tab, the **Audio source** group (`aud.source.lbl`) at the top. The label reads *"Audio source (only one active; the other options disable themselves)"*. **Step by step** 1. Open the **Audio / Voice** tab. 2. Under **Audio source**, click one radio button: - **Native LTX audio (the model generates the audio)** (`aud.rb.native`) - **Audio I provide (ready file)** (`aud.rb.user`) - **Clone voice (use a saved voice with its sample)** (`aud.rb.clone`) - **Mute (no audio)** (`aud.rb.mute`) 3. The unrelated controls grey out automatically so you only see what applies to your choice. 4. Optionally tick **Speak but don't sing** (`aud.no_sing`) to force spoken delivery of dialogue with no singing. 5. Go to the **Video** tab and generate — the video inherits this audio source.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Audio source	Radio group	Native LTX audio	Native / I provide / Clone voice / Mute	Selects the audio pipeline for the next video.
Speak but don't sing	Checkbox	Off	On / Off	Forces spoken pronunciation of dialogue and suppresses singing. Turn on when the model tends to sing lines you want spoken.

Specializations — In **Clone voice** mode, the decision tree is: text present in the TTS box → the text is synthesized in that voice; text empty → only the timbre is passed to LTX, which speaks your video prompt. **Tips for best results**

> - Use **Audio I provide** when you already have a clean recording — it is the most predictable path to perfect lip-sync in the Video tab's Talk mode.
> - **Native LTX audio** is best for ambience and background; use TTS or cloning for clear dialogue.

Provide a ready-made audio track (file or microphone)

> **Edition: Full+** · **Capability:** `cap.audio_tts_clone`

What it does — Loads an existing audio file or records a fresh clip from your microphone to use as the video's soundtrack. **Where to find it** — **Audio / Voice** tab, the **Ready audio track** group (`aud.file.group`), active when **Audio I provide** is selected. **Step by step** 1. Select **Audio I provide (ready file)** in the source group. 2. Click **Load audio file...** (`aud.file.pick.btn`) and pick a file, **or** 3. Click ● **Record** (`aud.file.rec.btn`) to capture from the microphone; the button changes to ■ **Stop** (`aud.rec.stop`) while recording, showing *"recording..."* (`aud.rec.recording`). 4. When done, the status shows the loaded/recorded clip; an empty slot reads **(no audio)** (`aud.file.none.lbl`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Load audio file…	Button	—	Audio files	Opens a file picker for a ready track.
● Record	Button	—	Toggle record/stop	Records from the default microphone. Disabled and labelled *"Record (microphone unavailable)"* (`aud.file.rec_na.btn`) when no mic is detected.

Tips for best results

> - Record 5–15 seconds of clean, single-speaker audio for best downstream results.
> - You can also fill this slot by synthesizing speech with TTS and clicking **Use as video audio** (below).

Generate speech from text (multi-engine TTS)

> **Edition: Full+** · **Capability:** `cap.audio_tts_clone`

What it does — Converts typed text into spoken audio using any of seven speech engines, in the voice you select from the library. **Where to find it** — **Audio / Voice** tab, the **TTS — Generate audio from text** group (`aud.tts.group`). **Step by step** 1. Choose an **Engine** (`aud.tts.engine_lbl`) from the dropdown. 2. If the engine's runtime isn't installed yet, click 📥 **Install** (`aud.tts.install.btn`) — it clones the required node and installs dependencies. 3. Pick a **Voice** (`aud.tts.voice_lbl`) from the library dropdown (or create one — see *Voice library*). 4. Type your line in the **Text** field (`aud.tts.text_lbl`). Type `\` to open the SSML tag palette (see *SSML markers*). 5. Click 🎙 **Generate TTS** (`aud.tts.gen.btn`). 6. Click ▶ **Play** (`aud.tts.play.btn`) to audition, and ⤴ **Use as video audio** (`aud.tts.use.btn`) to send the clip to the video's audio slot.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Engine	Dropdown	Edge	**Edge**, **Kokoro**, **F5-TTS**, **XTTS2**, **Fish**, **VibeVoice**, **CosyVoice2**	The synthesis backend. Edge uses Microsoft preset voices; the others (F5-TTS, XTTS2, Fish, VibeVoice, CosyVoice2) are cloning-capable and use a voice sample. Change to a cloning engine when you need a custom timbre.
📥 **Install**	Button	—	—	Installs the selected engine's node and dependencies on first use.
Voice	Dropdown	—	Library voices	The voice used for synthesis. Manage entries with the ➕ / ✎ / ⧉ / 🗑 buttons beside it.
Insert	Dropdown/menu	—	SSML markers	Inserts an SSML tag (pause / emotion / prosody) at the cursor.
Text	Multi-line text	(empty)	free text	The words to speak. Type \ for the SSML tag palette.
🎙 **Generate TTS**	Button	—	—	Synthesizes the text into audio.
▶ Play	Button	—	—	Plays the last synthesized clip.
⤴ Use as video audio	Button	—	—	Sends the clip to the **Audio I provide** slot for the next video.

Specializations — The engine selector's REST equivalent (`POST /audio/tts`) accepts `engine` (`edge | kokoro | f5tts | xtts2 | fish | vibevoice | cosyvoice2`), `voice_id` (default `it-IT-ElsaNeural`), `language` (default `it-IT`), plus `sample_path` and `sample_transcript` for cloning. See REST API. **Tips for best results**

> - **Edge** is the fastest, needs no sample, and has a large catalogue of preset languages/voices — ideal for narration.
> - **F5-TTS**, **XTTS2**, **CosyVoice2**, **Fish** and **VibeVoice** reproduce a specific timbre from a sample — use them for cloned characters.
> - A short, readable preview updates under the text field (`aud.tts.preview.lbl`) as you edit, so you can see how SSML tags will be interpreted before generating.

Build and manage the voice library

> **Edition: Full+** · **Capability:** `cap.audio_tts_clone`

What it does — Creates, edits, duplicates, and deletes reusable voices — including cloned voices built from a reference sample — shared across the whole app. **Where to find it** — **Audio / Voice** tab, the buttons beside the **Voice** dropdown open the voice editor dialog (`ui/voice_dialog.py`). **Step by step** 1. Click the

➕ **new** button (tooltip "*Save a new custom voice, e.g. cloned*", `aud.tts.new.tt`) to open the **New voice** dialog (`voice.new_title`). 2. Enter a **Name** (`voice.name_label`), e.g. "*Calm Alice*". 3. Choose an **Engine** (`voice.engine_label`). Cloning-capable engines are tagged "*clonable*" (`voice.cap_clonable`); Edge is "*preset only*" (`voice.cap_preset_only`). 4. For Edge, pick a **Base voice** (`voice.base_voice_label`). For a cloning engine, attach a **Voice sample** (`voice.sample_header`) — 5–15 seconds of clean, single-speaker speech — with 📁 **Load file...** (`voice.load_file_btn`) or ● **Record** (`voice.record_btn`). 5. Optionally add a **Sample transcript** (`voice.transcript_label`) — writing exactly what is heard improves the clone. 6. Add free-form **Notes** (`voice.note_label`) if you like, then click 💾 **Save** (`voice.save_btn`).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Name	Text	(empty)	free text	The voice's display name. Required.
Engine	Dropdown	—	Edge / CosyVoice2 / F5-TTS / XTTS / ...	Synthesis engine for this voice.
Base voice	Dropdown	—	Edge preset voices	Only for Edge (preset) voices.
Voice sample	File / recording	(none)	wav/mp3/flac/m4a	5–15 s reference clip for cloning engines. Required for clonable engines.
▶ Preview	Button	—	—	Auditions the attached sample.
Sample transcript	Text	(empty)	free text	Exact transcript of the sample; optional but improves clone accuracy.
Notes	Text	(empty)	free text	Free notes for your own reference.

Specializations — The row of small buttons next to the **Voice** dropdown are: ➕ new (`aud.tts.new.tt`), ✎ edit (`aud.tts.edit.tt`), ⧉ duplicate (`aud.tts.dup.tt`), and 🗑 delete (`aud.tts.del.tt`). The dialog validates that a name is present and that cloning engines have a sample attached (`voice.warn_sample_text`). **Tips for best results**

- Voices you create here are available everywhere — the Video Talk/lip-sync flow, Music video, and the AI Film Studio all read from this one library.
- Keep samples dry (no background music), mono, and consistent in tone for the most faithful clone.

Voice cloning

Edition: Full+ · **Capability:** `cap.audio_tts_clone`

What it does — Reproduces a real person's timbre from a short reference clip, then speaks any text (or lends the timbre to LTX) in that voice. **Where to find it** — Two entry points: the **Clone voice** radio in the audio source group, and any cloning-capable engine in the TTS voice editor. **Step by step** 1. Create a cloning voice as described above (a cloning engine + a 5–15 s sample + optional transcript). 2. To synthesize dialogue in that voice: select it in the TTS **Voice** dropdown, type text, and click 🎙 **Generate TTS**. 3. To lend only the timbre to a video: select **Clone voice** in the audio source group, choose the voice, and leave the TTS text empty — LTX will speak your video prompt in that timbre.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Reference sample	File / recording	(none)	5–15 s clean speech	The timbre source. Longer isn't better — clean and consistent is.
Transcript	Text	(empty)	free text	The exact words in the sample; raises clone fidelity for engines that use it.

Specializations — Via REST, cloning is `POST /audio/tts` with `sample_path` (and optionally `sample_transcript`) set on a cloning engine. See REST API. **Tips for best results**

- CosyVoice2, F5-TTS and XTTS2 give the most natural clones; try each on your sample and keep the best.
- Always fill the transcript for cloning engines that support it — it noticeably improves pronunciation and prosody.

SSML markers (pauses, emotion, prosody)

Edition: Full+ · **Capability:** `cap.audio_tts_clone`

What it does — Inserts inline SSML-style tags into your text to control pauses, emotion, and speaking rate/pitch (prosody). **Where to find it** — In the **Text** field, type `\` to open the filtering tag palette, or use the **Insert** dropdown (`aud.tts.insert_lbl`). **Step by step** 1. Place the cursor where you want a marker. 2. Type `\` — a palette opens and filters progressively as you keep typing (`aud.tts.text.ph`). 3. Pick a marker category (pause, emotion, prosody) and value; it is inserted at the cursor. 4. Watch the readable preview below the field update to confirm the effect.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
\ tag palette	Inline picker	—	pause / emotion / prosody markers	Inserts SSML tags; filters as you type.
Insert	Dropdown	—	Marker categories	Menu alternative to the \ palette.

Tips for best results

- Use short pauses between sentences for a natural cadence; over-tagging can make delivery robotic.
- Not every engine honours every marker — Edge and the cloning engines interpret prosody differently; audition after tagging.

Troubleshooting

- **The Record button says "microphone unavailable."** No input device was detected. Connect a microphone and restart the app; the button re-enables (`aud.file.rec_na.btn`).
- **An engine won't synthesize.** Click 📥 **Install** next to the **Engine** dropdown to fetch its node and dependencies; the first run of a cloning engine also downloads its model.
- **The clone doesn't sound like the person.** Use a cleaner 5–15 s single-speaker sample, fill in the **Sample transcript**, and try a different cloning engine (CosyVoice2 / F5-TTS / XTTS2).
- **My video has no dialogue even though I typed text in TTS.** Make sure the **Audio source** is set to **Clone voice** (with a selected voice) or that you clicked ↻ **Use as video audio** after generating — otherwise the video uses the native or muted source.

Music

Compose full songs and instrumentals, add a soundtrack or foley to any video, and clean up the result — all locally, with no cloud services.

The **Music** tab groups every audio-generation function of AI Creative Center. It has two main generators — **Song / Instrumental** (sung or instrumental music from lyrics and style tags) and **Sonify video (MMAudio)** (a matching soundtrack or sound effects for a video) — plus three finishing panels: audio post-production, vocal pitch correction, and a Music LoRA trainer.

> **Edition:** All music functions require **Full+**. Studio Pro adds nothing extra to this tab except model import. Adult (18+) content follows the same rules as elsewhere in the app.

The audio engines run as isolated background processes, so generation never freezes the interface. The first time you use an engine, install it with the dedicated button (it downloads the engine, its PyTorch runtime, and several gigabytes of model weights).

Song / Instrumental

> **Edition:** Full+ · **Capability:** `cap.music_song`

What it does — Turns a set of style tags and (optionally) lyrics into a complete audio track: a sung song with vocals, or a purely instrumental piece. **Where to find it** — Music tab → **Song / Instrumental** panel (top of the tab).

Step by step 1. Give the track a **Name** — this becomes the output file name. 2. Type **Style tags** (comma-separated), or click the 🎨 palette button to pick them, or the 🎲 button to load a complete random example (tags, lyrics and metadata). 3. Optionally enable ✨ **Enhance style (AI)** so a local language model expands your tags into a rich style description before generating. 4. Write your **Lyrics**, using the section chips to insert structure markers, or click ✨ **Generate lyrics with AI** to write them from a theme. 5. Choose the **Engine**, **Model**, **Duration** and the other parameters below. 6. If this is your first run, click **Install HeartMuLa** (or **Install ACE-Step**) and wait for the download to finish. 7. Click **Generate music**. Progress appears in the Progress panel at the bottom.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Name	Text	`song`	free text	Output file name (`.mp3` for HeartMuLa; `.wav` for ACE-Step).
Style tags	Text	`pop, upbeat, female vocals, synth, catchy`	comma-separated	Genre, mood, instruments, vocal type. Put the genre first — it carries the most weight.
🎨 (palette)	Button	—	—	Opens the **Musical styles palette** to pick tags visually.
🎲 (random example)	Button	—	—	Fills tags, lyrics and metadata with a random, always-different style example.
✨ **Enhance style (AI)**	Checkbox	Off	On / Off	Rewrites your tags into a full style caption via a local LLM before generating. Invisible and optional.
Lyrics	Multi-line text	empty	free text with `[Section]` markers	The words to sing. Disabled when **Instrumental** is on.
Section chips	Buttons	—	Intro · Verse · Pre-Chorus · Chorus · Bridge · Instrumental · Guitar Solo · Outro	Insert the corresponding `[Section]` marker at the cursor.
Song theme + ✨ **Generate lyrics with AI**	Text + button	empty	free text; language dropdown (15 languages)	Writes lyrics from a theme in the chosen language using a local Gemma-3 model, then inserts them into the Lyrics box.
Instrumental (no vocals)	Checkbox	Off	On / Off	Produces a purely instrumental track and disables the Lyrics box.
Engine	Dropdown	HeartMuLa	HeartMuLa · ACE-Step · ACE-Step 1.5	Chooses the generation model (see Engines). Available controls change with the engine.
Duration	Number (s)	120	5–360	Target length of the track in seconds.
Model	Dropdown	3B	3B · 7B (whichever are downloaded)	HeartMuLa checkpoint size. Larger = higher quality, more VRAM. HeartMuLa only.

Control	Type	Default	Range / Options	What it does & when to change
Temperature	Number	1.0	0.1–2.0	Creative randomness. Higher = more variety, less predictable. HeartMuLa only.
CFG	Number	2.5 (HeartMuLa) / 15.0 (ACE-Step) / 1.0 (1.5)	HeartMuLa 1.0–8.0 · ACE-Step 1.0–30.0 · 1.5 1.0–10.0	Adherence to your tags/lyrics. The scale differs per engine and is reset automatically when you switch engine. On HeartMuLa keep it around 2.5 (beyond ~4–5 risks artifacts).
Low VRAM (lazy load)	Checkbox	Off	On / Off	Offloads weights to system RAM/CPU during generation — much less VRAM, but slower.
Style adherence	Number	3.0	0.0–10.0	*ACE-Step only.* Weights the style tags via Double Condition Guidance. Only active above 1.0; 3–5 = tags weigh more; 0/≤1 = off.
Lyrics adherence	Number	3.0	0.0–10.0	*ACE-Step only.* Weights the lyrics. Together with Style adherence enables double guidance — both must be above 1.0 to take effect.
Planner 5Hz (think)	Checkbox	On	On / Off	*ACE-Step 1.5 only.* Runs a language-model planner that reasons over metadata and structure before generating, improving coherence.

The estimate badge — Below the parameters, a colored dot estimates the VRAM load (green = OK, yellow = near the limit, red = offload/OOM risk) based on duration, batch, steps and quantization. The time estimate is indicative and depends on your GPU.

Presets — Save the current style configuration under a name with 💾, reload it from the **Preset** dropdown, delete it with 🗑, or pull settings back from a previously generated track with **From track…**. The ↺ **Reset default** button restores the optimal values for the selected engine.

Engines

- **HeartMuLa** — The default engine. Produces complete songs (vocals + lyrics) or instrumentals. Offers the **Model** (3B/7B) and **Temperature** controls; its only adherence lever is **CFG**.
- **ACE-Step** — An alternative engine with finer control. Adds **Style adherence** and **Lyrics adherence** (Double Condition Guidance), an **Advanced (ACE-Step)** panel (scheduler, CFG type, omega, guidance interval/decay, ERG, output format, seed, batch, quantized model, overlapped decoding, torch.compile, LoRA), and an **Operation (ACE-Step)** panel for editing tasks (Retake, Repaint, Edit, Extend, Audio→Audio). Only ACE-Step supports Music LoRA training.
- **ACE-Step 1.5** — A turbo DiT model with a 5 Hz planner. Metadata are passed as structured parameters (see below) rather than appended to the caption. CFG stays near 1.0; adherence comes from the planner and the metadata.

Musical metadata

Collapsible panel: 🎼 Musical metadata (BPM, key, time signature, language).

Control	Type	Default	Range / Options	What it does & when to change
BPM	Number	(auto)	0–300 (0 = auto)	Tempo. 60–80 slow, 90–120 medium, 120–150 pop/dance/rock, 150–180 very energetic.
Key	Dropdown	(auto)	(auto) + common major/minor keys	Key/scale. Minor keys sound darker; major keys brighter.
Time signature	Dropdown	(auto)	(auto) · 4/4 · 3/4 · 6/8 · 2/4	Meter. 4/4 for most pop/rock, 3/4 waltz, 6/8 ballads.
Vocal language	Dropdown	(auto)	(auto) + 12 languages	Singing language. Auto is detected from the lyrics; set it if detection is wrong or the lyrics mix languages.

For HeartMuLa and ACE-Step these values are appended to the style tags; for ACE-Step 1.5 they are passed as structured fields.

Specializations — For the ACE-Step editing operations (Retake / Repaint / Edit / Extend / Audio→Audio), see the **Operation (ACE-Step)** panel. Selecting a region on the waveform player after generation automatically loads that time range into the Repaint operation.

Outputs — A generated song is saved as an audio file (`.mp3` with HeartMuLa) in the output folder, accompanied by a `.lyrics.txt` and a `.tags.txt` file holding the exact inputs, plus a metadata sidecar so the parameters can be reused later. Use 📝 **Generate LRC** in the Progress panel to produce synchronized karaoke lyrics from the audio.

Tips for best results

- Put the genre first in your tags — it dominates the style.
- Leave metadata on **(auto)** unless you have a specific tempo/key in mind; the model usually picks well.
- For the strongest tag/lyric adherence on ACE-Step, raise both **Style adherence** and **Lyrics adherence** above 1.0.
- If you hit VRAM limits (red badge), enable **Low VRAM (lazy load)** or, on ACE-Step, the **Quantized model** option in the Advanced panel.
- Use `{option a|option b|option c}` wildcards in tags or lyrics to randomize a choice at generation time.

Sonify video (MMAudio)

Edition: Full+ · **Capability:** `cap.music_sonify`

What it does — Generates an audio track that is synchronized to a video (soundtrack, sound effects, or both). With no video, it works as a text-to-audio generator. **Where to find it** — Music tab → **Sonify video (MMAudio)** panel.

Step by step 1. Set the **Video** field: click **Browse...**, or **Last video** to reuse the last video generated in the app. Leave it empty for text-to-audio. 2. Describe the sound you want in **Prompt** (e.g. "rain and thunder, footsteps on gravel"). 3. Optionally add a **Negative prompt** to exclude unwanted sounds. 4. Adjust **Duration**, **Steps**, **CFG** and **Seed** as needed. 5. If this is your first run, click **Install MMAudio**. 6. Click **Generate audio**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Video	File path	empty	video file, or empty	The video to sonify. Empty = text-to-audio.
Last video	Button	(disabled until available)	—	Loads the most recent video generated in the app.
Prompt	Text	empty	free text	Describes the sound to generate.
Negative prompt	Text	empty	free text	Sounds to avoid.
Duration	Number (s)	8	1–120	Length of the generated audio.
Steps	Number	25	10–100	Denoising steps. More = cleaner audio, slower.
CFG	Number	4.5	1.0–10.0	Adherence to the prompt.
Seed	Number	42	0–2,000,000,000	Fix it for reproducible results.

Tips for best results

- Describe concrete sound events ("thunderclap", "waves crashing") rather than abstract moods for tighter sync.
- Keep the duration close to the video length; longer requests take more VRAM and time.
- For a full musical score, describe the genre and instruments; for foley, describe the physical events on screen.

Post-processing

Collapsible panel: 🎚 Audio post-production (normalize · trim · fade · convert). No download required — it runs on any existing audio file via FFmpeg.

What it does — Cleans up and standardizes a finished track: loudness normalization, silence trimming, fades, and format conversion. **Where to find it** — Music tab, below the Sonify panel.

Step by step 1. Set the **Source**: drag an audio file in, **Browse**, or click **Last track** to use the last generated track. 2. Enable the operations you want. 3. Click 🎚 **Process**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Normalize loudness (EBU R128, -14 LUFS)	Checkbox	On	On / Off	Brings perceived volume to a uniform streaming level without clipping.
Trim silence (start/end)	Checkbox	Off	On / Off	Removes leading/trailing silence and long internal gaps.
Fade in	Number (s)	0.0	0.0–15.0	Fade-in duration (0 = none).
Fade out	Number (s)	0.0	0.0–15.0	Fade-out duration (0 = none).
Format	Dropdown	(unchanged)	(unchanged) · wav · mp3 · flac · ogg	Output format; (unchanged) keeps the source format.

Pitch correction (DiffPitcher)

Collapsible panel: 🎤 Vocal pitch correction (DiffPitcher).

What it does — Corrects an out-of-tune vocal by transferring the pitch curve from an in-tune guide vocal singing the same phrase. **Where to find it** — Music tab, below Post-production.

Step by step 1. Set **Off-key vocal** (the take to fix): drag in, Browse, or **Last track**. 2. Set **Guide vocal** — an in-tune recording of the same phrase. 3. Adjust **Steps** and **Transpose** if needed. 4. Click 🎤 **Correct pitch**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Steps	Number	50	10–100	Diffusion steps. More = finer correction, slower.
Transpose	Number (semitones)	0	-12 to +12	Shifts the reference pitch in semitones (0 = none).

> This module is a preview feature; results may vary depending on your material.

Music LoRA trainer

Collapsible panel: 🎓 Train Music LoRA (ACE-Step). Only available with the ACE-Step engine — with HeartMuLa the panel shows a note and stays disabled.

What it does — Trains a custom LoRA on your own audio dataset so the ACE-Step engine can reproduce a specific style, voice or instrument. **Where to find it** — Music tab, standalone panel between the Sonify panel and the Progress panel.

Step by step 1. Select **ACE-Step** as the engine so the panel becomes active. 2. Drag audio files or whole folders into the **Audio dataset** list. If a file has a matching `<name>_prompt.txt` / `<name>_lyrics.txt` next to it, those are used; otherwise the global fields below apply. 3. Fill in **Global tags** and, optionally, **Global lyrics**. 4. Set the training parameters. 5. Click 🎓 **Train LoRA** and confirm.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
LoRA name	Text	`music_lora`	free text	Output folder name for the trained LoRA.
Global tags	Text	empty	free text	Style tags applied to files without their own prompt file.
Global lyrics	Multi-line text	empty	free text	Lyrics applied to files without their own lyrics file (optional).
Max steps	Number	2000	100–100000	Training steps. More = more learning and more time.
Learning rate	Number	1e-4	1e-6–1e-2	A good starting point is 1e-4.
Repeat dataset	Number	2000	1–100000	Dataset repetitions: high for few files, lower for many.
Precision	Dropdown	bf16-mixed	bf16-mixed · 16-mixed · 32	bf16-mixed is fast and light on recent GPUs.
Batch (accum)	Number	1	1–16	Gradient accumulation (effective batch); raise it if you have lots of VRAM.

Once trained, apply the LoRA from the **Advanced (ACE-Step)** panel's LoRA field when generating.

Progress, playback and results

The **Progress** panel at the bottom of the tab shows a progress bar, a status line, and a live log. After generation you can:

- **Play** the result in your system player, or use the in-app **waveform player** (drag-select a region to send it to the ACE-Step Repaint operation).
- **Open folder** to reveal the output files.
- 📊 **Spectrum / A-B** to inspect the spectrogram and compare two tracks by spectral centroid.
- 📚 **Library** to browse your generation history — replay, reuse the parameters, or delete tracks.
- 📝 **Generate LRC** to transcribe the audio into synchronized karaoke lyrics.
- **Stop** to cancel a running generation.

Related

- To turn a song into a full video clip, see Music Video.
- For cinematic, scripted projects with dialogue and scoring, see Cinema — AI Film Studio.
- For voice synthesis and cloning used by other modules, see the Audio & voice chapter.

Music Video

Turn a song into a finished music video: AI Creative Center writes a shot list from your lyrics, generates an opening image and a video clip for each shot, then concatenates everything and adds the audio track.

The **Music Video Studio** is a project-based workflow. Each project is saved to disk, so you can build a video in stages, close the app, and come back to exactly where you left off. The Studio walks through four phases — Song, Shots/Images, Videos, Assembly — each of which you can review and edit before moving on.

> **Edition:** Full+ · **Capability:** `cap.music_video`. Studio Pro adds nothing extra here except model import.

The whole pipeline runs locally with native engines (no cloud): a local language model plans the storyboard from your lyrics, the image engine renders each opening frame, and the LTX-2 video engine animates it into a clip.

How the workflow fits together

1. **Song** — Generate (or regenerate) the track with HeartMuLa. All songs made for the project are listed; you pick the active one.
2. **Shots / Images** — A local LLM reads your lyrics and builds a shot list, distributing shots across the song's duration. You edit the list — reorder, add, delete, adjust timing and prompts — then generate an opening image per shot.
3. **Videos** — Each opening image is animated into an image-to-video clip. You can regenerate, reorder or delete individual clips.
4. **Assembly** — The clips are concatenated in order and the active song's audio is muxed in, producing the final `music_video.mp4`.

Because the shots are separate image-to-video clips, scene changes are clean hard cuts by design; optional transitions can soften them at assembly time.

Starting a project

You can launch the Studio from the Music tab:

- 🎬 **Generate music video** — Creates a new project from the current song fields (name, lyrics, style tags, starting image, scene pace, video style, duration) and opens the Studio at the Song phase.
- 📁 **Music video projects** — Opens the projects manager.

The projects manager (**Music video projects** dialog) lists every saved project with its current phase and song count. From here you can 📁 **Open**, 📋 **Duplicate**, or 🗑 **Delete** a project. Opening one launches the Studio at its saved phase.

When the Studio opens, it inherits the image model from the Images tab and the video model and resolution from the Video tab, so generated frames and clips match the models you were already using.

Phase 1 — Song

> **Where to find it** — Music Video Studio → **1 · Song** tab.

What it does — Generates the song for the project and lets you keep several takes, choosing which one is active.

Step by step 1. Set **Temperature** and **CFG** for the song generation. 2. Click 🎵 **Generate/Regenerate song**. Each run is added to the list. 3. Select a track and use ▶ **Play** to audition it, ✓ **Use this** to make it the active song, or 🗑 to delete it. 4. Click → **Go to shots** to continue.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Temperature	Number	1.0	0.1–2.0	Creative randomness of the song generation.
CFG	Number	1.5	1.0–10.0	Adherence to the project's tags and lyrics.

The lyrics, tags and target duration come from the project (set when you created it). For full control over the song itself, generate it in the Music tab first.

Phase 2 — Shots / Images (storyboard)

> **Where to find it** — Music Video Studio → **2 · Shots / Images** tab.

What it does — Builds and edits the shot list, then generates the opening image for each shot.

Step by step 1. Click **1) Build shot list (LLM)**. The language model reads the project's lyrics and produces a list of shots spread across the song, each with an image prompt and a video prompt. 2. *(Optional)* Click 🎯 **Align to vocals** to snap each shot's start time to the real sung onsets (requires speech-to-text; if unavailable you are prompted to install it). 3. Edit the list on the left: ↑ **/** ↓ to reorder, ➕ **end** / ➕ **after** to add a shot, 🗑 to delete. 4. Select a shot to edit it in the **Shot detail** panel: adjust **Start (s)**, rewrite the **Opening image prompt** and **Video prompt**, then 💾 **Apply changes**. 5. Regenerate that shot's image with 🖼 **Regenerate image**, or 📁 **Replace…** it with your own file. 6. When the list is ready, click **2) Generate images** to

render every missing opening image. 7. Click → **Go to videos** to continue.

Shot-detail controls

Control	Type	Default	Range / Options	What it does & when to change
Start (s)	Number	(auto)	0 and up	Start time of the shot; lengths recompute automatically so each shot runs to the next one.
Length:	Read-only	(auto)	—	Duration of the shot (distance to the next shot's start).
Opening image prompt	Multi-line text	(from LLM)	free text	Prompt for the still frame that opens the shot.
Video prompt	Multi-line text	(from LLM)	free text	Prompt for the motion in the clip.

Batch memory — A **Batch memory** control (**Auto** / **Sticky (fast batch)**) sits above the shot list. *Sticky* keeps the image model resident in VRAM for the whole image batch instead of reloading it per shot — much faster with large models. The default is Sticky and your choice is remembered.

> If the starting image you supplied when creating the project exists, it is reused as the opening image of the first shot.

Phase 3 — Videos

> **Where to find it** — Music Video Studio → **3 · Videos** tab.

What it does — Animates each opening image into a clip, then assembles the final video with audio.

Step by step 1. Click **3) Generate videos** to animate every shot that has an image but no clip yet. 2. Manage clips in the list: ↑ / ↓ to reorder, 🎬 **Regenerate video** to redo one, 🗑 to delete. Thumbnails show the first frame of each clip. 3. Choose a **Transition** for the joins. 4. Click **4) Assemble + audio** to concatenate the clips in order and mux the active song. The result is saved as `music_video.mp4`.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Transition	Dropdown	Hard cut	Hard cut · Crossfade · Fade · Dissolve · Wipe	How consecutive clips are joined at assembly. Hard cut is the clean default; the others soften the join with an FFmpeg crossfade.

Exporting to an editor (NLE)

> **Where to find it** — Music Video Studio → **3 · Videos** tab → format dropdown + 📦 **Export NLE**.

What it does — Exports the shot list as an editing timeline you can open in a professional non-linear editor, keeping each clip on its own segment for further editing.

Step by step 1. Generate the video clips first. 2. Pick a format from the dropdown: **FCPXML**, **OTIO**, **AAF** or **EDL**. 3. Click 📦 **Export NLE**. The timeline file is written to the project's `export` folder.

Use FCPXML for Final Cut Pro, AAF for Avid/Premiere, EDL for broad compatibility, and OTIO as an open interchange format.

Parameters used across the pipeline

When driven through the REST API (`/music/music_video`) or when a project is created, these values control the whole pipeline:

Parameter	Default	What it does
`lyrics`	(required)	The song lyrics; drive both the song and the LLM storyboard.
`song_path`	empty	An already-generated song; empty means generate it first.
`image_path`	(required via API)	Starting/background image reused as the first shot's opening frame.
`width × height`	704 × 512	Frame size of every clip (inherited from the Video tab in the Studio).
`fps`	25	Frame rate.
`scene_seconds`	6.0 (API) / from pace preset	Average shot length = how often scenes change. Smaller = faster cuts.
`video_style`	empty	A look applied to every shot (e.g. "cartoon", "black & white, sad").
`image_hint`	empty	Extra guidance for the LLM storyboard.
`steps`	8	Denoising steps per video chunk.

In the Studio, the **Scene-change pace** you chose when creating the project maps to `scene_seconds`: Slow (~12s), Medium (~7s), Fast (~4s, default) or Very fast (~2.5s).

Tips for best results

- Write lyrics with clear `[Verse]` / `[Chorus]` sections — the storyboard follows the order of the lyrics, so structure improves shot-to-lyric alignment.
- Review and edit the shot list *before* generating images; regenerating a single shot later is cheap, but re-planning the whole list resets your edits.
- Set a **Video style** to keep every shot visually consistent (e.g. "anime", "film noir").
- Use 🎯 **Align to vocals** for lyric-driven videos so cuts land on the sung phrases.
- Faster **scene-change pace** suits energetic tracks; slower pace suits ballads.

Related

- Generate or fine-tune the song itself in Music.
- The clips use the same engine as the Video tab, and the opening frames use the Images engine.

- For scripted, dialogue-driven films, see Cinema — AI Film Studio.

3D

Build a complete 3D pipeline from a single image: reconstruct a textured 3D model, rig it with a skeleton, convert it between formats, animate it from your webcam, and even generate an explorable 3D world.

AI Creative Center's 3D suite is a set of tabs that chain together naturally: generate a model in **Image → 3D model**, send it to **Rig 3D** to add a skeleton, animate it in **Animation / Mocap**, convert it for your target engine in the **Converter**, and build a surrounding scene in **3D World**. Each tab has a built-in interactive 3D viewer on the right (orbit, pan, zoom).

> **Edition:** The entire 3D suite requires **Full+**. This covers Image → 3D model (`cap.model3d` / `cap.threed`), Rig 3D (`cap.rig3d`), Mocap animation (`cap.mocap`), the Converter, and 3D World / WorldGen (`cap.world_gen`). Studio Pro adds nothing extra to these modules except model import.

Some tabs download large model weights on first use, and 3D World and rigging require a local Blender installation (auto-installed when needed). Each tab notes its requirements below.

Image → 3D model (TRELLIS.2)

> **Edition:** Full+ · **Capability:** `cap.model3d / cap.threed`

What it does — Reconstructs a full 3D model — mesh plus PBR materials — from a single photo or generated image. **Where to find it** — **3D Model** tab.

Step by step 1. Set the **Source image**: **Browse...**, **Use generated image** (the current image in the Images tab canvas), or **Use masked subject** (only the subject isolated by your canvas mask, background excluded). 2. Choose the **Quality / model** settings (resolution, export format, background removal). 3. Adjust the **Geometry** and **PBR texture** settings if you need more or less detail. 4. Set the **Seed** (or leave **Random** on). 5. Click **Generate 3D model**. On first use the weights (~25 GB) are downloaded; 24 GB VRAM is recommended. 6. When done, inspect the model in the viewer and use its buttons to **Open folder**, **Open in system viewer**, or **Save as...**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Resolution	Dropdown	1024_cascade	512 · 1024 · 1024_cascade · 1536_cascade	Reconstruction quality. 512 = fast; 1024_cascade = best; 1536_cascade = highest (more VRAM/time).
Export format	Dropdown	glb	glb · obj · ply · stl	File format of the exported model.
Remove background (BiRefNet)	Checkbox	On	On / Off	Isolates the subject before reconstruction. Recommended for generated images.
Geometry detail	Number	49152	16384–262144	Sparse-structure token budget. Higher = more detail, more VRAM and time.
Image adherence	Number	6.5	0.0–20.0	How closely the shape follows the image (typical 6.5).
Sampling steps	Number	12	1–50	Geometry sampling steps.
Mesh faces (max)	Number	500000	1000–5000000	Final mesh decimation (triangle count).
Generate PBR texture	Checkbox	On	On / Off	Generates and bakes PBR materials (color, roughness, metallic). Off = geometry only.
Texture size	Dropdown	2048	1024 · 2048 · 4096	Texture map resolution. Enabled only when textures are on.
Texture adherence	Number	3.0	0.0–20.0	How closely the texture follows the image.
Texture steps	Number	12	1–50	Texture sampling steps.
Seed	Number	0	0 to $2^{31}-1$	Fix it for reproducible results. Disabled while **Random** is on.
Random	Checkbox	On	On / Off	Uses a fresh random seed each run.

Advanced *(collapsible)*

Control	Type	Default	Range / Options	What it does & when to change
Precision	Dropdown	auto	auto · bf16 · fp16 · fp32	Numeric precision of the model.
Attention backend	Dropdown	auto	auto · flash_attn · xformers · sdpa · sageattn	Attention implementation; leave on auto unless you know your GPU favors a specific one.
Background color	Dropdown	black	black · gray · white	Background used during reconstruction.

Tips for best results

- Use a clean, well-lit image of a single subject; keep **Remove background** on for generated art.
- Start at **1024_cascade**; only step up to **1536_cascade** if you have the VRAM and need extra detail.
- Lower **Mesh faces (max)** for lighter models destined for real-time engines.

Rig 3D (SkinTokens)

Edition: Full+ · **Capability:** `cap.rig3d`

What it does — Automatically predicts a skeleton and skinning weights for a character mesh and exports a rigged GLB/FBX, ready to animate. **Where to find it** — **Rig 3D** tab.

Step by step 1. Set the **Source mesh**: **Browse...** or **Use last 3D model** (the last model made in the 3D Model tab). 2. Choose the **Skeleton / export** options. 3. Adjust **Skinning** if needed. 4. Click **Generate rig**. On first run ~4 GB of weights are downloaded, and Blender 4.0+ is required — the app offers to install it automatically via winget (~350 MB) if it isn't found. 5. Inspect the rigged model in the viewer.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Bone naming	Dropdown	Keep model names	Keep model names · Mixamo · Unreal Engine 5	Naming convention for the skeleton. Mixamo = Mixamo retarget; UE5 = UE5 Mannequin; Keep = original names.
Export format	Dropdown	glb	glb · fbx	glb = preview in viewer; fbx = for Mixamo / game engines.
Origin at feet (bottom-center)	Checkbox	Off	On / Off	Places the skeleton origin at the character's feet (recommended for Unreal/Unity).
Transfer rig onto original mesh	Checkbox	On	On / Off	Transfers the predicted rig onto the original high-detail mesh, preserving detail (recommended).
Voxel postprocessing	Checkbox	Off	On / Off	Refines skinning weights with a voxel pass (slower).
Influences per vertex	Number	4	1–32	Max bones influencing each vertex; higher = smoother deformation.
Shade smooth angle	Number (°)	55.0	0.0–180.0	Smoothing angle for shading.

Advanced (sampling) *(collapsible)*

Control	Type	Default	Range / Options	What it does & when to change
Beam search	Number	10	1–20	More beams + lower temperature = more deterministic skeleton.
Temperature	Number	1.0	0.1–2.0	Randomness of the skeleton prediction.
Top-K	Number	5	1–200	Sampling top-K.
Top-P	Number	0.95	0.1–1.0	Sampling top-P (nucleus).
Repetition penalty	Number	2.0	0.5–3.0	Discourages repeated tokens in the predicted rig.

Tips for best results

- Feed a character mesh, ideally in an A- or T-pose, for the cleanest skeleton.
- Keep **Transfer** on to preserve the high-detail original geometry.
- Choose the **Bone naming** that matches your target: Mixamo for Mixamo animations, UE5 for Unreal.

Converter

Edition: Full+ · **Capability:** 3D suite (`cap.model3d / cap.threed`)

What it does — Imports a 3D file in almost any format and re-exports it in the format you need. Runs headless via Blender. **Where to find it** — **Converter** tab.

Step by step 1. Set the **Source 3D file** with **Browse...**. 2. Pick the target **Format**. 3. Adjust **Scale** and the texture/animation

options. 4. Click **Convert**. Progress and messages appear in the log below. Output goes to `output/converted/`.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Format	Dropdown	glTF Binary (.glb)	glb · gltf · fbx · usd · usdz · obj · dae · stl · ply · abc · bvh · x3d · dxf	Output format. GLB for web/Unity, FBX/USD/glTF for Unreal, DXF for AutoCAD (geometry only), STL for 3D printing.
Scale	Number	1.0	0.001–1000.0	Uniform scale factor applied on export.
Embed textures in file	Checkbox	On	On / Off	Packs textures into the output file where the format supports it.
Include animations	Checkbox	On	On / Off	Preserves animation tracks in formats that support them.

> USD/USDZ/Alembic/DAE/DXF availability depends on the modules of your installed Blender build. Unreal projects: use FBX/USD/glTF (there is no `.uasset` export).

Animation / Mocap

> **Edition:** Full+ · **Capability:** `cap.mocap`

What it does — Captures motion from your webcam (or imports a mocap file) and retargets it onto a rigged model's existing skeleton. **Where to find it** — **Animation / Mocap** tab.

Step by step 1. Set the **Rigged model**: **Browse...** or **Use last rig**. 2. Choose a motion source: - **Webcam capture** — pick the **Webcam** index, click **Start capture**, move in front of the camera (a live preview updates), then **Stop capture**. The mocap environment (MediaPipe) is required and can be installed with the provided script. - **Import FreeMoCap session** — **Browse...** for a `.bvh` or `.fbx` you recorded/processed elsewhere. 3. Set the **Retarget / export** options. 4. Click **Apply animation**. The result appears in the viewer.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Webcam	Number	0	0–8	Camera index for capture.
Target skeleton	Dropdown	auto	auto · Mixamo · Unreal Engine 5	Bone-naming convention; must match the template chosen in the Rig tab. auto tries names as-is.
Export format	Dropdown	fbx	fbx · glb · bvh	Format of the animated output.

> The animation is applied to the rig's existing skeleton (no new skeleton is created). If the model has no rig, the BVH armature is kept. Single-camera mocap gives indicative quality.

3D World (WorldGen)

> **Edition:** Full+ · **Capability:** `cap.world_gen`

What it does — Generates an explorable 3D scene from a photo (or from text): a mesh (GLB) and/or a photorealistic Gaussian splat, navigable with orbit and WASD controls. **Where to find it** — **3D World** tab.

Step by step 1. Give the scene a **Name**. 2. Set an **Image** (starting photo) with **Browse...**, or leave it empty and write a **Prompt** to generate from text. (If the image is empty, the current Images-tab canvas image is used when available.) 3. Choose the **Output** mode, **Resolution**, and low-VRAM option. 4. If FLUX.1-dev weights are gated, paste your **HF token** (used only for the one-time download). 5. If this is your first run, click **Install module**, then **Generate scene**. Watch progress in the log; **Stop** cancels. 6. Explore the result in the **Viewer (orbit / 🚶 WASD)**, or open the standalone **Web player**. 7. Export the mesh with **Export FBX** or **Export OBJ**, or **Open folder** to reveal the files.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Name	Text	`world`	free text	Project/output name.
Image	File path	empty	image file, or empty	Starting photo. Empty = generate from the prompt (or the canvas image).
Prompt	Text	empty	free text	Scene description / guidance.
Output	Dropdown	Mesh (GLB, exportable)	Mesh · Splat (photorealistic) · Both	What to generate. Mesh is exportable; Splat is photorealistic; Both produces mesh then splat.
Resolution	Number	1600	512–4096	Generation resolution.
Low VRAM	Checkbox	Off	On / Off	Reduces VRAM use for smaller GPUs.
HF token	Password	empty	token	Hugging Face token for the gated FLUX.1-dev weights (one-time download only).

Specializations — In **Both** mode the mesh is generated first, then the splat automatically. A generated mesh (GLB) becomes available to the **Rig 3D** tab via *Use last 3D model*, so a scene can flow into the same rig → animate chain as a TRELLIS model.

Tips for best results

- Use a photo with clear depth cues (a room, a landscape) for a convincing scene.
- Choose **Splat** for photorealistic look-around, **Mesh** when you need to export to a game engine, **Both** to keep your options open.
- Enable **Low VRAM** if generation runs out of memory; expect it to be slower.

The 3D viewer

Every 3D tab embeds an interactive viewer on the right. Orbit, pan and zoom the model, and use the buttons to **Open folder**, **Open in system viewer**, or **Save as...**. If your GPU or driver cannot run the embedded WebGL viewer, the app falls back to a simple buttons-only viewer so generation still completes and you can open the file externally.

Related

- Feed the Images tab's output straight into **Image → 3D model** with *Use generated image*.
- The 3D chain flows tab to tab: 3D model → **Rig 3D** → **Animation / Mocap**, with the **Converter** ready for any format handoff.

Cinema — AI Film Studio

Turn a book, a short story, a one-line idea, or a random theme into a finished film — screenplay, storyboard, generated shots, voices, music and a final master or NLE project — all inside one guided, non-destructive workspace.

The **AI Film Studio** (🎬 **AI Film Studio**) is the flagship module of **AI Creative Center**. It orchestrates every generative engine in the app — image, video, voice, music, sound effects, lip-sync and LoRA training — behind a single, linear, film-production workflow: **Script** → **Set design** → **Montage** → **Master**. Every stage is re-enterable, every artifact is regenerable, and nothing is ever silently overwritten: each regeneration creates a new **version** you can restore.

> **Edition: Full+** · **Capability:** `cap.filmstudio`
>
> The whole module, and every sub-module inside it, requires the **Full** edition or higher. Individual sub-capabilities are noted per section: Script `cap.filmstudio_script`, Asset/Characters `cap.filmstudio_asset`, Set design/Storyboard `cap.filmstudio_scenografia`, Montage `cap.filmstudio_montaggio`, Voice `cap.filmstudio_voice`, LoRA Studio `cap.filmstudio_lora`.

Opening the Film Studio

The Film Studio opens in its own window titled 🎬 **AI Film Studio** (in the Pro interface, 🎬 **AI Film Studio — Pro** ✨). Across the top runs the **project bar**; below it are the workflow tabs, in order:

1. **Script** (*Copione*)
2. **Set design** (*Scenografia* — storyboard)
3. **Montage** (*Montaggio*)
4. **Asset Manager**
5. **Voice**
6. **LoRA Studio**

A beginner **Flow** banner reminds you of the path: *① Script (idea → Story → screenplay) → ② Storyboard (keyframe → video) → ③ Editing. Use @Name for asset consistency.* (The banner is hidden in the Pro interface, which shows a compact flow strip instead.)

Two global run-modes govern how much the app asks you before proceeding:

- **Supervised** — every stage waits for your explicit approval (the professional default).
- **Auto** — the app advances stage to stage with sensible defaults, stopping only on strong errors, then presents the result for final review. Even in Auto, every artifact stays regenerable.

Projects

Create, open and manage projects

> **Edition: Full+** · **Capability:** `cap.filmstudio`

What it does — A project is one film (or one episode of a series). It stores the input, screenplay, scenes, shots, continuity, timeline and every generated artifact.

Where to find it — The **project bar** at the top of the Film Studio window, and the 📁 **Projects** dialog (*Projects — AI Film Studio*).

Step by step 1. Type a name in the **New project...** field. 2. Set the **Duration**, the **Coherence** path, and the run-mode. 3. Press ➕ **Create**. The project is added to the selector as *Name (#ID)*. 4. To reopen or manage projects later, press 📁 **Projects**, select one, and use 📂 **Open**, 📋 **Duplicate** or 🗑 **Delete**. 5. Use the small 🖍 **Rename**, ⧉ **Duplicate**, 🗑 **Delete** buttons in the project bar for quick actions on the current project. 6. Optionally attach the project to a **Series** (choose from the selector or press + to create one) so it can share cast and LoRAs with other episodes.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
New project...	Text	—	Free text	The project name.
Duration	Number (seconds)	**60 s**	5 – 7200 s	Target film length. Scenes and shots are constrained to sum to this.
Coherence (path_mode)	Dropdown	**Fast**	**Fast**, **LoRA**, **Mixed**	How character/asset consistency is achieved (see below).
Run mode	Dropdown	**Supervised**	**Supervised**, **Auto**	Whether the app pauses for approval at each stage.
Series	Dropdown	— none —	Existing series or +	Groups episodes that share cast/locations/LoRAs.
forbidden (negative)	Text	—	Free text	Project-wide negative elements injected into every prompt.
must-have (always)	Text	—	Free text	Project-wide required elements injected into every prompt.

Specializations — the three coherence paths - **Fast** — no training. Consistency comes from the asset's **reference images** plus **PuLID** for faces and multi-reference conditioning. Instant, zero training cost; ideal for drafts, animatics, minor assets and short projects. - **LoRA** — a trained **LoRA** (image and video) per asset gives strong, repeatable identity. `@Name` resolves to the LoRA's **trigger** token. Best for a main cast, series and final quality. - **Mixed** *(recommended)* — the main cast on **LoRA**, extras on **Fast**, in the same job. Each asset can declare its own path.

Tips for best results

> - Start every serious production in **Supervised** mode so you can approve the screenplay before spending GPU time on shots.
> - Use **Series** for anything episodic — the cast LoRAs you train once carry over automatically.
> - Fill **must-have** and **forbidden** early (e.g. an aspect look you always want, or artifacts you never want) — they are injected everywhere.

Script (Copione) tab

Write or generate the screenplay

> **Edition: Full+** · **Capability:** `cap.filmstudio_script`

What it does — Turns any of four input kinds into a cinematic **screenplay**, opening with a short **plot summary**, respecting your **duration**, **genre(s)** and **style(s)**, and supporting `@Name` asset references throughout.

Where to find it — The **Script** (*Copione*) tab.

Step by step 1. Pick an **Input type** and enter your material in the input box (*Paste the story, idea or theme… (@Name for assets)*). 2. Choose one or more **Style** and **Genre** values (both fields accept comma-separated multiples and are editable). 3. (Optional) Open the 🎲 **Writers' table (branches/beats)** to co-write interactively: the LLM proposes story beats with alternative branches; pick a branch per beat and it weaves them into the Story. 4. (Optional) Press 📖 **Generate Story** to produce an editable cinematic **prose** treatment first. 5. Press ✍️ **Generate script** to produce the screenplay. It appears with a **Plot summary** on top and the editable **Script** below. 6. Refine freely by hand, right-click a passage to **regenerate a segment** (similar, or "what to change"), or press ↻ **Regenerate** for a fresh full pass with a new seed. 7. Press 💾 **Save**. The screenplay is versioned (**Script saved (v{n})**).

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Input type	Dropdown	Full story	**Full story**, **Mini-idea**, **Random theme**, **Interactive**	How the LLM treats your input (reduce / expand / invent / build step by step).
Style	Editable dropdown	—	~60 visual styles (comma-separated)	Visual look (e.g. noir, minimalist, hyperreal).
Genre	Editable dropdown	—	~23 genres (comma-separated)	Genre(s); can be combined (e.g. comedy + horror).
Story input	Multi-line text	—	Free text, supports `@Name`	Your source material.

Specializations — the four input kinds - **Full story** — a book/short story: the LLM reduces it to cinematic format. - **Mini-idea** — a short premise: the LLM expands it. - **Random theme** — just a theme: the LLM invents and expands. - **Interactive** — build the story conversationally; the 🎲 **Writers' table** proposes branching beats you weave into the Story.

The tab exposes three output areas: **Cinematic story** (editable prose), **Plot summary**, and the editable **Script**. The **@Name** system works in every text area — type `@` to open the asset picker; type `#` after a name to jump to its variants.

Tips for best results

> - Reference recurring characters and places with `@Name` from the first draft — the breakdown and continuity engine rely on them.
> - Use 📖 **Generate Story** before ✍️ **Generate script** when you want to lock the narrative tone before the technical treatment.
> - Regenerate *segments* rather than the whole screenplay when you only want to fix one beat — you keep everything else intact.

Set design / Storyboard (Scenografia) tab

Break down, storyboard and shoot the film

> **Edition: Full+** · **Capability:** `cap.filmstudio_scenografia`

What it does — Turns the approved screenplay into **Scenes**, each split into **Shots**, then generates the **keyframe** images and the **shot videos**, all governed by the continuity engine. This is where the film is actually rendered.

Where to find it — The **Set design** (*Scenografia*) tab.

Step by step 1. Press 🎬 **Scene/shot breakdown (LLM)**. The scene/shot tree fills with **S1, S2…** scenes and their shots, each with a duration. 2. Select the **Image model** and **Video model** (both default to **(auto)**). 3. Press 🖼 **Generate keyframes** to render the opening (and intermediate/final) reference images for every shot. 4. Press 🎥 **Generate videos** to render the shot clips (LTX-2.3), conditioned on the keyframes and, in the LoRA path, on the subjects' video LoRAs. 5. Or press 🎬 **Generate full film** to queue the entire keyframe + video pipeline at once (a cost estimate is shown; then go to **Montage** → **Master**). 6. Use the per-shot/scene panel to edit descriptions (with `@Name`), set the **Seed**, and regenerate keyframe or video individually.

Key controls

Control	Type	Default	Range / Options	What it does & when to change
Image model	Dropdown	**(auto)**	Installed image models	Which checkpoint renders keyframes.
Video model	Dropdown	**(auto)**	Installed video (GGUF) models	Which transformer renders shot videos.
🔗 **Continuous match**	Checkbox	Off	On / Off	Anchors each shot to the previous shot's last frame (same scene) for cuts without a hard jump.
Seed (0=auto/lock)	Number	0	0 – $2^{31}-1$	Per-shot seed; lock a value for reproducible regenerations.
Shot type	Dropdown	(auto)	wide, close-up, medium, insert, …	Shot grammar for the selected shot.
Camera move	Dropdown	(auto)	pan, tilt, dolly, zoom, …	Camera motion for the selected shot.
Transition	Dropdown	cut →	cut, fade, in black, dissolve, wipe	Transition to the next shot.
In / Out (trim)	Number (s)	0	0 – 3600 s (step 0.1)	Trim the shot's usable range for the timeline.

Specializations — creative assistants - 🎨 **Style Bible** — define the film's **emotional logline**, **color palette**, **look (grain/optics/light)** and **reference directors/DoP**. Once saved, it is injected into every prompt. - 🧐 **Continuity supervisor** — re-reads the breakdown and flags continuity jumps (180°/eyeline, props, time of day, conditions). - 🎥 **DoP: suggest shot** — the director of photography proposes camera + lighting per shot for the selected scene, drawn from the Style Bible; apply the ones you like (writes `shot.camera`). - 🎬 **Expand dialogue coverage** — turns a multi-line dialogue shot into proper shot-reverse-shot coverage (one shot per line, alternating over-the-shoulder / clean single angles). - 🩹 **Take history** — every keyframe/video regeneration is kept as a numbered take you can restore; nothing is lost. - 📄 **Shot-list** — export the breakdown as a production shot-list.

Chain-linking for smooth cuts — With 🔗 **Continuous match** on, the app extracts the last frame of shot *N* and feeds it as the frame-0 conditioning of shot *N+1* (at a low strength, so motion still evolves), giving raccord cuts within a scene.

Tips for best results

- Set the **Style Bible** before generating anything — it is the single biggest lever on a consistent look.
- Run the 🧐 **Continuity supervisor** after the breakdown and again after editing shots, before committing GPU time to videos.
- Keyframe first, review, *then* video. Regenerating a keyframe is far cheaper than a bad video.
- Turn on 🔗 **Continuous match** only within continuous action; across a hard cut you usually want a fresh frame.

Continuity engine

The continuity engine makes consistency an **explicit state**, not a lucky side effect of prompting.

Per-scene / per-shot state fields

Field	Default	Options / Notes
Time of day	day	day, night, dawn, dusk, golden hour
Weather	clear	clear, cloudy, rain, snow, fog, storm
Lighting	natural soft	natural soft, hard sun, low key, high key, candlelight, neon
Color grade	neutral	narrative color grade
Mood	neutral	neutral, tense, joyful, melancholic, epic, romantic, scary
Lens	*(empty)*	24mm wide, 35mm, 50mm, 85mm portrait, anamorphic — *persists across scenes*
Film look	*(empty)*	warm cinematic, cool teal-orange, bleach bypass, vintage film grain, high contrast noir, pastel — *persists*
Per-asset state	—	for each `@character`: **variation**, **wardrobe**, **condition** (e.g. wounded, wet), **emotion**

How it works - **Forward propagation** — state propagates forward from scene to scene and shot to shot. Photographic choices (lens, film look) persist; character changes (e.g. an actor gets muddy in scene 2) carry until you change them. - **Prompt injection** — the relevant state is composed into a phrase (time, weather, lighting, color grade, mood, lens, film look, plus each asset's condition/emotion/wardrobe) and injected into every image/video prompt, so you never re-type it. - **Warnings** — the engine flags incoherences (e.g. *time of day changes: day → night*, or *a character's condition lost without reason*) so you can catch jumps before rendering.

To edit continuity, select a **scene** in the storyboard, set the fields in the **Scene continuity** panel, and press **Apply + propagate**. Use 👤 **Character state…** to set wardrobe / condition / emotion per character; it is saved and propagated to following scenes.

Montage tab

Assemble, score and export the film

Edition: Full+ · **Capability:** `cap.filmstudio_montaggio`

What it does — Builds an OpenTimelineIO (OTIO) timeline from the generated shots, lets you reorder/trim clips and add audio, then either renders a finished **Master** or exports a **project** for an external editor.

Where to find it — The **Montage** (*Montaggio*) tab.

Step by step 1. Press ↻ **Refresh timeline** to (re)build the timeline from the generated clips. 2. Drag clips to reorder; trim in/out on the timeline; ripple-delete is reversible (↩ **Restore excluded clips**). 3. Generate audio: 🔊 **Generate audio (dialogue+music)**, 🎵 **Soundtrack**, 🌧 **Ambient/SFX**, 💥 **Action SFX**. 4. Set the **Output** format: **Aspect**, default **Transition** and its duration, and the **NLE format**. 5. Press 🎞 **Master (A)** to render the finished video (all audio mixed, transitions applied), or 📦 **Export NLE (B)** to export the timeline for an external editor.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Aspect	Dropdown	**16:9**	16:9, 2.39:1 Cinemascope, 9:16 Vertical, 1:1	Output aspect ratio (letterbox/pillarbox, no crop).
Transition (default)	Dropdown	**Cut**	Cut, Dissolve, Dissolve to black, Dissolve, Wipe ←, Slide →, Circle	Default transition between clips.
Transition duration	Number (s)	**0.5 s**	0.1 – 3.0 s (step 0.1)	Length of transitions.
NLE format	Dropdown	**FCPXML**	FCPXML (Filmora/Resolve), OpenTimelineIO, AAF (Premiere/Avid), EDL	Interchange format for the NLE export.

Specializations - **Audio tracks** — the timeline carries separate **dialogue**, **music** and **ambient** tracks, mixed with ducking. Dialogue is synthesized from the shot's `@Name` →voice bindings and lip-synced; music and ambient are generated from the scene mood; **Action SFX** are anchored per shot. - **Master (A)** — renders a single mixed file at the chosen aspect/transitions. The delivered master is the one point where the invisible provenance signature and the visible brand watermark are applied (see Provenance signature). - **Export NLE (B)** — emits **FCPXML / AAF / EDL / OTIO** via OpenTimelineIO, with a self-contained media + manifest package. If OTIO is unavailable, a JSON manifest + minimal EDL fallback is written. Import into **Premiere** (AAF/XML), **DaVinci Resolve** (FCPXML), Filmora and others to finish. - **Review assistants** — 🎞 **Animatic** (fast keyframe-sequence preview with dialogue), 🔍 **Second look (critic)** (an AI critic's final creative notes), 📉 **Emotional rhythm** (tunes shot durations to a per-scene tension curve while keeping scene totals), and 🌐 **Dub to language…** (regenerate dialogue in another language, optionally re-lip-syncing the videos).

Tips for best results

- Render an 🎞 **Animatic** first to judge pacing before spending GPU on full videos.
- Produce **both** outputs when in doubt: a quick **Master** to share, and an **NLE export** to finish properly.
- Use 📉 **Emotional rhythm** to shorten cuts in high-tension scenes automatically — it preserves each scene's total duration.

Asset Manager

Create reusable actors, props, locations and architectures

> **Edition: Full+** · **Capability:** `cap.filmstudio_asset`

What it does — A CRUD library of first-class assets that stay consistent across scenes and projects, each with a generated **Character/Asset Sheet**, **variations**, and (for actors) **wardrobe**.

Where to find it — The **Asset Manager** tab.

Step by step 1. Press ➕ **New**, pick the **Type**, and fill **Name**, **Description** (supports `@Name`), **Tags** and an optional **Base photo**. 2. Press 💾 **Save**, then 🎭 **Generate Character Sheet** to produce a coherent multi-angle turnaround (front, three-quarter, side, close-up, full body, detail). 3. Curate the gallery: ✓ **Accept/Remove** individual images, or ↻ **Regenerate this angle** (the old version is kept in take history). 4. Add **Variations / Sub-locations** (e.g. *wounded*, *dirty*, an interior clearing) and press 🎭 **Generate set** for each. 5. For actors, add **Wardrobe** items and press 👗 **Wardrobe** to auto-generate a six-angle outfit turnaround.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Type	Dropdown	Character	**Character**, **Object** (prop), **Place** (location), **Architecture**	The asset category.
Name	Text	—	Free text	Used as the `@Name` reference and (in LoRA path) the trigger alias.
Description	Multi-line text	—	Free text, supports `@Name`	The prompt the sheet is generated from.
Tags	Text	—	comma-separated	Search tags (LIKE with AND/OR/NOT).
Base photo	File	—	Image	Optional real photo to anchor the appearance.

Specializations — reference resolution by path - **Fast** — `@Name` resolves to the asset's **reference images** plus **PuLID** (faces) or Kontext (props/places). - **LoRA** — `@Name` resolves to the asset's trained **LoRA + trigger**. - **Mixed** — per-asset choice within the same job.

Variations are addressed as `@Name#Variant`; deleting `#Variant` from a reference reopens the variant picker, deleting `@Name` reopens the asset picker. Replacing an asset in a *scene* description propagates to every shot in that scene.

Tips for best results

> - Anchor important characters with one or two clean **Base photos** for a stronger, more stable identity.
> - Build the variations you know the story needs (wounded, wet, night) up front — the continuity engine can then switch to them by name.
> - Accept/curate the Character Sheet carefully: it doubles as the **training dataset** for the asset's LoRA.

Voice tab

Give characters voices and sync their lips

> **Edition: Full+** · **Capability:** `cap.filmstudio_voice`

What it does — Creates voice profiles — from a **description**, from **parameters**, or **cloned** from a sample — and binds them to actors so their dialogue is synthesized automatically and lip-synced onto their shots.

Where to find it — The **Voice** tab.

Step by step 1. **From description:** name the voice, describe it (e.g. *old man, baritone, authoritative tired tone, northern accent*) and press 🗣 **Create voice from description**. 2. **From parameters:** name it, set gender/age/language/emotion/rate/pitch and press 🎚 **Create voice from parameters**. 3. **Cloned:** name it, load a **Sample…** (`.wav` / `.mp3` / `.flac`) and press 🎙 **Register cloned voice**. 4. **Bind:** pick an **Actor** and press 🔗 **Bind**. Dialogue attributed to that `@Name` will use this voice, with lip-sync applied to the shot.

Parameters (voice from parameters)

Control	Type	Default	Range / Options
Gender	Dropdown	Female	Female, Male
Age	Dropdown	Adult	Child, Young, Adult, Old
Language	Editable dropdown	it	it, en, es, fr, de, pt, ru, zh, ar
Emotion	Dropdown	Neutral	Neutral, Happy, Sad, Angry, Whisper, Calm
Rate	Dropdown	Normal	Very slow, Slow, Normal, Fast, Very fast
Pitch	Dropdown	Medium	Very low, Low, Medium, High, Very high

Specializations - **Dialogue timing** — the app checks that a line fits the shot's duration and warns if a line is longer than its shot (extend the shot or trim the text). - **Dubbing** — from Montage you can regenerate all dialogue in another language while keeping each character's bound voice, then re-lip-sync.

Tips for best results

> - Bind a voice to every speaking actor before generating dialogue — unbound `@Name`s fall back to a generic voice.
> - For cloning, use a clean 10–30 second sample with no background noise or music.

LoRA Studio

Train and manage character LoRAs (image → video)

> **Edition: Full+** · **Capability:** `cap.filmstudio_lora`

What it does — Hosts the shared LoRA library for the project and drives the **character pipeline** that turns an asset's photos into a trained image LoRA and then a video LoRA — the backbone of the **LoRA** coherence path.

Where to find it — The **LoRA Studio** tab. Actual training happens in the app's **Video LoRA** and **Photo LoRA** tabs (see Training your own LoRA); trained LoRAs appear here automatically.

Step by step 1. Select an asset in **Character pipeline (image → video)**; optionally point **photos folder** at a folder (empty = use the asset's reference images). 2. Choose **Supervised (gate)** or **Automatic**, then press ▶ **Start pipeline**. 3. In Supervised mode, a **gate** appears — **Pick the good photos** that represent the character well, then continue. 4. When finished, the LoRA is **registered** in the library. Set its **Trigger**, **Type**, **Scope** and **Asset**, then 💾 **Save**.

The pipeline state machine

State	Meaning
new	Pipeline created, not started.
img_training	Generating the dataset and training the **image** LoRA.
review	Paused at the photo-selection gate (Supervised) — waiting for your picks.
vid_training	Training the **video** LoRA from the selected material.
registered	Both LoRAs trained and registered in the library.

(An **error** state captures failures; the pipeline is idempotent and resumable.)

Parameters (LoRA details)

Control	Type	Default	Range / Options	What it does & when to change
Trigger	Text	—	Free text	The rare token that summons the subject in prompts.
Type (flavor)	Dropdown	—	image, video	Whether the LoRA is for image or video generation.
Scope	Dropdown	**Episode**	Episode, Series	Episode-only vs. shared across the whole series.
Asset	Dropdown	—	Project assets	The asset this LoRA represents.
Weight	—	balanced default	per-LoRA	How strongly the LoRA is applied.

Specializations - **Base arch & flavor** — each asset gets an **image** LoRA (FLUX-family) and a **video** LoRA (LTX-2.3), trained from the same image base and sharing the same trigger, so `@Name` works identically in stills and motion. - **Multi-LoRA warning** — when a shot uses several LoRAs at once, the app warns above a small threshold (⚠ … *simultaneous LoRA — interference risk, consider splitting*) and suggests splitting the shot or using the Fast path for minor subjects.

Tips for best results

> - Train the **image LoRA first**, review its samples, then let the pipeline train the **video LoRA** — the still identity anchors the motion one.
> - Keep no more than ~3 LoRAs active in a single shot; beyond that, faces start to bleed.
> - Scope recurring series cast as **Series** so every episode reuses them with pinned versions.

Job Scheduler

Every heavy generation — images, videos, LoRA training, audio — passes through a **priority queue**. The queue does not block the UI: you keep working on the script or assets while it runs, and it recovers orphaned jobs after a crash.

Job types and time estimates (used for the cost estimate shown before **Generate full film**):

Job type	Estimate
`image`	25 s
`video`	90 s
`lora_train`	1800 s (30 min)
`audio_voice`	20 s
`audio_music`	120 s
`audio_sfx`	30 s
`lipsync`	60 s

Jobs carry a **priority** (default 5; lower runs first) and a status of *pending → running → done / error / retry*. Before launching a batch, the app sums these estimates so you can decide whether to proceed. The Pro interface shows a live 🎞 **Job** indicator (*{r} running · {w} waiting*).

Generation service

Under the hood, two native functions do the actual rendering. You never call them directly, but their parameters explain what the UI controls map to.

`gen_image` — renders one image: - `prompt`, `out_png`, `width` (**1024**), `height` (**1024**), `seed` (**7**) - `face_ref` → **PuLID** face conditioning (strength 0.85) · `ref_image` / `ref_images` → **Kontext** multi-reference (used when there is no face_ref) - `loras` (optional) · `arch` (**flux**) · `model` (checkpoint) · `negative` (extends the default negative with continuity's forbidden elements) - Internal defaults: 22 steps, guidance 3.5.

`gen_shot_video` — renders one shot clip (LTX-2.3 GGUF): - `prompt`, `out_mp4`, `width` (**768**), `height` (**512**), `fps` (**25**), `duration_s` (**4.0**), `seed` (**7**) - `keyframes` = `[{path, frame, strength}]` (opening/intermediate/final conditioning) · `loras` (optional) · `audio_path` (for lip-sync) · `model_gguf` (project's chosen transformer) · `negative` - Internal defaults: 8 steps, guidance 3.0, CPU offload; long shots are chunked automatically. If a persistent video worker is running, the model stays loaded in VRAM between shots.

Style Bible

The 🎨 **Style Bible** (in the Set design tab) is the project's visual constitution: an **emotional logline**, a **color palette**, a **look** (grain / optics / light) and **reference directors/DoP**. Once saved, it is injected into *every* image and video prompt, giving the whole film a coherent identity without you repeating yourself. Combine it with the continuity engine's persistent **lens** and **film look** for a locked cinematographic style.

A full worked example: idea → finished film

1. **Create the project.** In the project bar, name it "*The Last Lighthouse*", set **Duration** to 120 s, **Coherence** to **Mixed**, **Run mode** to **Supervised**, and press ➕ **Create**.
2. **Cast.** In **Asset Manager**, create an actor `@Mara` (description + one base photo), generate her Character Sheet, accept the best angles, and add a *storm-soaked* variation. Create a location `@Lighthouse`.
3. **Voices.** In **Voice**, create a weary female voice from parameters and 🔗 **Bind** it to `@Mara`.
4. **Script.** In **Script**, pick **Mini-idea**, type "*On the last night before the lighthouse is decommissioned, @Mara keeps the lamp burning through a storm.*", set Style "*moody, teal-orange, film grain*" and Genre "*drama*". Press 📖 **Generate Story**, then ✍️ **Generate script**, review the summary, and 💾 **Save**.
5. **Style Bible.** In **Set design**, open 🎨 **Style Bible**: logline "*solitude versus duty*", palette "*cold blues, amber lamp*", look "*35mm grain, anamorphic flares*", reference "*Deakins*". Save.
6. **Breakdown.** Press 🎬 **Scene/shot breakdown**, then 🧐 **Continuity supervisor** to catch jumps. Set scene continuity to **night / storm / low key**.
7. **Storyboard.** Press 🖼 **Generate keyframes**, curate, then 🎥 **Generate videos** with 🔗 **Continuous match** on for the action beats.
8. **Coverage & LoRA (optional).** For the close dialogue, use 🎬 **Expand dialogue coverage**; for maximum consistency, train `@Mara`'s LoRAs from **LoRA Studio**.
9. **Montage.** In **Montage**, press ↻ **Refresh timeline**, generate 🗣 **dialogue**, 🎵 **Soundtrack** and 🌬 **Ambient**, set **Aspect** to **2.39:1**, then render 🎞 **Master (A)** — or 📦 **Export NLE (B)** as FCPXML to finish in Resolve.

Troubleshooting

Symptom	Likely cause	Fix
"LLM unavailable (pick Gemma or connect LM Studio)"	No text engine connected.	In the LLM engine panel choose **Local Gemma**, or select **LM Studio**, set the **Port** and press **Connect**.
"Create/select a project."	No active project.	Create or open a project in the project bar / 📁 **Projects**.
"Generate the script first."	Trying to break down with no screenplay.	Generate and 💾 **Save** the script in the **Script** tab first.
Scene/film durations don't add up (⚠ warnings)	Shot durations ≠ scene total, or scenes ≠ film total.	Adjust shot/scene durations; the warning shows the exact mismatch.
Dialogue longer than the shot	A line exceeds the shot duration.	Extend the shot or trim the line (the app flags which shots overflow).
"⚠ … simultaneous LoRA — interference risk"	Too many LoRAs in one shot.	Split the shot, or move minor subjects to the **Fast** path.
Continuity supervisor flags jumps	Time-of-day / eyeline / condition mismatch.	Open the flagged scene, fix continuity, **Apply + propagate**.
Master render is slow / reloads the model each shot	No persistent video worker.	Start the persistent LTX-2 worker so the model stays loaded between shots; otherwise expect per-shot reloads.
NLE export won't open in my editor	Wrong interchange format.	Choose the matching **NLE format** (FCPXML for Resolve/Filmora, AAF for Premiere/Avid, or OTIO/EDL as fallback).
Faces drift in video	Fast path over a long sequence.	Switch the character to the **LoRA** path (train image + video LoRA) for strong, repeatable identity.

> **See also:** Images for the keyframe engine, Video for shot generation and LTX controls, Training your own LoRA for the training tabs the LoRA Studio drives, and Import models to bring in your own checkpoints and community LoRAs.

Training Your Own LoRA

Teach the app new faces, characters, objects, places and styles by training your own LoRA adapters — from a simple images folder, or from a dataset the app builds for you automatically.

A **LoRA** is a small "add-on" file that steers a base model toward a specific subject or look. Once trained, a LoRA plugs into image and video generation everywhere in **AI Creative Center**: you type its **trigger word** in a prompt and the subject appears, consistently, take after take.

This chapter covers the three training tools the app offers, from the most hands-on to the most automated:

Tool	Where	What you supply	Output	Edition
Image LoRA training (`tab.train_tab`)	*Train LoRA* tab	Your own images folder + captions	A FLUX image LoRA	**Studio Pro**
Photo LoRA (auto-dataset) (`lora_image_tab`)	*Photo LoRA (FLUX.2)* tab	A subject description + a few reference photos	A FLUX.2 image LoRA	**Full+**
Video LoRA (auto-dataset) (`lora_video_tab`)	*Video LoRA (LTX-2.3)* tab	A subject + reference photos	An LTX-2.3 video LoRA	**Full+**

> All three train **locally**, inside isolated environments. The heavy generic trainer and both auto-dataset trainers use the **ai-toolkit** engine under the hood. The first time you use any of them, you must run its one-time **Install** step (several GB of download).

The two auto-dataset tools are the same ones the Cinema — AI Film Studio LoRA Studio drives for character consistency; anything you train here appears in the shared LoRA library and in the Film Studio's LoRA list.

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1. Image LoRA training (bring your own images)

Image LoRA training

> **Edition: Studio Pro** · **Capability:** `cap.train_lora`

What it does — Trains a professional-grade image LoRA from a folder of images you already have, with full control over the base architecture, captioning, rank, steps, learning rate and multi-resolution training. This is the "power user" trainer: it does not generate a dataset for you — you provide the pictures.

Where to find it — The **Train LoRA** tab (`tab.train_tab`). This tab is only present in the **Studio Pro** edition.

Step by step 1. Open the **Train LoRA** tab. 2. Press **Install training module** the first time. Confirm the prompt — *ai-toolkit + PyTorch (cu128)* download into an isolated environment (several GB). Wait for **Training module installed** ✓. 3. Under **Dataset**, click **Browse…** and pick the **Folder** that holds your images. The hint reminds you it should be a *folder with images + matching .txt caption files*. The tab reports **{n} images · {c} captions**. 4. Optionally set a **Trigger word** (for example `ohwx`) — a rare activation token you will later type to summon the subject. 5. If your images have no captions, generate them: choose a **caption engine**, choose a **caption mode**, tick **Overwrite existing** only if you want to replace captions already present, then press **Generate captions**. Use **Review caption** to read and hand-edit each caption in a non-blocking window (it auto-saves). 6. Under **Base model**, pick the **Architecture** and, if needed, fill the **HF repo / folder** and **HF token** (required for gated models such as FLUX.1/2-dev). 7. Set the training **Parameters** (see table). The defaults are production-safe. 8. Enter one **Preview prompt** per line so the trainer renders validation samples as it learns. 9. Press **Start training**. Watch the **Progress** bar, the step counter (**Step {s}/{t}**), the live log and the sample-preview image (refreshed every few seconds). Press **Stop** to abort. 10. When you see **Training complete** ✓, your `.safetensors` LoRA is saved and ready to select in Images and Video.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Folder (Dataset)	Folder picker	—	Any folder	Your source images; each image may have a same-named `.txt` caption.
Trigger word	Text	*(empty)*	Free text (e.g. `ohwx`)	The rare token you type to invoke the subject. Leave blank for a pure style LoRA.
Caption engine	Dropdown	First available	**JoyCaption**, **Florence-2**	JoyCaption = best for training (uncensored, tags/descriptive); Florence-2 = lightweight and fast.
Caption mode	Dropdown	**Descriptive**	**Descriptive**, **Booru tags**, **SD prompt**	JoyCaption caption style. Ignored by Florence-2. Tags suit character/style LoRAs; descriptive suits scenes.
Overwrite existing	Checkbox	Off	On / Off	Replace captions that are already present. Leave off to only fill in missing captions.
Architecture (Base model)	Dropdown	First entry	Installed base architectures (FLUX.1, FLUX.2, …)	The model your LoRA attaches to. Must match the model you will generate with.
HF repo / folder	Text	—	HF id or local path	Where to fetch the base weights.
HF token	Password	—	Your token	Required only for gated models (FLUX.1/2-dev).
Rank (dim)	Number	**16**	2 – 256	Capacity of the adapter. Higher = more detail but bigger file and more VRAM; 16–32 is a sweet spot.
Alpha	Number	**16**	1 – 256	Scaling factor; commonly set equal to rank.
Steps	Number	**2000**	100 – 100000 (step 100)	Training length. More steps = stronger learning but risk of overfitting.
Learning rate	Editable dropdown	**1e-4**	1e-4, 2e-4, 5e-5, 1e-3, 8e-5 (or type your own)	How fast it learns. Lower is safer for faces; higher trains faster but can destabilize.
Batch size	Number	**1**	1 – 16	Images per step. Raise only if you have spare VRAM.
Seed	Number	**42**	0 – 2,000,000,000	Fixes randomness for reproducible runs.
Save every	Number	**250**	50 – 10000 (step 50)	How often a checkpoint is written, so you can pick an earlier, less-overfit version.
Preview every	Number	**250**	50 – 10000 (step 50)	How often validation samples are rendered.
Resolutions	Checkboxes	**512, 768, 1024 all on**	512 / 768 / 1024	Multi-scale training. Keep all on for versatility; drop the largest to save VRAM.
Quantize (less VRAM)	Checkbox	**On**	On / Off	Trains in a memory-light mode. Turn off only if you have abundant VRAM and want maximum fidelity.
Low VRAM (CPU offload)	Checkbox	Off (auto-set on low-VRAM machines)	On / Off	Offloads to system RAM. Slower but fits big models on small GPUs.
Also train the text encoder	Checkbox	Off	On / Off	Learns the trigger more strongly at the cost of flexibility. Leave off for most subjects.
Preview prompts	Multi-line text	—	One prompt per line	Prompts used to render validation samples during training.

Specializations - **Caption workflow** — You can train with your own hand-written captions, auto-generate them, or auto-generate then refine in the 📝 **Review caption** window. For characters, JoyCaption in **Booru tags** mode with a trigger word usually gives the cleanest identity. - **Checkpoint picking** — With **Save every** at 250, a 2000-step run leaves you several intermediate LoRAs; if the final one looks "baked", pick an earlier checkpoint.

Tips for best results

> - Aim for a clean, varied dataset: many angles, expressions and lighting conditions of the *same* subject; remove blurry or duplicated shots.
> - Always set a **rare trigger word** so the concept attaches to the token, not to a common word.
> - Start from the defaults (rank 16, 2000 steps, LR 1e-4). Only push steps higher if the subject is complex.
> - Keep **Quantize** on unless you see quality issues — it dramatically lowers VRAM use.
> - Use the **Preview prompts** to test *poses your dataset does not contain* — that reveals whether the LoRA generalizes or just memorizes.

2. Photo LoRA — auto-dataset (FLUX.2)

Photo LoRA (auto-dataset)

> **Edition: Full+** · **Capability:** `cap.lora_image_train`

What it does — Trains a **FLUX.2** image LoRA without you having to assemble a dataset. You describe one or more subjects and add a few reference photos; the app **generates** a rich, best-practice image dataset (many views, expressions and lighting), captions it, and trains the LoRA for you.

Where to find it — The **Photo LoRA (FLUX.2)** tab (`ilora.train` — "LoRA training (FLUX.2)").

Step by step 1. Open the **Photo LoRA (FLUX.2)** tab. 2. Press **Install ai-toolkit** the first time and let the environment download. 3. In **LoRA subjects**, press ➕ **Add subject** and fill the **Subject details**: **Type** (Character / Place / Object / Concept), **Name/Trigger** (this name *is* the trigger token), and — for people — tick **Face focus** to learn the face rather than clothes or background. 4. Add **Reference photos**: for each photo choose its **Angle** and press **Add photo…**. More angles produce a more coherent subject. 5. Review the **Automatic coverage** checklist — the cases the app will generate (views, expressions, lighting). Tick or untick to taste. (Motion-only cases are hidden here because this trainer produces still images.) 6. Optionally add **Custom situations** (one per line, unlimited) and any **real material** (extra photos of the subject). 7. Set the **LoRA training** options: **LoRA name**, **Base model**, **Rank**, **Steps**, **Resolution** and **Captioner**. 8. Press 🎬 **Generate dataset** to build (and inspect) the dataset only, or 🎓 **Generate + Train FLUX.2 LoRA** to run the whole pipeline end to end. 9. Watch the log and the sample-preview image. When finished, the LoRA is saved under the app's `output/loras` folder and registered in the library.

Parameters

Control	Type	Default	Range / Options	What it does & wh change
Type	Dropdown	Character	**Character**, **Place**, **Object**, **Concept**	Chooses which bes practice coverage s applies.
Name/Trigger	Text	—	Free text (e.g. `Martina`)	The subject's name trigger token.
Face focus	Checkbox	Off	On / Off	Learns the face onl (varies clothes/backgroun across the dataset robustness). Use fo people.
Angle (per photo)	Dropdown	front	front, three_quarter, profile, back, any	Tags the reference so the right one an each generated vie
Automatic coverage	Checkboxes	Balanced default set	Per-type case list	Which views/expressions/ the dataset covers.
Custom situations	Multi-line text	—	One per line, unlimited	Extra scenes to ger (e.g. "runs on the b
LoRA name	Text	`my_flux_lora`	Free text	Output file name.
Base model	Dropdown	**FLUX.2 [dev]**	FLUX.2 [dev], FLUX.2 Klein 4B, FLUX.2 Klein 9B, FLUX.1 [dev]	The base the LoRA attaches to. Klein is lighter/faster; [dev] highest quality.
Rank	Number	**16**	4 – 128	Adapter capacity.
Steps	Number	**2000**	200 – 20000 (step 200)	Training length.
Resolution	Dropdown	**1024**	768, 1024, 1280	Dataset/training resolution. Higher detail, more VRAM
Captioner	Dropdown	**Procedural (trigger+tags)**	Procedural, JoyCaption, Florence-2	How dataset image described. JoyCapt realistic VLM captio (best, uncensored); Procedural = trigg tags, no model.
Caption mode	Dropdown	Descriptive	Descriptive, Booru tags, SD prompt	JoyCaption style (w JoyCaption is chos

Specializations - **Klein vs [dev]** — FLUX.2 Klein 4B/9B train and run faster and lighter; FLUX.2 [dev] gives the highest fidelity but is the heaviest to train. - **Dataset-only mode** — 🎬 **Generate dataset** lets you build and review the generated images before committing to a full training run.

Tips for best results

- Give **3–6 varied reference photos** at different angles; even for **Face focus**, variety prevents a "frozen" look.
- Keep **Procedural** captioning for a clean, trigger-anchored identity; switch to **JoyCaption** when you want the LoRA to also understand context and clothing.
- Prefer **FLUX.2 Klein** for quick iterations, then re-train on **[dev]** for the final asset.

3. Video LoRA — auto-dataset (LTX-2.3)

Video LoRA (auto-dataset)

> **Edition: Full+** · **Capability:** `cap.lora_video_train`

What it does — Trains an **LTX-2.3** *video* LoRA so a subject stays consistent in motion — talking, turning, moving through a shot. The app auto-generates short training **clips** covering many views, expressions and camera moves from your reference photos, then trains the video LoRA.

Where to find it — The **Video LoRA (LTX-2.3)** tab (`tlora.train` — "LoRA training").

Step by step 1. Open the **Video LoRA (LTX-2.3)** tab. 2. Press **Install trainer** the first time. 3. In **LoRA subjects**, press ➕ **Add subject** and set **Type**, **Name/Trigger** and, for people, **Face focus**. 4. Add **Reference photos** (more angles = more coherent video), choosing an **Angle** for each. 5. Tune the **Automatic coverage** checklist — for video this includes motion cases (talking, turning the head, camera pans/dollies, turntables). Add **Custom situations** and optional **real material** (subject videos or photos). 6. Set **LoRA training** options: **LoRA name**, **Rank**, **Steps**, **Clip length**, **PuLID**, **Low VRAM** and **Captioner**. 7. Press 🎬 **Generate dataset** to build the clips only, or 🎓 **Generate + Train LoRA** for the full run. 8. Watch the log and sample previews. The finished video LoRA is saved and registered in the library.

Parameters

Control	Type	Default	Range / Options	What it does & when
Type	Dropdown	Character	**Character**, **Place**, **Object**, **Concept**	Selects the coverage s
Name/Trigger	Text	—	Free text (e.g. `Martina`)	Subject name and trig
Face focus	Checkbox	Off	On / Off	Learns only the face; v backdrop across clips.
Angle (per photo)	Dropdown	front	front, three_quarter, profile, back, any	Anchors the correct re generated case.
Automatic coverage	Checkboxes	Balanced default set (incl. motion)	Per-type case list	Cases to generate; vid pan/dolly/orbit/turntal head.
Custom situations	Multi-line text	—	One per line, unlimited	Extra actions to film (e.
Optional real material	File list	—	Videos/photos	Real subject footage t dataset.
LoRA name	Text	`my_lora`	Free text	Output file name.
Rank	Number	**16**	4 – 128	Adapter capacity.
Steps	Number	**2000**	200 – 20000 (step 200)	Training length.
Clip length	Number	**3.0 s**	2.0 – 6.0 s (step 0.5)	Duration of each gene Longer clips teach mo more.
PuLID (strong face likeness)	Checkbox	**On**	On / Off	Locks facial identity du generation. Keep on fo
Low VRAM	Checkbox	**On**	On / Off	Memory-light mode. T speed if you have the
Captioner	Dropdown	Procedural	Procedural, JoyCaption, Florence-2	How clips/frames are c

Specializations - **Motion coverage** — The video coverage set adds movement cases (camera **pan**, **dolly**, **orbit**, object **turntable**, and for people **talking**, **nod**, **turn-head**, plus a multi-angle **rotation** clip). These are what make the LoRA hold up *in motion*, not just in stills. - **Shared subject editor** — The subject editor (types, multi-angle photos, coverage, custom situations, real material, face-focus) is identical to the Photo LoRA tab, so a subject you define one way transfers to the other.

Tips for best results

> - Keep **PuLID** on and provide clear frontal + three-quarter reference photos — faces drift most in motion.
> - Start with the default **3.0 s** clip length; raise it only if the subject's action needs more time to read.
> - Leave **Low VRAM** on unless training is too slow and your GPU has headroom.
> - Train a **Photo LoRA first** for a rock-solid still identity, then a **Video LoRA** for motion — this "image → video" order is exactly what the Film Studio LoRA Studio automates.

Where trained LoRAs go

All three trainers save `.safetensors` files and register them in the app's shared LoRA library. From there you can:

- Select them in the **LoRA** picker on the Images and Video tabs (filtered by architecture).
- Manage, re-scope and re-train them from the **LoRA Studio** inside the AI Film Studio.
- Import externally trained LoRAs, or download community LoRAs, via Import models.

> **Reminder** — Match the LoRA to the base model it was trained on. A FLUX.2 photo LoRA belongs on FLUX.2 image generation; an LTX-2.3 video LoRA belongs on LTX-2 video generation. The pickers already filter by architecture to keep you on the right track.

Import models

Bring your own models into AI Creative Center: discover and download community models from Civitai, import files and folders, curate a searchable library, and manage catalog paths — all with automatic classification and content rating.

The **Import Models** tab is your model manager. It searches Civitai, ingests LoRAs, checkpoints, VAEs, embeddings, and ControlNets from links or local files, sorts them into the right folders by type and architecture, enriches them with metadata, and keeps a browsable library. Behind a single **Import all** button sits a full classification and cataloging pipeline, so most of the time you just paste a link or drop a file and let the app do the rest.

> **Edition:** The Import Models tab requires **Studio Pro**. Within it, **Discover** is gated by `cap.civitai_import` and **Advanced** by `cap.import_models`; in a lower edition these two modes appear with a 🔒 lock badge and an upsell panel. **Import** and **Library** are available whenever the tab is.

Where to find it — **Advanced** ▸ 📦 **Import Models**. The tab opens on the **Import** mode by default.

The tab is organized into four modes, selectable from the mode buttons at the top:

Mode	Icon	What it's for
Discover	🔍	Search and browse Civitai; add results to the import queue.
Import	⬇	Paste a Civitai link/ID or drag files/folders; stage and ingest them.
Library	🗃	Browse, filter, and edit your imported models.
Advanced	⚙	Configurable paths, compatibility patch, conversion, and the full ingest pipeline.

A shared operations **log** and a **progress bar** run along the bottom (visible in Import and Advanced modes).

Discover

> **Edition:** Studio Pro · **Capability:** `cap.civitai_import`

What it does — Searches the Civitai community catalog and lets you add any result straight to your import queue.

Where to find it — **Import Models** ▸ **Discover**.

Step by step 1. Type into **Search on Civitai...** and press **Search** (or leave it blank to browse). 2. Narrow results with the **type** and **base-model** filters. 3. (Optional, Unlocked 18+) tick the adult-content checkbox to include mature results. 4. On any result card, click **+ Add** to place it in the import queue. 5. Switch to **Import** to review the queue and click **Import all**.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Search on Civitai...	Text field	empty	Any query	Free-text search. Empty = browse.
Type filter	Dropdown	**All**	All · LoRA · Checkpoint · VAE · Embedding · ControlNet	Restricts results to one model type.
Base-model filter	Dropdown	**All base models**	All base models · Flux.1 D · SDXL 1.0 · Pony · Illustrious · SD 1.5 · SD 3.5	Restricts results to one base architecture.
18+ content (adult content)	Checkbox	Off	On / Off	Includes adult results. Requires the **Unlocked (18+) mode**; otherwise it is disabled and shows an "adult content feature not enabled" notice.
Search	Button	—	—	Runs the query.
+ Add	Button (per card)	—	—	Adds that model to the import queue.

When there are no matches the panel reads "*No results.*" An empty search shows the hint "*Search for models and add them to the import queue.*"

Specializations — Classic vs. Pro In the **Pro** variant, thumbnails are **lazy-loaded**: results appear instantly and previews download only when you click 🖼 **Load previews**, keeping large searches responsive. The Pro Discover page also hosts the Civitai **link/ID** field (moved here from Import).

Tips for best results

> - Use the base-model filter to avoid importing a LoRA that doesn't match the checkpoint you generate with — a Flux LoRA won't work on an SDXL model.
> - Browse with an empty query plus a type filter to see what's trending for a given architecture.

Import

> **Edition:** Studio Pro · **Capability:** `cap.import_models`

What it does — Takes Civitai links/IDs and local files or folders, recognizes each model, categorizes it, and moves it to the correct folder ready to use.

Where to find it — **Import Models** ▸ **Import**.

Step by step 1. Add models to the queue by any of these: - Paste a link or ID into the Civitai field and click **Add from Civitai**. - Drag files or folders onto the drop area. - Click **Choose files...** or **Choose folder...**. 2. Review each queued item's badges (type, family/architecture, content rating) and destination. 3. Remove

unwanted items with ✕, or empty the whole list with **Clear queue**. 4. Click ⬇ **Import all** to stage and catalog everything.

Parameters

Control	Type	Default	Options	What it does
Civitai link/ID	Text field	empty	`https://civitai.com/models/…` or an ID	The model to fetch from Civitai.
Add from Civitai	Button	—	—	Resolves the link/ID and queues the model.
Drop area	Drag target	—	—	Drop files or folders of models here.
Choose files…	Button	—	—	Pick model files from disk.
Choose folder…	Button	—	—	Pick a folder of models.
⬇ Import all	Button	—	—	Stages every queued item (download/copy → classify → catalog).
Clear queue	Button	—	—	Empties the import queue.

Recognized model file types are `.safetensors`, `.ckpt`, `.pt`, `.pth`, and `.gguf`. Each queued card shows a destination (`→ …`) and badges for type, family, and content rating; when empty the queue reads "*The queue is empty — add models above.*"

Staging — When you click **Import all**, each item is either downloaded from Civitai or copied into your inbox, with live status: *downloading…* → *copying…* → *ready* (or *error*). Items are then cataloged automatically.

Specializations — Classic vs. Pro The **Pro** variant moves the Civitai link/ID field to the **Discover** page and shows a dashed-outline empty state: "*No models queued. Drag a file or add from «Discover».*" The four-mode header uses **segmented buttons** with 🔒 lock badges on the gated modes.

Tips for best results

> - You can queue Civitai links and local files together and import them in one pass.
> - Dropping a whole folder is the fastest way to bulk-import a downloaded model pack.

The classification pipeline

Every imported model is analyzed automatically so it lands in the right folder with the right metadata. The analysis fills in these fields:

Field	Meaning
type	LoRA · Checkpoint · VAE · TextEncoder · Embedding · ControlNet · Upscaler …
family	Flux · SDXL · SD1.5 · SD3 · LTX · Wan21 · Wan22 · Z-Image · Ernie …
dest_subfolder	Where the file is placed, e.g. `LoRA/Flux`.
sha256	The file's hash, used for a Civitai by-hash lookup.
triggers	Trained trigger words (activation keywords).
preview	A preview image URL.
base_arch	The base architecture reported by Civitai.

How type and family are determined — The app first inspects the file itself (by structure and size) to classify type and family. If that is inconclusive, it falls back to the native model detector, which maps internal codes (for example `flux1` / `flux2` → **Flux**, `sdxl` → **SDXL**, `ltxv` → **LTX**, `wan21` → **Wan21**) to standard names. For `.safetensors` / `.ckpt` / `.pt` files it also computes the SHA-256 and queries Civitai **by hash**; a match overrides the guessed type/family and supplies triggers, a preview, and the base model.

Content rating (adult-content gating)

AI Creative Center classifies each model's content rating through a **cascade**, stopping at the first confident source. This is what drives whether a model is treated as 18+/adult content:

1. **Civitai metadata (authoritative)** — if the model comes from Civitai, its rating flags are used directly.
2. **Name/tag heuristic** — otherwise, the filename and tags are scanned for adult-content hints.
3. **Preview classifier** — if a preview image is available, an on-device classifier inspects it.
4. **Conservative default** — if still uncertain, the model is treated as adult content (the safe default).

The **Unlocked (18+) mode** (`cap.adult_mode`) governs what happens next. When it is **off**, any model rated as adult content is **skipped** during import (the log notes "*Skipped (adult content, feature not enabled)*"), and the adult-content checkboxes in Discover and Library are disabled. When it is **on**, adult-rated models import and browse normally. You can always correct a model's rating by hand in the Library detail view.

Library

> **Edition:** Studio Pro

What it does — Browses your imported models as cards, filtered by type, architecture, and content rating, and lets you edit each model's trigger words, notes, and rating.

Where to find it — **Import Models ▸ Library**.

Step by step 1. Filter the catalog with **Filter the library…**, the type dropdown, the architecture dropdown, and the adult-content checkbox. 2. Click a card's **Details** to open its editor. 3.

Edit **Trigger words** and **Description / notes**, adjust the content-rating checkbox, and save. 4. Use ↻ to reload the library after importing new models.

Parameters

Control	Type	Default	Options	What it does
Filter the library...	Text field	empty	Any text	Filters cards by name.
Type filter	Dropdown	**All**	All · LoRA · Checkpoint · VAE · Embedding · ControlNet	Restricts to one type.
Architecture filter	Dropdown	**All archs**	All archs · (families found in your catalog)	Restricts to one architecture.
18+ content (adult content)	Checkbox	Off	On / Off	Shows adult-rated models. Requires Unlocked (18+) mode.
↻	Button	—	—	Reloads the library.
Details	Button (per card)	—	—	Opens the model editor.

The detail editor exposes: **Trigger words (comma-separated)**, **Description / notes**, and an **18+ content** (content rating) checkbox. Your edits are stored as manual overrides and re-applied on every re-scan, so a metadata refresh never wipes them. When the library is empty it reads "*No models in the library (import some).*"

Specializations — Classic vs. Pro The **Pro** variant adds per-card quick actions: a ★/☆ favorite toggle and an 📁 **Open folder** button (opens the model's folder in your file manager), alongside **Details**.

Tips for best results

> - Fill in **Trigger words** for LoRAs that aren't on Civitai — the generation tabs surface these so you remember how to activate them.
> - Use the architecture filter to quickly find all LoRAs compatible with the checkpoint you're about to use.

Advanced

> **Edition:** Studio Pro · **Capability:** `cap.import_models`

What it does — Exposes the catalog's configurable paths, the ComfyUI compatibility patch, model-format conversion, and the full manual ingest pipeline (analyze → place → scan → enrich → build HTML).

Where to find it — **Import Models ▸ Advanced**.

Configurable paths

A collapsible group where you can point the catalog at the right locations. Each has a text field, a browse button, and a tooltip:

- **Models folder** — where classified models are placed.
- **ComfyUI folder** — your ComfyUI installation root.
- **Dictionary root (dict_root)** — where `catalog.json`, `manual_enrichment.json`, and the HTML catalog live.
- **Inbox** — the staging folder that Import copies into and Place moves from.
- **Tools folder** — where the catalog scripts live.

Buttons: **Save paths** (persists and syncs to the tools' `paths.json`), **Reset to defaults**, and **Suggest from ComfyUI** (proposes paths from a detected ComfyUI install — the final choice stays yours). The defaults match your current installation; change them only if you've moved ComfyUI, your models, or the dictionary.

Compatibility check (ComfyUI patch)

Installs a compatibility patch as a ComfyUI custom node so LTX-2.3 runs on any ComfyUI version **without editing core files** (it survives updates). The group shows the current install status and offers **Install / Update patch** and **Uninstall patch**. A checkbox unlocks the LTX-2 prompt-enhancer so it describes subjects without refusals (for adult content). After installing, restart ComfyUI to apply.

Model conversion (ComfyUI → native)

One-time converters that turn ComfyUI/quantized models into the native backend format: **Gemma-3 → bf16** and **Text encoder → HF**.

The ingest pipeline

The Advanced mode gives you each stage of cataloging as its own button, plus a one-click full run. On the source folder they act as follows:

Operation	What it does	Network
Analyze	Classifies source files by type/architecture **without moving them** (dry run).	No
Place / Move	Moves files from the source into the correct folders (LoRA/Checkpoints/VAE/... or workflows).	No
Scan	Updates `catalog.json` with base metadata (architecture, triggers, content rating).	No
Enrich	Civitai by-hash lookup plus re-applies your manual overrides.	Yes
Overrides	Re-applies only your manual overrides (`manual_enrichment.json`).	No
Build HTML	Regenerates the static HTML catalog (index + entry pages).	No
Full ingest	Runs Place → Scan → Enrich → Build HTML in sequence.	Yes (Enrich step)

You can also set the **source folder**, add files/folders to a working list, **copy to the inbox**, and clear the list.

Tips for best results

- Use **Analyze** first as a dry run to confirm files will be classified correctly before **Place** actually moves them.
- Run **Enrich** while online to pull triggers and previews from Civitai; run **Overrides** offline to re-apply just your hand edits.
- If video generation stops working after a ComfyUI update, re-run **Install / Update patch** and restart ComfyUI.

Related: Getting started for the Classic/Pro interface switch and the model-overrides setting, Advanced settings for where Import Models lives, and the Appendix for the full capability→edition table and file locations.

Advanced settings

The power-user control room: quality presets, the exact sampler/scheduler/SHIFT levers that drive video quality, text-encoder selection, the prompt enhancer, fixed seed, unlimited-length auto-extend, VRAM calibration, and the gallery/workflow/analyzer tools.

The **Advanced** tab is where AI Creative Center exposes the low-level dials that the rest of the interface keeps sensible for you. Nothing here is required for everyday work — the defaults are tuned to the verified high-quality LTX-2.3 pipeline — but when you want deterministic runs, faster drafts, sharper detail, or truly long videos, this is the tab you reach for.

> **Edition:** The **Advanced** tab as a whole requires **Full+**. It groups the sampler/seed controls (which also live inline in the Video tab), the long-video engine, the VRAM calibration tool, and the gallery/workflow utilities. The **Import Models** sub-tab shown next to **Tools** is a separate module — see Import models — and requires **Studio Pro**.

The tab is split into two sub-tabs: 🛠 **Tools** (all the settings and utility buttons documented below) and 📦 **Import Models** (documented in its own chapter).

How presets work in this app (read this first)

There is one rule that governs every "preset" and "optimize" action in AI Creative Center, and it will save you confusion:

> **Presets write into the GUI controls — they do not silently reconfigure the pipeline.**

When you pick a quality preset, the app fills in the **Steps** and **CFG** fields (and, for the Video-tab presets, resolution / FPS / duration). Those fields then remain fully editable by hand. Nothing is hidden: what you see in the controls is exactly what the model receives. This means you can start from a preset and then fine-tune a single value without fighting an invisible override.

The Sampler / Seed group

> **Edition:** Full+ · **Capability:** `cap.video` (video sampling)

What it does — Controls how many denoising steps run, how strongly the prompt is enforced (CFG), which sampler and scheduler are used, the LTX SHIFT quality lever, which text encoder reads your prompt, whether the prompt is auto-enhanced, and the random seed.

Where to find it — This group (internally the **Sampler / Seed** group) is embedded directly in the Video tab, and its settings are documented here because they are the "advanced" heart of video generation. The heading reads **Sampler / Seed**.

Step by step 1. (Optional) Choose a **Quality (preset):** entry to fill Steps and CFG with a known-good combination. 2. Adjust **Steps** and **CFG** if you want finer control. 3. Leave **Sampler / Sched.** on the defaults unless you have a specific reason to change them. 4. Tune the **Shift (LTX) base** / **max** values — this is the highest-impact, zero-cost video quality lever. 5. Choose a **Text encoder** (leave on **Auto** to let content mode decide). 6. Optionally enable **Migliora prompt (LTX enhancer)** (prompt enhancement) for short prompts. 7. Enable **Seed fisso** (Fixed seed) and set a **Seed** value if you need reproducible output.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Steps	Number	32	1–60	Denoising steps. More = more detail and coherence, slower. Selecting a distilled model lowers this automatically (e.g. 8).
CFG	Number	1.0	0.0–15.0 (step 0.1)	Classifier-free guidance strength. Kept at **1.0** by default because negative guidance is supplied by NAG; raise it only if the prompt is being ignored. High CFG (2.5–4.5) tends to "over-cook" LTX video.
Quality (preset):	Dropdown	**Custom**	Custom · **Fast (8 steps)** · **Pro (balanced, 30 steps)** · **HQ (max, 50 steps)**	Shortcut that writes Steps + CFG per the official LTX-2.3 modes. **Custom** changes nothing. Fast = 8 steps / CFG 1.0; Pro = 30 / 3.0; HQ = 50 / 4.0. Values remain hand-editable afterward.
Sampler	Dropdown	**euler_ancestral_cfg_pp**	euler_ancestral_cfg_pp · euler_cfg_pp · euler · dpmpp_2m	The sampling algorithm. The default matches the AuthorStorm reference and gives the best quality on LTX. See the note below on RES samplers.
Scheduler	Dropdown	**LTXVScheduler (shift)**	LTXVScheduler (shift) · linear_quadratic · bong_tangent · karras · beta	Noise schedule. The default is LTX's own flow-/shape-aware scheduler and enables the SHIFT lever below. Picking a classic scheduler routes the sampler through a BasicScheduler and **ignores** the LTX SHIFT values.
base (Shift)	Number	0.95	0.10–6.00 (step 0.05)	LTXVScheduler **base shift**. Higher = more steps spent on structure and **motion** (temporal coherence, less flicker); lower = finer detail but
max (Shift)	Number	2.05	0.50–12.00 (step 0.05)	LTXVSc **max sh** resolut depend bound shift. R longer clips wi motion uncerta
Text encoder:	Dropdown	**Auto**	Auto (by content mode) · Gemma 3 12B — full (max quality) · Gemma 3 12B — abliterated	Which reads y prompt prompt **adhere** (compo subject far mo texture 12B, so no spe differen speciali
Migliora prompt (LTX enhancer)	Checkbox	Off	On / Off	Prompt enhanc When c LLM (G expand prompt detaile descrip before genera for sho leave o your pr already
Seed fisso (Fixed seed)	Checkbox	Off	On / Off	When c **Seed** v is used giving reprod results. a fresh seed is run.
Seed	Number	777	0 to $2^{31}-1$	The ran Only ap when **F** is on.

> **Note on samplers:** The RES family (`res_2s`, `res_multistep`, …) is deliberately **not** offered. Although excellent in theory for flow-matching, in practice these break LTX-2.3 and produce green/cyan noise (verified in A/B testing). Stick to the four samplers listed.

Specializations — text encoders - **Auto** — the content mode picks the encoder for you: standard content uses the **full** Gemma encoder; Unlocked (18+) mode uses the **abliterated**

(uncensored) encoder. - **Gemma 3 12B — full** — maximum semantic fidelity; softens explicit content. - **Gemma 3 12B — abliterated** — an uncensored variant, much better on explicit prompts. An encoder marked absent (⛔) means the weight file has not been downloaded yet.

Specializations — model-aware suggestions When you switch models in the Video tab, the app may suggest steps, CFG, sampler, and scheduler values tuned for that model. Suggestions only override controls you have not manually changed, so your explicit choices are respected.

Tips for best results

> - Leave **CFG at 1.0** and **Scheduler on LTXVScheduler (shift)** unless you have a concrete problem to solve — this is the verified reference pipeline.
> - To sharpen a soft result at **zero extra cost**, adjust the SHIFT **base**: lower it slightly for more fine detail, raise it for steadier motion and less flicker.
> - Use **Fixed seed** while dialing in a shot so that each change you make is the only variable.
> - Turn the **prompt enhancer** off once your prompts are detailed; it can steer short prompts helpfully but adds its own interpretation.

Quality presets

> **Edition:** Full+

The **Quality (preset):** dropdown in the Sampler / Seed group is the fast way to set a coherent Steps + CFG pair:

Preset	Steps	CFG	Use it for
Custom	— (unchanged)	— (unchanged)	Full manual control; the preset makes no changes.
Fast (8 steps)	8	1.0	Quick drafts to validate an idea.
Pro (balanced, 30 steps)	30	3.0	Everyday balanced quality.
HQ (max, 50 steps)	50	4.0	Final, highest-quality renders (slowest).

The Video tab additionally offers richer named presets (for example ⚡ **FAST**, 🎯 **PRO**, 💎 **HQ**, plus aspect-ratio presets such as 📺 **Broadcast 16:9**, 📱 **Social 9:16**, 🎬 **Cinemascope**) that also set resolution, FPS, and duration. Those live in the Video tab; the same "presets write into the controls" rule applies.

Long-video auto-extend (unlimited length)

> **Edition:** Full+ · **Capability:** `cap.video`

What it does — Produces videos of **unlimited length** at constant VRAM by generating a base pass and then chaining automatic image-to-video extension segments, stitched together with overlap.

Where to find it — **Advanced** ▸ **Tools**, the 🎞 **Long videos (unlimited length)** group.

Step by step 1. Leave auto-extend enabled (it is on by default). 2. Set the **single-pass threshold** — videos longer than this trigger automatic segmenting. 3. Tune segment length, overlap, and context to trade continuity against motion. 4. Request your target duration in the Video tab and generate; the app handles the rest.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Auto-generate videos longer than the threshold (in segments)	Checkbox	On	On / Off	Master switch. When on, any Video-tab duration above the threshold is produced as a base pass plus automatic extensions.
Single-pass threshold	Number (s)	20.0	4–30	Maximum duration of a single LTX pass. Beyond this, auto-extension kicks in. LTX-2.3 reaches ~20 s natively, so 20 is the recommended value.
Extension segment length	Number (s)	10.0	4–15	Length of each chained image-to-video segment. Longer = fewer joins = fewer transitions. 10 s recommended.
Overlap/drop (frame):	Number	24	0–48	Frames dropped at the start of each segment to hide the image-to-video transition from the static frame. 24 = 1 s at 24 fps. Higher = cleaner transitions but more discarded content.
Contesto (frame): (context frames)	Number	16	8–64	Tail frames from the previous segment used as context for the next one.
Forza contesto: (context strength)	Number	0.5	0.0–1.0 (step 0.05)	How strongly the context constrains the next segment: 1.0 = heavily anchored (less drift, less motion), 0.0 = free (more motion, more drift).
Scene anti-drift (photo):	Number	0.25	0.0–1.0 (step 0.05)	Re-injects the original photo's appearance into every segment to curb cumulative degradation (blur, over-exposure, artifacts) in very long videos. 0 = off; 0.25 recommended; high values freeze the look and reduce motion.
Re-anchor face every (segments):	Number	0	0–20	Every N segments, restarts from the original photo instead of the last frame (resets face drift). 0 = never (maximum continuity). Use 2–4 only if the face drifts too much.
Resume after crash	Checkbox	On	On / Off	Saves every generated segment to disk; if generation is interrupted, on restart it regenerates only the missing segments instead of starting over.

Tips for best results

- Keep the **single-pass threshold at 20 s** — that is LTX-2.3's native ceiling; pushing it higher degrades the base pass.
- If faces slowly drift over a long clip, first raise **Scene anti-drift** to ~0.25; only if that is not enough, set **Re-anchor face every** to 2–4.
- Raise **context strength** toward 1.0 for locked-down, low-motion scenes; lower it for dynamic action where you want more movement.
- Leave **Resume after crash** on for long overnight renders so a single interruption doesn't cost you the whole video.

VRAM calibration

Edition: Full+ · **Capability:** `cap.video` / general memory management

What it does — Analyzes all your local image and video models, measures and caches each model's **real VRAM footprint**, and reuses those measurements to decide what to keep on the GPU and what to offload — so you get the largest models your card can run without out-of-memory errors.

Where to find it — **Advanced** ▸ **Tools**, the **VRAM memory management** group, button **Calibrate model VRAM...**.

Step by step 1. Open **Advanced** ▸ **Tools**. 2. Click **Calibrate model VRAM...**. 3. Let the calibration run — it is incremental, so subsequent runs only re-measure new or changed models. 4. The cached footprints are then used automatically for placement decisions on every job.

Tips for best results

- Run this once after installing new large models (or after a driver change) so placement decisions stay accurate.
- Calibration is incremental with invalidation, so re-running it is cheap.

Tools: Navigator, Workflow presets, and Settings analyzer

Edition: Full+

These three buttons live in **Advanced** ▸ **Tools** and open utility dialogs.

Navigator Gallery

What it does — Browses your previous generations with previews, full metadata, and one-click parameter reuse. **Where to find it** — **Advanced** ▸ **Tools** ▸ **Navigator Gallery...**, shortcut **Ctrl+G**. **Tips**

- Use it to recover the exact seed and settings of a shot you liked, then reuse them for a variation.

Workflow presets

What it does — Opens a library of 35 optimized workflows (for example Talking Head, Cinematic Landscape, Product Showcase, Action Scene, Cyberpunk, Fashion, and more) as ready-made starting points. **Where to find it** — **Advanced** ▸ **Tools** ▸ 🎬 **Workflow presets...**.

Settings analyzer

What it does — Checks your current steps, CFG, resolution, LoRA strength, and any content-mode / architecture conflicts, and suggests automatic fixes where possible. **Where to find it** — **Advanced** ▸ **Tools** ▸ 🔍 **Analyze settings...**. **Tips**

> - Run the analyzer before an expensive render if you have been experimenting — it catches mismatched settings that would waste a long generation.

Related chapters: Video for where these sampler/seed controls appear inline, Getting started for the app-wide Settings dialog and VRAM limit, and the Appendix for the full capability→edition table.

Invisible Provenance Signature

How AI Creative Center invisibly signs everything it produces so you can later prove it was AI-generated — and how to control it.

Every image and every video frame produced by **AI Creative Center** carries an invisible **provenance signature**. It is a robust, imperceptible watermark that embeds the brand mark **AI Tools Forge** directly into the pixels, so the origin of a file can be verified even after it has been screenshotted, resized, compressed, or re-photographed. The signature does not alter the picture in any way you can see.

> **Edition: All editions** · **Capability:** `cap.stego` (default **ON**, user-disableable)

This chapter explains what the signature is, how to turn it on or off, how to add an optional short text tag, and how the standalone **ATF Detector** reads it back. See also Getting started and Advanced settings.

What it is

The provenance signature is a **spread-spectrum watermark in the DCT (frequency) domain**. Instead of writing a visible mark in one corner, it spreads a very low-amplitude, pseudo-random pattern across the whole picture. That pattern is generated from a secret key, so only a matching detector can find it — no visible logo, no metadata that a re-export would strip.

It is designed to be:

- **Invisible** — the signal sits in the mid-frequency band, below the threshold of human perception.
- **Redundant** — it is repeated over an 8×8 grid of tiles, so it survives partial cropping.
- **Robust** — each tile is normalized to a canonical size before analysis, so the mark survives resizing, JPEG compression, screenshots, and photos of a screen.
- **Anti-averaging (video)** — the pattern changes on every frame, so averaging frames together does not reveal a fixed mark to remove.
- **Blind** — verification needs only the secret key, never the original file.

> **Honest note.** No invisible watermark is literally impossible to remove. This one is strongly resistant to random edits, frame averaging, and screenshots; a determined attacker with the secret key could still target it. The signature is a provenance aid, not a DRM lock.

Turn it on or off (and add a text tag)

Where to find it — Settings → Provenance (`ui/settings_dialog.py`, the **Provenance** group).

Step by step

1. Open **Settings**.
2. Find the **Provenance** group.
3. Toggle **"Invisible provenance signature (recommended)"** — checked by default.
4. Optionally, type up to five characters into **"Signature text (max 5)"** (for example, your own short handle or code). Leave it empty for a signature with no text — it is still fully valid.
5. Save. The setting takes effect on the next generation.

Parameters

Control	Type	Default	Range / Options	What it does & when to change
Invisible provenance signature (recommended)	Checkbox	On	On / Off	Embeds the invisible signature into every generated photo and video frame. Turn off only if you have a specific reason not to sign your outputs.
Signature text (max 5)	Text field	Empty	0–5 characters	Optional short tag stamped invisibly alongside the presence mark and read back by the ATF Detector. Empty = signature without text. Five characters is the physical limit for a screenshot-robust invisible mark; for longer text, use the file's own metadata instead.

> **Tips for best results** - Leave the signature **on**. It is the only reliable way to later prove one of your renders came from this software. - Keep the text tag short and meaningful — an initials handle or short ID. Anything longer will not survive compression and is silently truncated to five characters. - The text tag is optional. If you only need to prove "this was AI-generated with AI Tools Forge," the presence mark alone (empty text) is enough.

Technical parameters

For advanced users and integrators, these are the fixed constants of the watermark (`core/stego.py`). You never set these in the app; they are listed here for transparency and for anyone integrating the ATF Detector.

Constant	Value	Meaning
Grid	**8 × 8** (64 tiles)	Redundant impression sites; survives partial crops.
Canonical tile size	**80 px**	Each tile is normalized to this size before DCT — the source of resize robustness.
Nonce bank	**24**	Number of per-frame carrier variants the detector tries.
Presence amplitude (alpha)	**~9.0**	Tuned for invisibility (~PSNR 39, max pixel delta ~9). Higher values would be visible ("blotchy").
Detection z-threshold	**5.0**	Presence is declared above this z-score (false-positive rate ~1e-6).
Payload	**5 chars + CRC8** = 48 bits	Optional text tag with an integrity checksum.
Payload amplitude (beta)	**1.2**	Kept low to preserve invisibility; the priority is a clean image.
DCT band	mid-band, r ∈ [4, 26]	Survives resize/JPEG/screenshot; DC and high frequencies excluded.
Perceptual mask	Laplacian-based, per-pixel	Hides the signal in local detail/texture (~100%) and nearly zeroes it on smooth areas like skin and sky (~8%).
Per-frame nonce	yes (video)	The carrier changes every frame so averaging frames reveals no fixed pattern.

The key is derived deterministically from an SHA-256 stream, so the same signature can be produced and verified identically across Python/NumPy and JavaScript — which is how the mobile companion and on-device detector can read the same mark.

Robustness

The signature is engineered to survive the ways files actually get shared and re-captured:

- **Resizing** — the grid-to-image ratio scales with the image, so the mid-band survives resampling.
- **JPEG / re-compression** — the mid-band is chosen precisely to survive lossy compression.
- **Screenshots** — normalization to the canonical tile size keeps the mark readable after a screen capture.
- **Photo of a screen** — the robust detector performs a geometric re-sync that compensates for a slightly rotated or perspective-skewed re-photograph.
- **Perspective / mild warping** — the robust path searches over small rotations (±15°) and scales (0.90–1.10) to re-align the grid.

The hardest case remains a strongly skewed photo-of-screen with moiré; even then the presence mark usually holds while the optional text tag may be lost.

The ATF Detector

The **ATF Detector** is the standalone tool that reads the signature back. It works **blind** — it needs only the secret key, never your original file — and it returns four values:

- **found** — whether the signature was detected.
- **z-score** — the confidence of the detection.
- **nonce** — which per-frame carrier variant matched.
- **text** — the optional 5-character tag, if one was embedded and its checksum verifies.

Two modes are available (`core/stego.py`):

Function	Use	Threshold
`detect_image`	Fast blind detection, no geometric re-sync — good for clean files.	z ≥ **5.0**
`detect_image_robust`	Full geometric re-sync (rotation × scale search) for skewed photos of a screen.	z ≥ **6.5**

The robust path uses a higher threshold (6.5) because searching over many geometric warps inflates the maximum correlation under the null hypothesis. For files, `detect_file` runs the robust detector on images and samples several frames of a video, keeping the strongest result.

> **Tips for best results** - Use the **robust** detector for anything that has been photographed off a screen or heavily edited; use the **fast** detector for original files. - A high z-score with no recovered text simply means the presence mark survived but the optional tag did not — the file is still proven AI-generated. - Because detection is blind and key-derived, you can verify third-party copies of your work without keeping the originals.

Setup & Guided Wizards

The three assistants that get you from a fresh install to your first render: the first-run model setup, the selective installer, and the guided creation wizard.

AI Creative Center ships with three wizards. Two are one-time helpers that prepare your models and disk; the third is an everyday assistant that turns a goal ("make a video from a photo") into a ready-to-run tab. This chapter documents all three. See also Getting started.

First-run Setup Wizard

> **Edition: All editions**

What it does — On the very first launch, this wizard links the AI models you already have on disk (without copying or re-downloading them) and then downloads only the weights your edition's modules actually need.

Where to find it — Appears automatically at first launch (`ui/setup/first_run.py`). It reuses the app's **Weights** tab inside a guided frame.

When it appears — Only when setup has never been completed *and* no base models are already present. It is skipped automatically once either condition changes.

Step by step

1. **Base folder** — Point the wizard at the folder where you already keep your AI models.
2. **Scan for existing** — The wizard recursively scans that folder for known model files.
3. **Link found / Import unknown** — For recognized files it offers **" 🔗 Link found"**, which creates a filesystem link (a junction or hardlink — **no copy, no re-download**, and no administrator rights required). For files it does not recognize, it offers **" + Import unknown"** so you can classify them.
4. **Select missing + Download** — Finally, tick the weights that are still missing and download only those required by your edition's modules.

Two buttons finish the wizard: **"Done ✓"** (which marks setup complete) and **"Skip for now"** (which also marks it done so it does not reappear on every launch).

> **Tips for best results** - Point it at your top-level models folder (for example `E:\AI\Models`) and let it link everything at once — linking is instant and uses no extra disk space. - To bypass the wizard entirely (for example on an automated or headless install), set the environment variable `LTX_NO_SETUP=1`. - Both **Done** and **Skip** set the internal `setup_done` flag, so the wizard will not nag you again. You can always reopen model management from the **Weights** tab.

Selective installer

> **Edition: All editions** (edition-aware download)

What it does — The selective installer links your local models and downloads **only** the weights for the modules included in your edition, then produces a native, signed installer. It avoids pulling gigabytes you will never use.

Where to find it — `installer/install.py`, driven by the declarative `installer/install_spec.json`.

How it works

1. **Detect the edition** — The installer reads your edition manifest (or an explicit `--edition` override) to learn which modules and adult-mode weights are allowed.
2. **Check prerequisites** — Python 3.10+, NVIDIA driver + CUDA, a compatible PyTorch, and ffmpeg are verified up front.
3. **Build the plan (only-wanted)** — From the weights catalog it selects only the models your edition wants: `editions` filters per tier, adult-sensitivity weights download only when your edition allows Unlocked (18+) mode, and `on_demand` weights are deferred to first use rather than downloaded now.
4. **Show a sized summary** — Before anything is downloaded you see the full plan with estimated sizes and are asked to confirm. `--dry-run` prints the plan without downloading.
5. **Install** — Base weights for your edition are fetched; large on-demand models (FLUX.2 encoder, FLUX Fill, TRELLIS.2, SkinTokens, SeedVR2, and so on) are pulled the first time you use the feature that needs them.

Native installers & signing — Distribution builds wrap the app in a native installer per platform — **Inno Setup** on Windows, a **DMG** on macOS, and an **AppImage** on Linux — each code-signed for its platform so it installs without security warnings.

> **Tips for best results** - Run with `--dry-run` first to see exactly what will be downloaded and how large it is. - Because on-demand weights are deferred, expect a one-time download the first time you open a heavy module (3D, video upscale, world generation). This is normal. - Your edition determines which weights are eligible: higher tiers unlock more modules and therefore more downloadable weights. See the capability table.

Guided creation Wizard

> **Edition: All editions** (individual goals still respect your edition's capabilities)

What it does — Takes you from "what do you want to create?" to the correct tab with the right fields pre-filled. It is the fastest way to start when you are not sure which tab to open.

Where to find it — `ui/wizard.py`. Open it from the main window; it applies your choices by switching to the correct tab (or opening AI Film Studio) and filling in the fields.

Page 1 — Choose your goal

Pick one of nine goals and set the content mode (a standard/Unlocked toggle). A contextual description explains each goal as you select it.

Goal (label)	What it produces	Needs a starting image?	Audio mode	Routes to
Text → Video	A video from a text description.	No	Native	Video tab
Image → Video	Animates a starting image into a video.	Yes	Native	Video tab
Image → Video + voice (lipsync)	A photo plus a cloned voice, with synced lips.	Yes	Clone	Video tab
Audio → Video	A video driven by an audio track.	No	User-supplied audio	Video tab
Text → Image	An image from a text description.	No	—	Images tab
Image → 3D model	Turns a photo into a 3D mesh/GLB.	Yes	—	3D Model tab
Generate music / song	Composes a song or an instrumental track.	No	Native	Music tab
Clone voice / speech synthesis	Synthesizes or clones a voice from a sample.	No	Native	Audio/Voice tab
AI Film Studio (text → film)	A full film: storyboard, scenes, editing.	No	Native	AI Film Studio

Page 2 — Prompt & duration

The second page adapts to the goal you chose:

1. A **prompt** box (shown only for goals that need a description — hidden, for example, for Image → 3D).
2. A **duration** field in seconds (shown only for goals where length matters — video and music). Its range is **1–60 s**, defaulting to **8 s**.
3. A blue **contextual hint** telling you where the choice will land (for example, "The prompt and length will be applied to the Video tab").

When you finish, the wizard returns your goal, category, content mode, whether an image is needed, the audio mode, your prompt, and the length — and the main window applies them.

> **Tips for best results** - The content-mode toggle (standard vs Unlocked 18+) is only effective if your edition includes `cap.adult_mode`; otherwise it stays in standard mode. - The wizard fills fields but does not lock them — after it hands you to a tab you can fine-tune everything before generating. - For **Image → Video + voice**, prepare your voice sample first (see Audio & voice); the wizard sets the clone audio mode but you still supply the sample in the Video tab.

Local REST API

Automate every creative function in AI Creative Center from your own scripts, the mobile companion, or an AI agent — all on your own machine, nothing in the cloud.

> **Edition: Studio Pro only** · **Capability:** `api.rest`

The **local REST API** exposes every native function of AI Creative Center — images, video, film shots, music, voice, 3D, world generation, and LoRA training — as **asynchronous jobs** that return download links. It is the same engine that powers the mobile companion app and the multi-agent skill packs. Everything runs on `http://127.0.0.1:8200`; no data leaves your computer.

See also Advanced settings and the capability table.

Starting and stopping the server

- **Autostart** — Run `python -m api.autostart`. It starts the server if it is down, waits for `/health`, and prints `READY http://127.0.0.1:8200` when ready. The agent skill packs call this at the start of a session.
- **Manual start** — `python -m api.app`.
- **Auto-shutdown** — After **30 minutes** with no requests *and* no jobs queued or running, the server shuts itself down. The next call simply re-launches it via autostart.
- **Interactive docs** — Swagger UI at `http://127.0.0.1:8200/docs`; the OpenAPI schema at `/openapi.json`.

Environment variables: `LTX_API_HOST` (default `127.0.0.1`), `LTX_API_PORT` (default `8200`), `LTX_API_IDLE_TIMEOUT` (default `1800` seconds), `LTX_API_PYTHON` (the Python interpreter that runs the API).

The asynchronous job model

Every creative action is a `POST` with a JSON body that returns a **job id** immediately; you then poll for the result and download the files. GPU jobs run **one at a time** — do not fire jobs in parallel.

1. **Submit** — `POST` an action → `{ "job_id": "...", "status": "pending" }`.
2. **Poll** — `GET /jobs/{job_id}` (back off 2–5 s) until `status` is `success` or `error`:
3. `running` → reports `progress` (0–1) and a `message`.
4. `pending` → reports `queue_position`.
5. `success` → returns `files: [{ path, name, download_url }]`.
6. `error` → returns an `error` message (often a missing prerequisite to install).
7. **Download** — `GET /download?path=<path>` (served only from inside `output/` and `output/api_uploads/`).

Chaining — Upload a file once with `POST /files/upload` (multipart, field `file`) and reuse the returned `path` in any `*_path` field (`image_path`, `video_path`, `audio_path`, `mask_path`, `mesh_path`). A job's output path is a local path on the server, so you can feed it straight into the next step with no intermediate download.

Gating: 403 with upsell

Each endpoint checks the same capability the desktop app checks. If your edition does not include a capability, the endpoint returns **HTTP 403** with upsell information (the minimum edition and how to unlock it) rather than running. The API itself is Studio Pro (`api.rest`); the tables below list the extra capability each endpoint requires.

Endpoint reference

Infra

Method	Path	Purpose
GET	`/health`	`{ api, native_backend_up }`
POST	`/files/upload`	Upload a file (multipart) → `{ path, download_url }`
GET	`/jobs/{job_id}`	Job status (plus `files[]` on success)
GET	`/download?path=`	Download a produced file

Images (`tag: images`)

Method · Path	Key parameters (defaults)	Capability
POST `/image/generate`	`prompt*`, `model*`, `arch=flux` (flux/flux2/zimage/ernie), `negative`, `width=1024`, `height=1024`, `steps=20`, `guidance=3.5`, `seed=0`, `loras`, `pulid_face`, `pulid_strength=0.8`	`cap.gen_images`
POST `/image/edit`	`prompt*`, `image_path*`, `model*`, `arch=kontext` (kontext/flux2_kontext), `steps=20`, `guidance=2.5`, `seed=0`, `max_side=1536`	`cap.gen_images`
POST `/image/recolor`	`image_path*`, `target*`, `subject`, `model*`, `steps=20`, `guidance=2.5`, `seed=0`	`cap.inpaint`
POST `/image/inpaint`	`prompt*`, `image_path*`, `mask_path*`, `model*`, `arch=inpaint` (inpaint/flux2_inpaint), `steps=20`, `guidance=30.0`, `seed=0`	`cap.inpaint`
POST `/image/outpaint`	As inpaint (mask marks the region to expand)	`cap.inpaint`
POST `/image/upscale`	`image_path*`, `model*` (e.g. `4x-UltraSharpV2.safetensors`), `target_scale=4.0`, `seed=0`	`cap.upscale_image`
POST `/image/segment`	`image_path*`, `text*`, `threshold=0.3`, `use_sam=true` → mask PNG	`cap.sam`
POST `/image/faceswap`	`target_path*`, `source_face_path*` (InsightFace)	`cap.gen_images`

Video (`tag: video`)

Method · Path	Key parameters (defaults)	Capability
POST `/video/generate`	`prompt*`, `negative`, `width=768`, `height=512`, `fps=25`, `length_s=5`, `steps=40`, `guidance=3.0`, `seed=42`, `loras`; **i2v**: `image_path` (+`ref_strength=1.0`, `i2v_method=adguide`); **a2v**: `audio_path` (+`audio_scale=3.0`)	`cap.video`
POST `/video/keyframes`	`prompt*`, `keyframes*` `[{path,frame,strength}]` + base params	`cap.video`
POST `/video/extend`	`prompt*`, `video_path*`, `add_seconds=5`, `seg_seconds=0`, `steps=8`	`cap.extend_retake`
POST `/video/retake`	`prompt*`, `video_path*`, `start_time`, `end_time`, `steps=40`	`cap.extend_retake`
POST `/video/v2v_edit`	`prompt*`, `control_video*`, `edit_lora*`, `control_strength=1.0`, `length_s=5`	`cap.video`
POST `/video/controlnet`	`prompt*`, `control_video*`, `control_type=canny` (canny/depth/pose), `control_strength=0.8`, `image_path?`	`cap.controlnet`
POST `/video/lipsync`	`image_path*`, `audio_path*`, `prompt`, `audio_scale=3.0`, `separate_vocals=false`, `talk_region` (mouth/face), `talk_mask_offset=0`	`cap.video`
POST `/video/enhance`	`video_path*`, `engine=seedvr2` (seedvr2/flashvsr), `seedvr2_model`, `resolution=1080`, `batch_size=5`, `block_swap=0`, `scale=4.0`	`cap.restore`
POST `/video/rife`	`video_path*`, `factor=2`	`cap.restore`
POST `/video/faceswap`	`video_path*`, `source_face_path*`	`cap.video`

> Width/height are rounded to multiples of **64** (required by LTX-2). Clips over 150 frames are segmented automatically (memory-bounded chunks, no out-of-memory on long clips).

Music & Audio (`tag: music`)

Method · Path	Key parameters (defaults)	Capability
POST `/music/song`	`tags*` (style), `lyrics`, `name=song`, `engine=heartmula` (heartmula/acestep), `duration_s=120`, `instrumental=false`, `version=3B`, `guidance=15.0` (acestep), `lazy_load=false`	`cap.music_song`
POST `/music/sonify`	`video_path` (empty = text→audio), `prompt`, `duration_s=8`, `cfg_strength=4.5`, `num_steps=25` (MMAudio)	`cap.music_sonify`
POST `/music/music_video`	`lyrics*`, `image_path*`, `song_path`, `width=704`, `height=512`, `fps=25`, `scene_seconds=6`, `video_style`, `steps=8`	`cap.music_video`
POST `/audio/tts`	`text*`, `engine=edge` (edge/kokoro/f5tts/xtts2/fish/vibevoice/cosyvoice2), `voice_id`, `language`, `sample_path?` (clone), `sample_transcript`	`cap.audio_tts_clone`
POST `/audio/separate_vocals`	`audio_path*` (demucs → vocals.wav)	`cap.audio_tts_clone`
POST `/audio/transcribe`	`audio_path*` (whisper → txt)	`cap.audio_tts_clone`

3D / Rig / World (`tag: 3d`)

Method · Path	Key parameters (defaults)	Capability
POST `/three d/image_to_3d`	`image_path*`, `name=model`, `seed=1`, `ss_steps=12`, `slat_steps=12`, `simplify=0.95`, `texture_size=1024` (TRELLIS → GLB)	`cap.three d`
POST `/three d/rig`	`mesh_path*`, `name=rig`, `template=Mixamo`, `file_format=glb` (SkinToken → skeleton + skinning)	`cap.three d`
POST `/three d/world`	`name=world`, `image_path` (empty = text→scene), `prompt`, `resolution=1600`, `low_vram=false`, `mesh=true` (WorldGen → GLB/PLY)	`cap.three d`
POST `/three d/convert`	`glb_path*`, `fmt=fbx` (fbx/obj, via Blender)	`cap.three d`

> Live webcam mocap and Blender retarget/animation remain interactive in the desktop GUI and are not exposed over the API.

Film (`tag: film`)

Method · Path	Key parameters (defaults)	Capability
POST `/film/script`	`subject*`, `style`, `max_tokens=2000` (LLM → treatment + scenes, .txt)	`cap.filmstudio`
POST `/film/asset_image`	`prompt*`, `width=1024`, `height=1024`, `seed=7`, `arch=flux`, `face_ref` (PuLID), `ref_image` (Kontext), `loras`, `negative`	`cap.filmstudio`
POST `/film/shot_video`	`prompt*`, `width=768`, `height=512`, `fps=25`, `duration_s=4`, `seed=7`, `keyframes`, `loras`, `audio_path`, `negative`	`cap.filmstudio`

> Full project orchestration (database, scheduler, editing, dubbing, Auto mode) stays in the stateful Film Studio GUI. The song→video music clip lives at `/music/music_video`.

Training (`tag: training`)

Method · Path	Purpose	Capability
POST `/train/lora_image`	Auto-dataset photo LoRA training	`cap.lora_image_train`
POST `/train/lora_video`	Auto-dataset video LoRA training	`cap.lora_video_train`

Catalog (`tag: catalog`)

Method · Path	Purpose
GET `/features`	Activation state per capability (identical to desktop) + upsell info
GET `/models?task=`	Available models per task (`image` / `video` / `upscale` / `restore`) + LoRA + recommended model
GET `/components?model=`	Auto components for a chosen model (arch / VAE / text encoder / steps / CFG / compatible LoRA)
GET `/loras?model=`	LoRA list (all, or only those compatible with a given model)

Prerequisites (gated features)

Some features need isolated virtual environments or weights. If they are missing, the job returns `error` with an install hint: SeedVR2, FlashVSR, TRELLIS, WorldGen, HeartMuLa, ACE-Step, MMAudio, TTS-clone engines (f5/xtts/fish/vibevoice), face-swap (InsightFace), and PuLID all resolve to a `scripts/install_*.py` helper.

Multi-agent skill packs

The `agents/` folder ships machine-readable contracts (`openapi.json`, `tools.openai.json`, `manifest.json`) plus a skill pack per agent — **Hermes** is the default, alongside `openclaw`, `claude_code`, `antigravity`, `openai_codex`, and `generic`. Each pack contains a `SKILL.md` and a `plugin.json`. Rebuild them with `python -m agents.build_packs` and install one with `python -m agents.install_skill [--agent <name>]`. These packs let an AI agent drive the whole API — submit jobs, poll, and download results — using the same async model described above.

Worked example — image then video

```
# 1. Generate an image (Z-Image)
curl -s -X POST http://127.0.0.1:8200/image/generate \
  -H "Content-Type: application/json" -d '{
    "prompt":"a humanized beet chasing a flying winged opossum, storybook",
    "arch":"zimage","model":"E:/AI/Models/diffusion_models/Z-Image/....safetensors",
    "width":1024,"height":768,"steps":10,"guidance":3.5,"seed":42 }'
# → {"job_id":"...","status":"pending"}   then poll:  GET /jobs/<job_id>

# 2. Animate the produced image (i2v, reuse its server-side path)
curl -s -X POST http://127.0.0.1:8200/video/generate \
  -H "Content-Type: application/json" -d '{
    "prompt":"the beet chases the flying opossum across a meadow, playful chase",
    "image_path":"E:/AI/ltxstudio/output/native_zimage_42.png",
       "width":768,"height":576,"fps":16,"length_s":10,"steps":10,"guidance":3.0,"seed":42 }'
```

Tips & Tricks

A skimmable, cross-module guide to getting the best results — the settings that matter, and the traps to avoid.

This chapter consolidates the practical advice scattered through the app's tooltips, presets, and internal notes. Each section is a domain you can jump to. For per-feature detail, follow the cross-links to Images, Quality & restore, Audio & voice, Music, 3D, and Advanced settings.

Images

- **Choose the scheduler deliberately — it changes image quality a lot.** For FLUX images, the flow-aware schedulers **bong_tangent / kl_optimal / beta57** are often the best (high quality). **karras** gives classic detail, **beta** suits distilled models, and **exponential** is softer.
- **Sampler:** euler and dpmpp are reliable. **res_2s** gives more detail on FLUX images — but never use it for LTX video, where it breaks (see Video below).
- **Face identity from a photo:** with PuLID, keep the strength around **0.8** (recommended range 0.6–0.9). If faces look plastic, use a **base bf16** model rather than a distilled/quantized one, and push the reference/image strength higher.
- **Generate at full resolution.** Leave any half-resolution / downscale option **off** for maximum native detail; it only hides the VAE decoder grid at the cost of softness.
- **Guidance:** FLUX text-to-image sits comfortably around guidance 3.5; inpaint/fill wants much higher guidance (~30) to fill cleanly.

Video

- **Use the verified sampler.** The default `euler_ancestral_cfg_pp` with `LTXVScheduler (shift)` and **CFG 1.0** is the tuned, verified LTX-2.3 pipeline — the guidance comes from NAG, not from a high CFG.
- **Avoid RES samplers on video.** `res_2s`, `res_multistep`, and similar look great on paper for flow but produce **green noise** on LTX-2.3 (verified A/B) — they were removed for exactly this reason. If you see cyan/green artifacts, you are on the wrong sampler.
- **The base shift is the #1 video quality lever.** In LTXVScheduler, a **higher base shift** (default 0.95) spends more steps on structure and motion (temporal coherence, less flicker); a **lower** shift gives finer per-frame detail but risks inconsistency between frames.
- **Image-to-video "image strength" controls motion feel.** Set the soft-reference **image strength to 0.45–0.60** for natural motion; **0.80+** sticks tightly to the first frame but makes the motion stiff.
- **Use the built-in presets as starting points:** ⚡ **FAST** (8 steps, reduced res — a draft to validate the idea, ~1 min), 🎯 **PRO** (24 steps, full 1280×704 — best balance), 💎 **HQ** (44 steps, no cache — maximum quality, slowest).
- **Long videos are chunked automatically.** Clips over ~150 frames are segmented into memory-bounded chunks and stitched, so you can request long durations without hitting out-of-memory.
- **TeaCache caveat.** With the raw L1 metric (no LTX-2.3 coefficients) the aggressive thresholds skip almost everything — fast but destructive (PSNR collapses). Presets ship it at **0.0**; only nudge the manual slider up if you have validated a low threshold on your own content.

Audio & Voice

- **Match the TTS engine to the job.** Use a lightweight engine (edge/kokoro) for fast narration and a clone-capable engine (f5tts/xtts2/fish/vibevoice/cosyvoice2) when you supply a voice sample. Provide the `sample_transcript` for the best clone fidelity.
- **Separate vocals before lip-sync** when your source audio has music under the speech — cleaner vocals give tighter lip motion.
- **Lip-sync strength:** aim for **1.5–3.0** on the sync scale. 1.0 is weak; above 3 the mouth motion becomes exaggerated.

Music

- **Sampler matters for lyric/style adherence.** For the music engine, **pingpong (SDE)** gives the best coherence and style/lyric alignment (recommended when you need the lyrics honored); **euler** is fast and standard; **heun** is more precise but slower.
- **Guidance:** the ACE-Step engine defaults to guidance ~15.0 — a strong pull toward your tags and lyrics.
- **Write descriptive tags.** The style tags (genre, instruments, mood) drive the result as much as the lyrics do.

3D

- **Feed a clean, well-lit single subject** for image→3D; busy backgrounds and occlusions hurt mesh quality.
- **Tune the mesh trade-offs:** more `ss_steps` / `slat_steps` (default 12) improves quality at the cost of time; `simplify` (default 0.95) and `texture_size` (default 1024) trade detail for file size.
- Rigging expects a clean mesh; convert to FBX/OBJ afterward via the built-in Blender step if your target tool needs it.

Film Studio

- Keep character **assets consistent** by reusing the same face reference (PuLID) and LoRA across shots.
- Generate **asset images first, then shots** — the shot video step can take keyframes and LoRA to keep continuity.

- Heavy project orchestration (scheduler, editing, dubbing, Auto mode) is stateful and lives in the GUI; use it rather than scripting each shot by hand.

Performance / VRAM

- **Pick a weight format for your GPU.** On 32 GB, **NVFP4 mixed** is fastest (~14 GB, ~2× faster than BF16, almost no visible difference). **FP8 Scaled** is the best universal compromise. For smaller cards, GGUF quants scale down: **Q8_0** (16 GB, 98–99% quality), **Q6_K** (12 GB, ~95%), **Q5_K_M** (8 GB sweet spot, ~90%), **Q4_K_M** (6–8 GB, 75–85%).
- **VRAM savings stack multiplicatively:** text-encoder offload after encoding, VAE offload except during decode, transformer block-swap, and unloading models between generations.
- **Use /free when a video job risks OOM** — it releases held VRAM before the next run.
- **Slow first run is normal.** The first time you open a heavy module the app downloads its on-demand weights; subsequent runs are fast.
- **Use the prompt enhancer** to expand terse prompts into richer descriptions before generating, especially for video where detailed motion cues help coherence.

> See the Appendix for the troubleshooting table and the full capability → edition matrix.

Appendix

Reference material: glossary, shortcuts, file locations, languages, the capability matrix, editions, and troubleshooting.

Glossary

Term	Meaning
LoRA	A small add-on file that adapts a base model toward a style, subject, or effect, at an adjustable strength.
Checkpoint	A full base model — the main set of weights that generates images or video.
VAE	Variational auto-encoder; converts between the compact "latent" space the model works in and real pixels.
Text encoder	The component that turns your prompt text into the conditioning the model understands (e.g. Gemma, Qwen3, Mistral).
CFG	Classifier-free guidance; how strongly the model follows the prompt. LTX video uses CFG 1.0 (guidance comes from NAG).
Steps	How many denoising iterations run; more steps = more refinement and more time.
Sampler	The algorithm that removes noise each step (e.g. `euler_ancestral_cfg_pp`).
Scheduler	The schedule of noise levels across steps (e.g. `LTXVScheduler (shift)`); strongly affects quality.
t2v / i2v / a2v	Text-to-video / image-to-video / audio-to-video generation modes.
Inpaint / Outpaint	Regenerate a masked region inside an image (inpaint) or extend the image beyond its edges (outpaint).
ControlNet	Guides generation with a structural reference (canny edges, depth, or pose).
Upscale	Increase resolution while adding plausible detail.
Quantization (fp8 / fp4 / gguf)	Compressed weight formats that cut VRAM use with small quality loss.
PuLID	A method that transfers a face's identity from a reference photo onto a generated image.
SAM2	Segment Anything 2; produces precise object masks, and tracks them across video frames.
Keyframe	A fixed image pinned to a specific frame/time to anchor a video.
Storyboard	A sequence of timed keyframes that produce one continuous video without manual editing.
Mocap	Motion capture; driving a rig from live webcam motion.
Rigging	Adding a skeleton to a 3D mesh so it can be posed and animated.
Skinning	Binding a mesh's surface to its skeleton so it deforms correctly when posed.
Splat	A Gaussian-splat representation of a 3D scene (points with color and shape), used by world generation.
OTIO	OpenTimelineIO; an interchange format for editing timelines.

Keyboard shortcuts

Shortcut	Action
Ctrl+G	Open the Navigator

Beyond this, AI Creative Center is driven through its buttons, tabs, and context menus rather than keyboard shortcuts.

File locations

What	Where
Generated outputs	`output/` (and API uploads in `output/api_uploads/`)
Models root	`E:\AI\Models` (organized by type: `diffusion_models`, `TextEncoders`, `LoRA`, `VAE`, `Checkpoints`, …)
GUI state (tab settings, scroll positions)	`data/gui_state.json`
User preferences (favorites, provenance signature settings, custom presets)	`data/user_prefs.json`
Weights configuration / first-run flag	`data/weights_config.json`
AI Film Studio project database	`core/film_studio/studio.db` (SQLite)
Hugging Face cache (one-time gated downloads)	`E:\hf_cache` (via `HF_HOME`)

> The app is 100% local at runtime. A Hugging Face token is used only for one-time downloads of gated weights, never for cloud inference.

Supported languages

The interface is fully translated into **15 languages**:

Arabic · German · Greek · English · Spanish · French · Hebrew · Hindi · Hungarian · Italian · Korean · Portuguese · Russian · Ukrainian · Simplified Chinese.

Switch language from the interface settings; the whole UI, including tooltips, is localized.

Capability → minimum edition

Feature	Capability	Min edition
Image generation (Demo capped 768 px/20; Supporter 1024 px/200; Full+ unlimited)	`cap.gen_images`	Demo
Invisible provenance signature	`cap.stego`	Demo (all)
Forced brand watermark on outputs	`cap.watermark`	Demo & Supporter only (removed Full+)
Image upscale	`cap.upscale_image`	Supporter
Audio / TTS / voice clone	`cap.audio_tts_clone`	Full
Video (t2v / i2v / a2v)	`cap.video`	Full
Extend video + ReTake	`cap.extend_retake`	Full
ControlNet	`cap.controlnet`	Full
Inpaint / Outpaint / Recolor	`cap.inpaint`	Full
Object segmentation (SAM)	`cap.sam`	Full
Enhance quality / restore / super-res / face	`cap.restore`	Full
Song / instrumental	`cap.music_song`	Full
Video sonification	`cap.music_sonify`	Full
Music video	`cap.music_video`	Full
Image→3D model	`cap.model3d` (`cap.threed`)	Full
Rig 3D	`cap.rig3d`	Full
Mocap animation	`cap.mocap`	Full
3D World (WorldGen)	`cap.world_gen`	Full
AI Film Studio (+ asset/script/scenografia/montaggio/voice/lora sub-caps)	`cap.filmstudio`	Full
Auto-dataset photo LoRA	`cap.lora_image_train`	Full
Auto-dataset video LoRA	`cap.lora_video_train`	Full
HDR/EXR frame export	`cap.exr_export`	Full
Unlocked (18+) mode	`cap.adult_mode`	Unlocked 18+ (premium)
Import own checkpoints/LoRA	`cap.import_models`	Studio Pro
Civitai import (search + download)	`cap.civitai_import`	Studio Pro
Per-node model overrides	`cap.node_overrides`	Studio Pro
Heavy generic FLUX LoRA training tab	`cap.train_lora`	Studio Pro
Local REST API	`api.rest`	Studio Pro

Editions

Edition	Price	Unlocked (18+)	Automation
Demo	€0	—	—
Supporter	$3	—	—
Full	€15	—	—
Unlocked 18+	€25	Yes	—
Studio Pro	€45	Yes	Local REST API

Each edition includes everything in the tiers below it. See the capability table above for exactly which features each tier unlocks.

Troubleshooting

Symptom	Cause	Fix
Out of memory (OOM)	Resolution/steps too high, or a large unquantized model.	Lower resolution or steps; switch to a quantized model (fp8/fp4/gguf); use **/free** to release VRAM before the next job.
Very slow first run	On-demand weights are downloading.	Wait for the one-time download to finish; later runs are fast.
Green / cyan noise in video	Wrong sampler for LTX.	Use **`euler_ancestral_cfg_pp`** with `LTXVScheduler (shift)`; avoid RES samplers on video.
Plastic-looking faces	Distilled/quantized model, or reference too loose.	Use a **base bf16** model and raise the image/reference strength (PuLID ~0.8).
A model does not appear in the dropdown	It is not linked or catalogued yet.	Rescan in the **Import Models / Weights** tab so the detector picks it up.

> For deeper tuning of samplers, schedulers, and per-module settings, see Tips & tricks and Advanced settings.